



<http://algs4.cs.princeton.edu>

PRIM'S ALGORITHM DEMO

- ▶ *Prim's algorithm*
- ▶ *lazy implementation*
- ▶ *eager implementation*



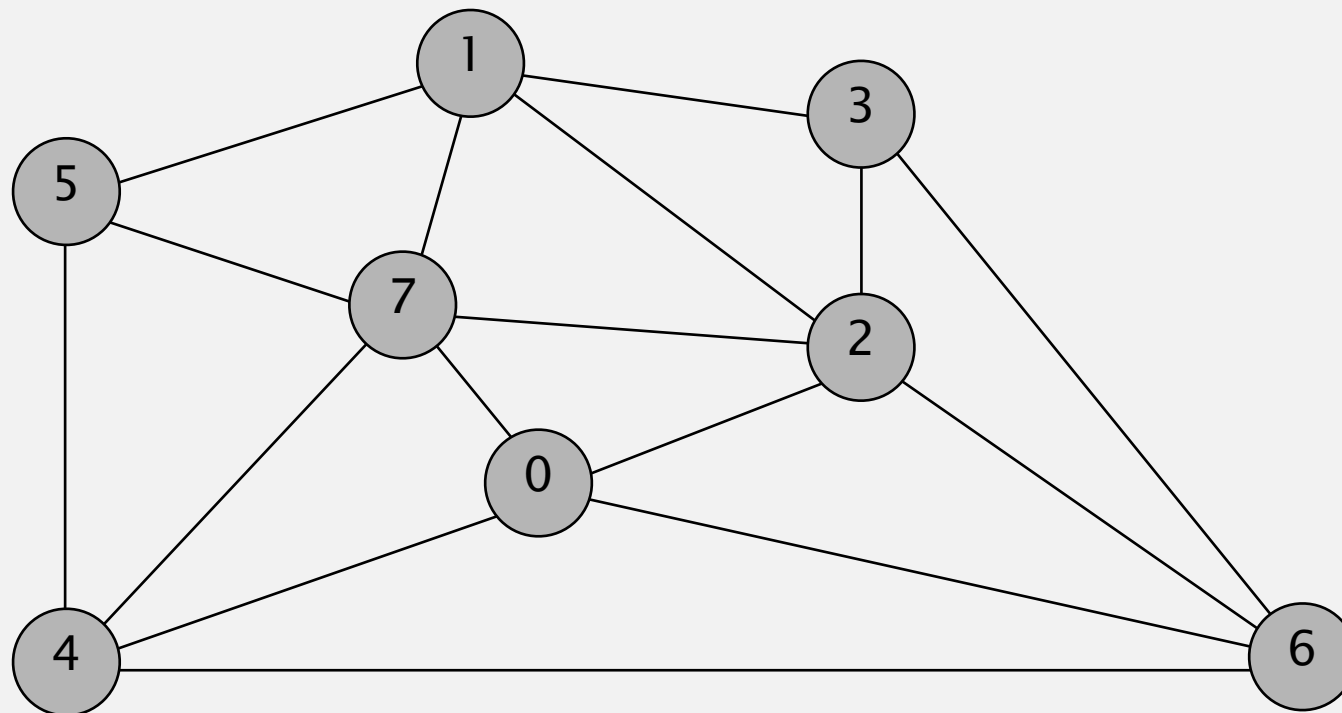
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PRIM'S ALGORITHM DEMO

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- ▶ *lazy implementation*
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Prim's algorithm demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

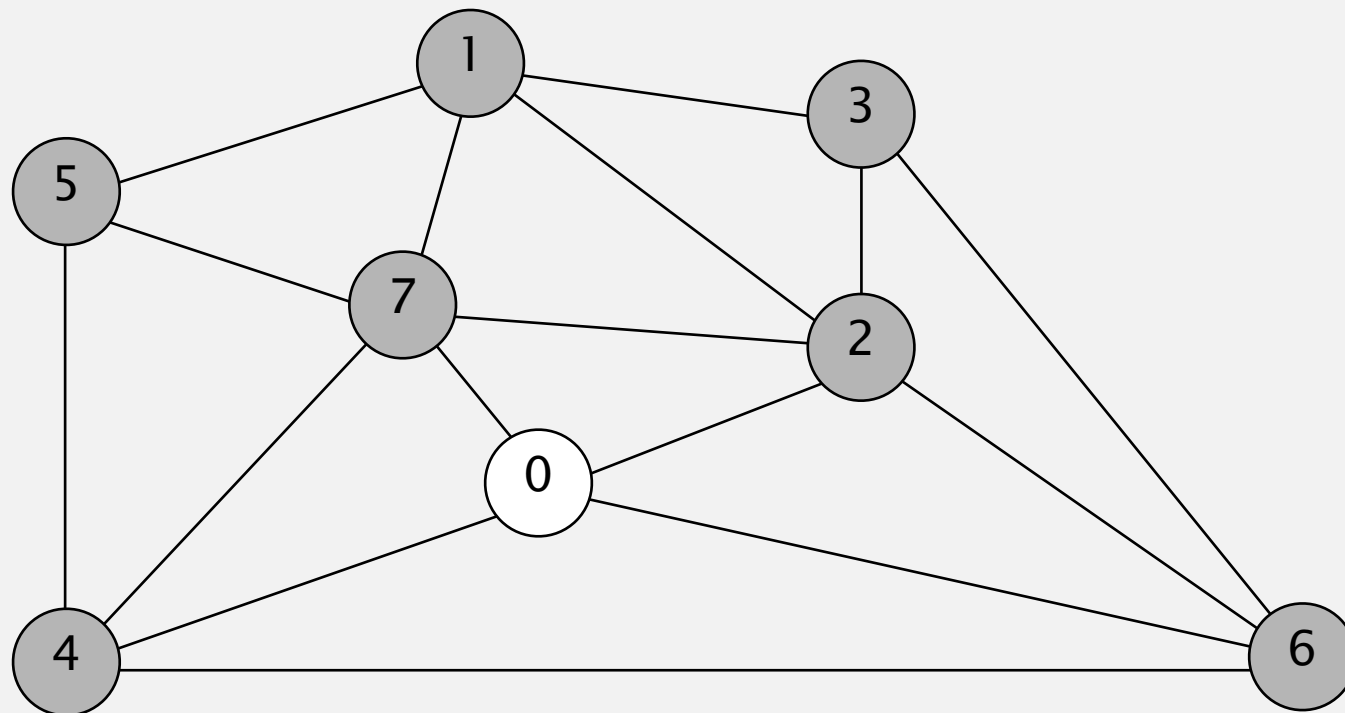


an edge-weighted graph

0-7	0.16
2-3	0.17
1-7	0.19
0-2	0.26
5-7	0.28
1-3	0.29
1-5	0.32
2-7	0.34
4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

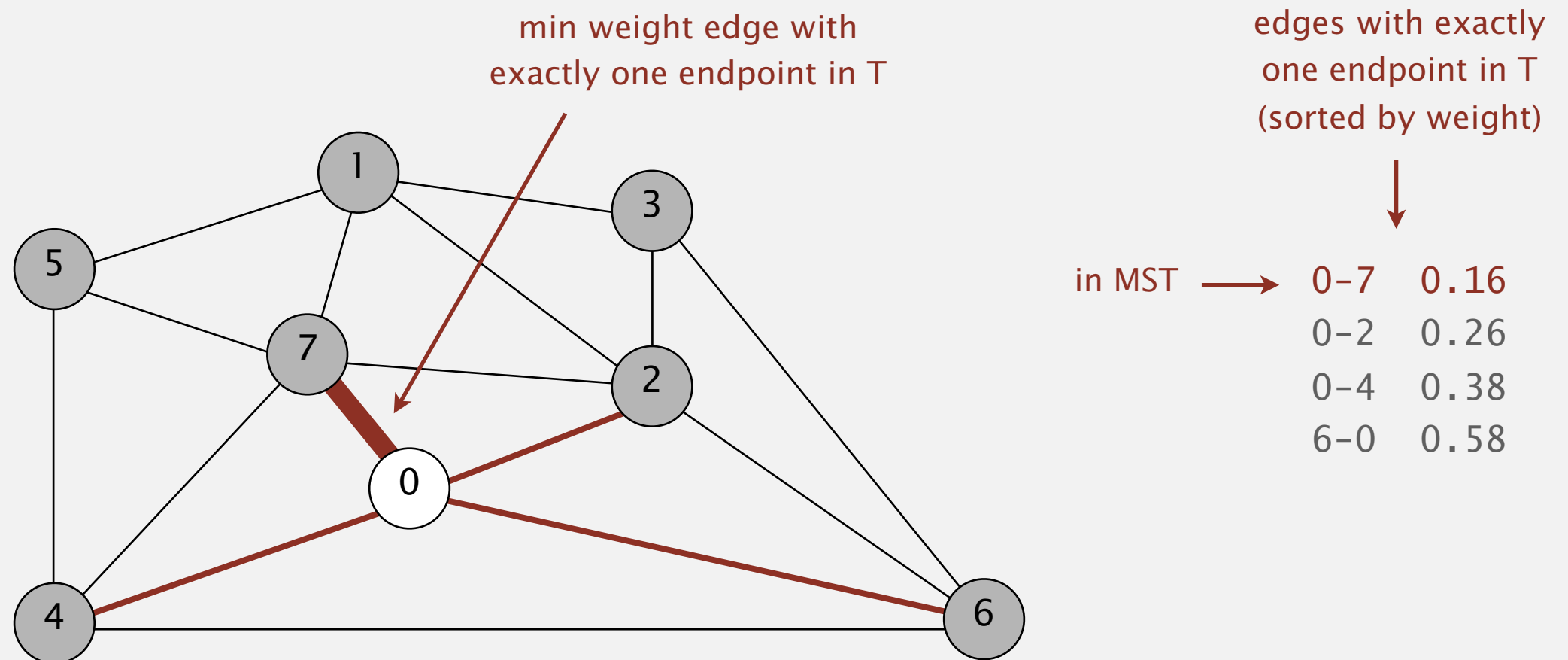
Prim's algorithm demo

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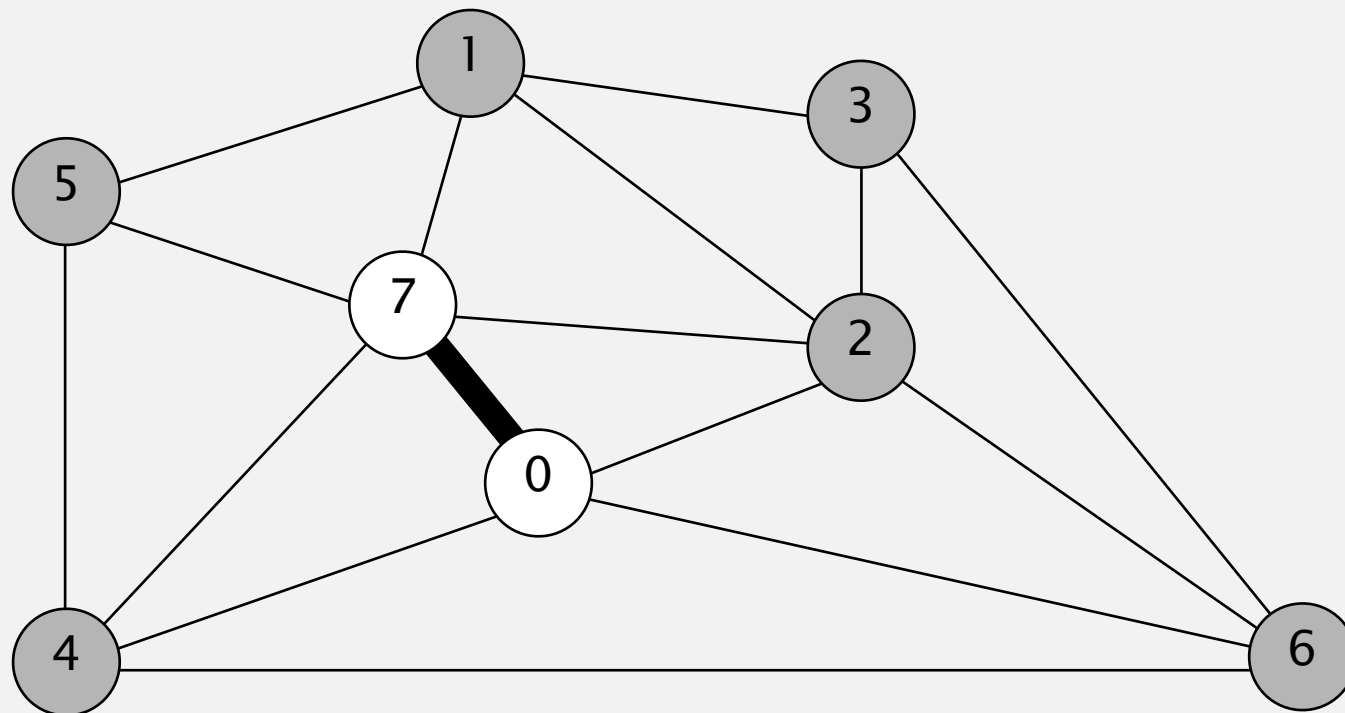
Prim's algorithm demo

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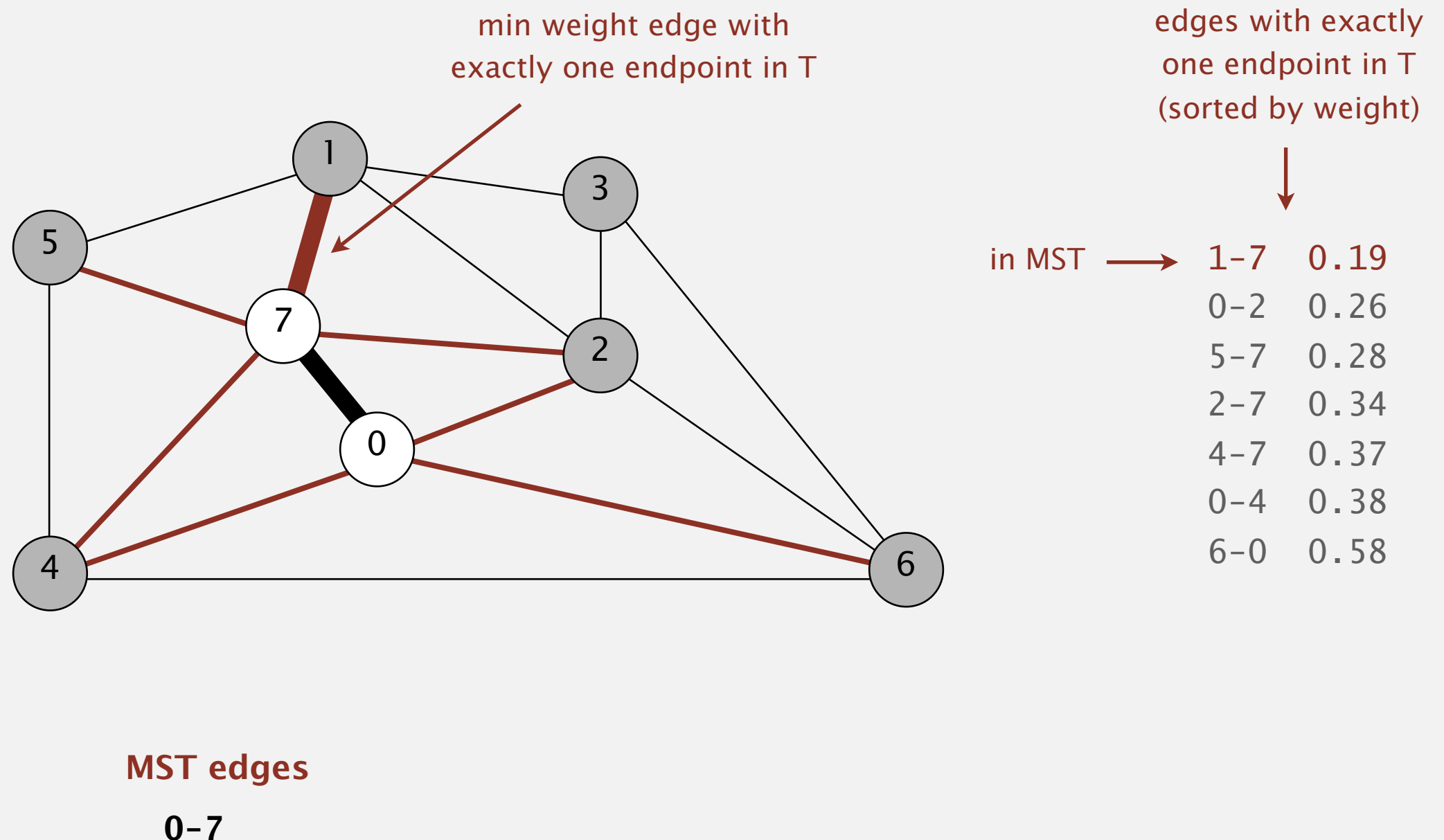


MST edges

0-7

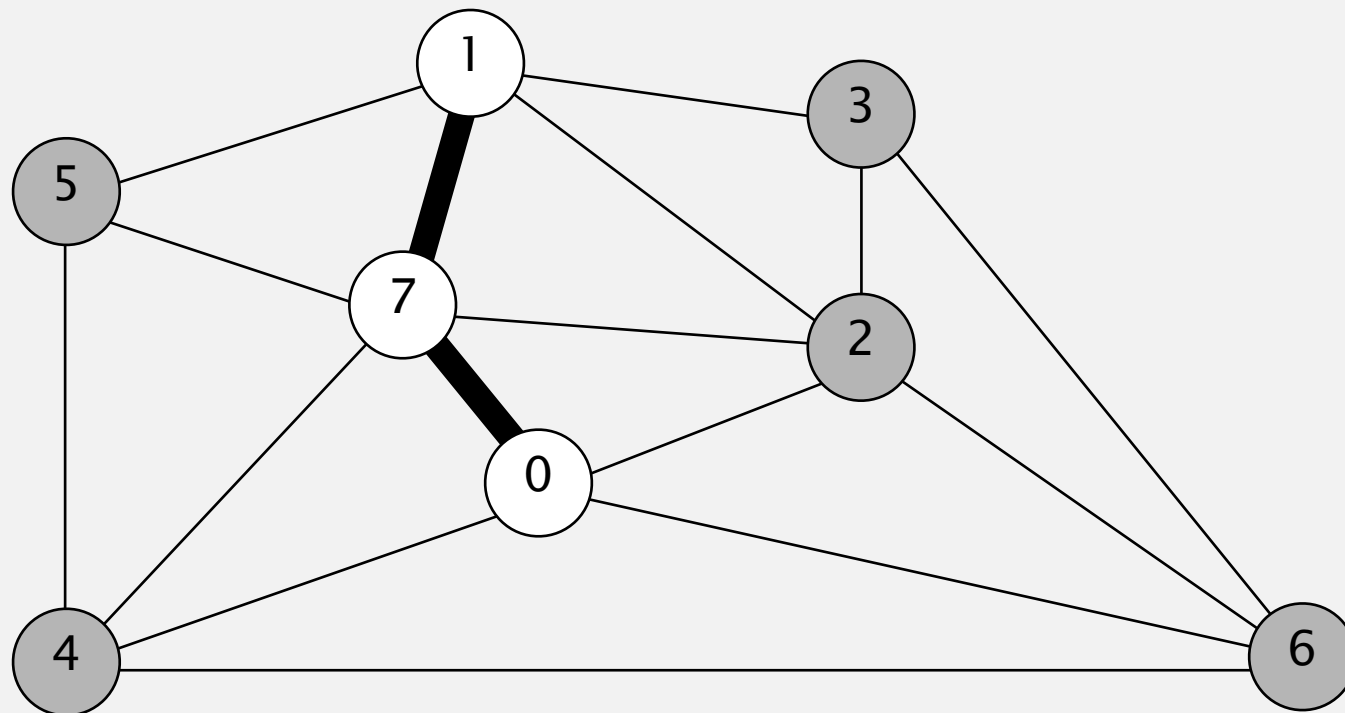
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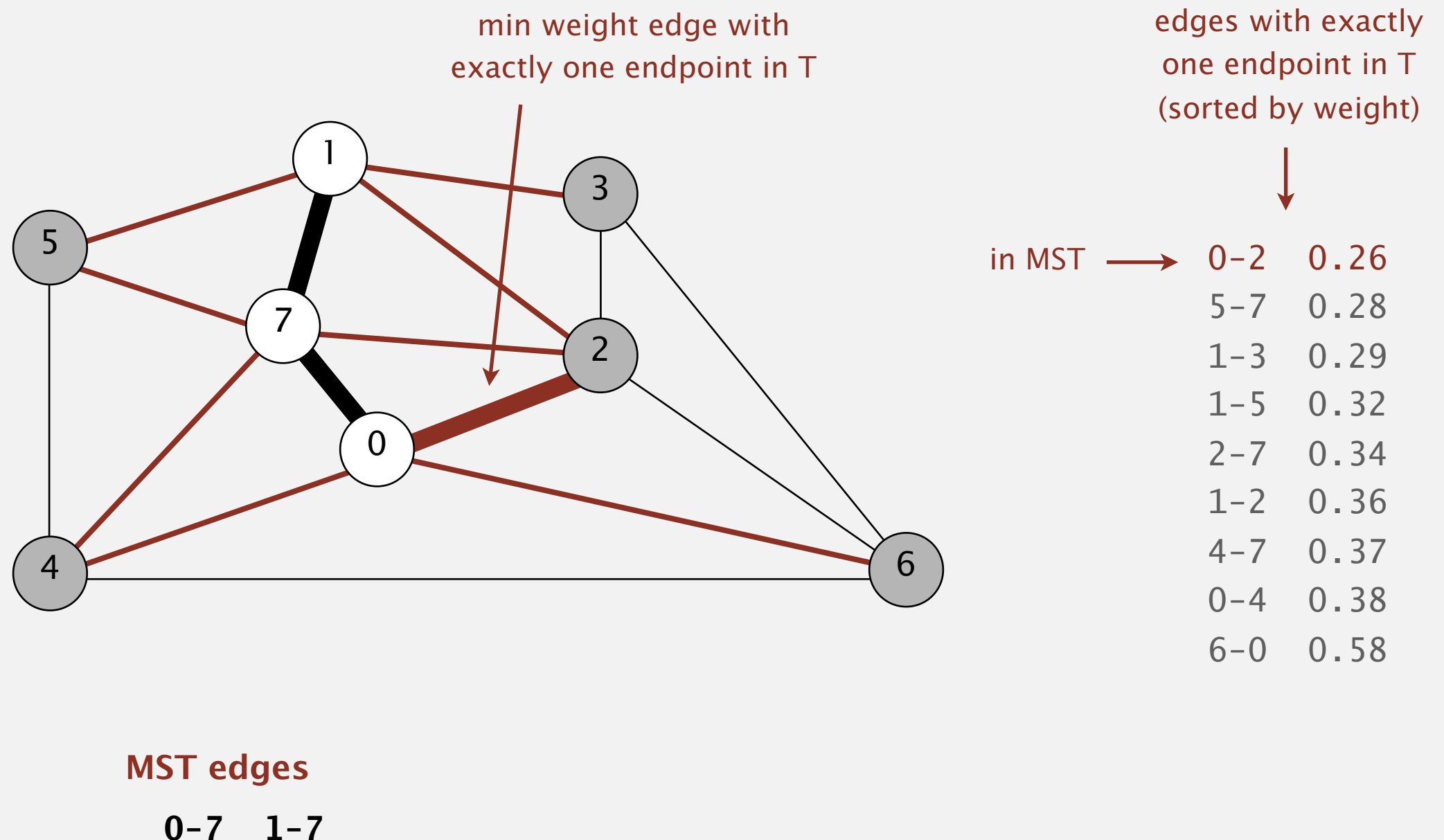


MST edges

0-7 1-7

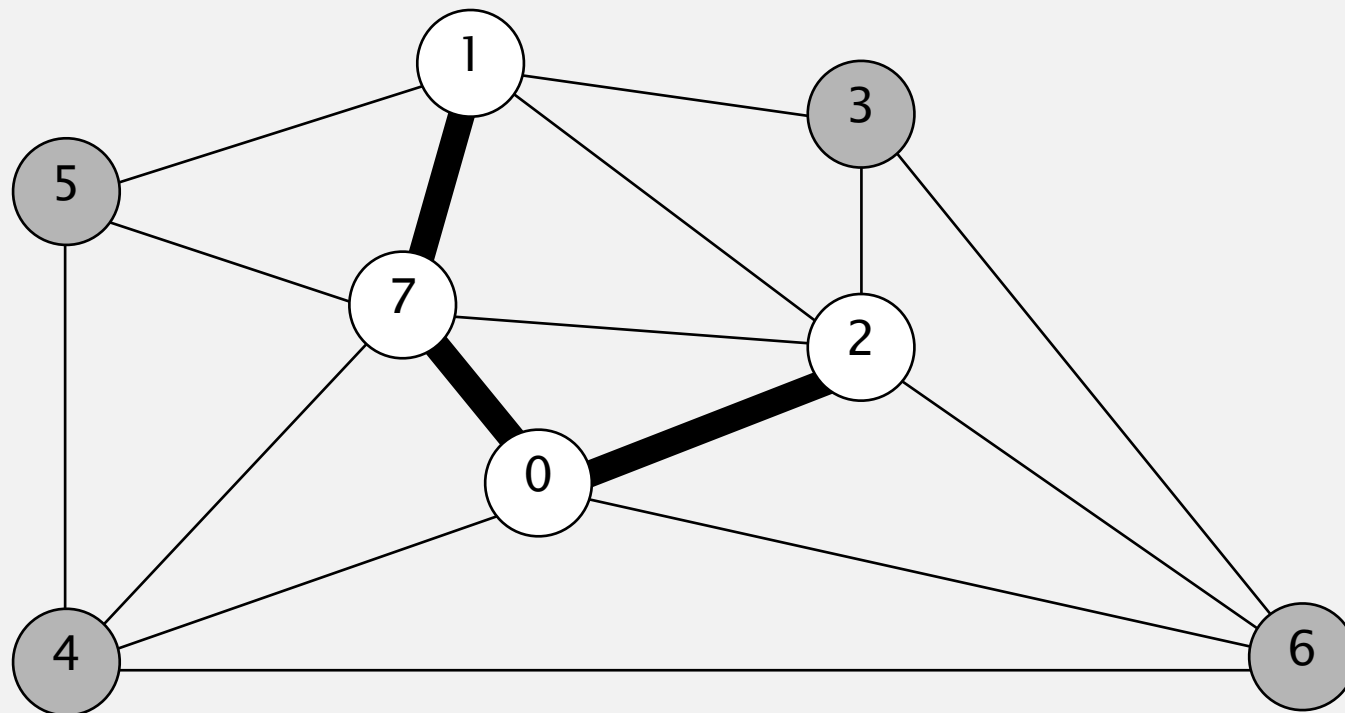
Prim's algorithm demo

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Prim's algorithm demo

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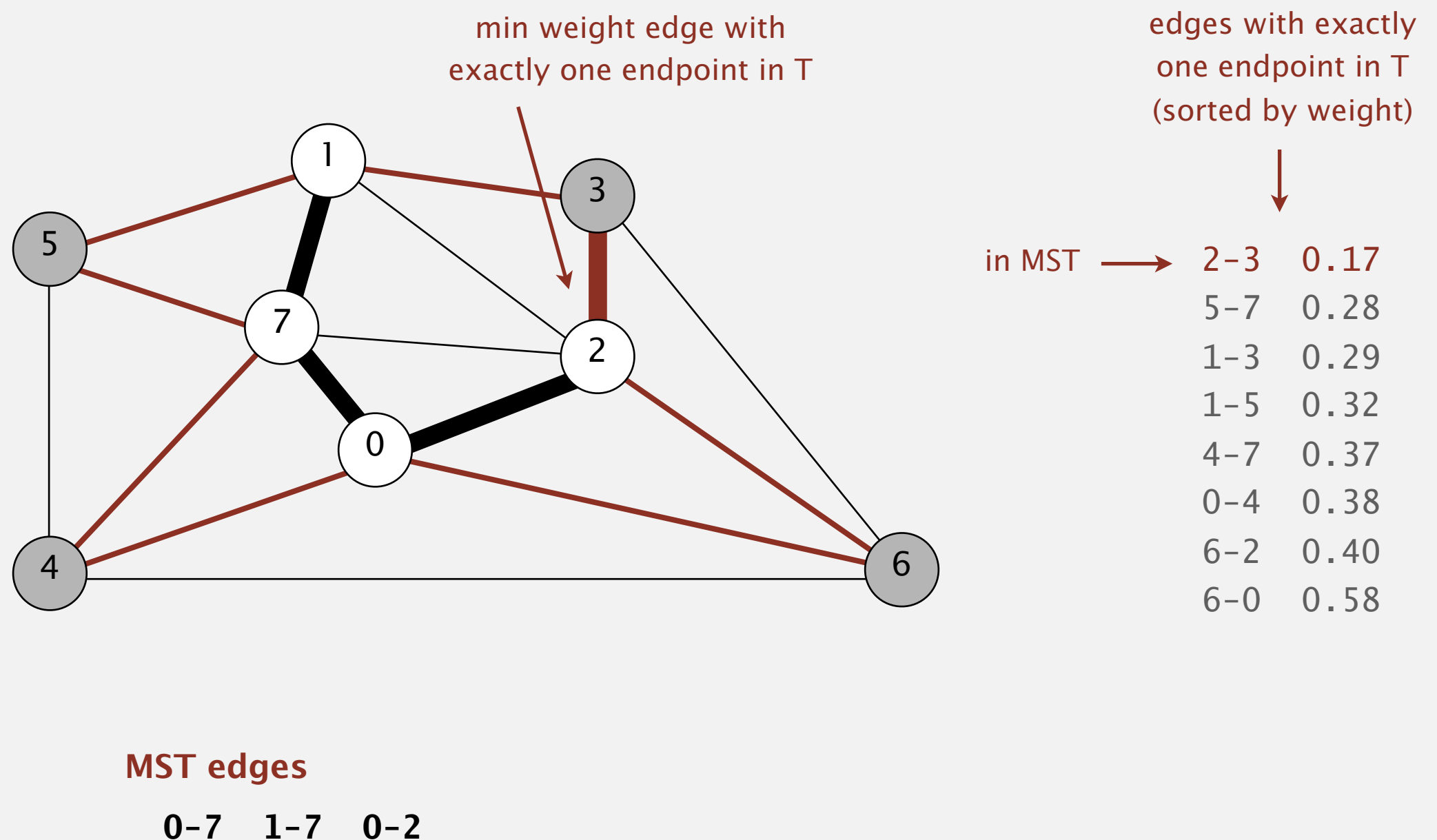


MST edges

0-7 1-7 0-2

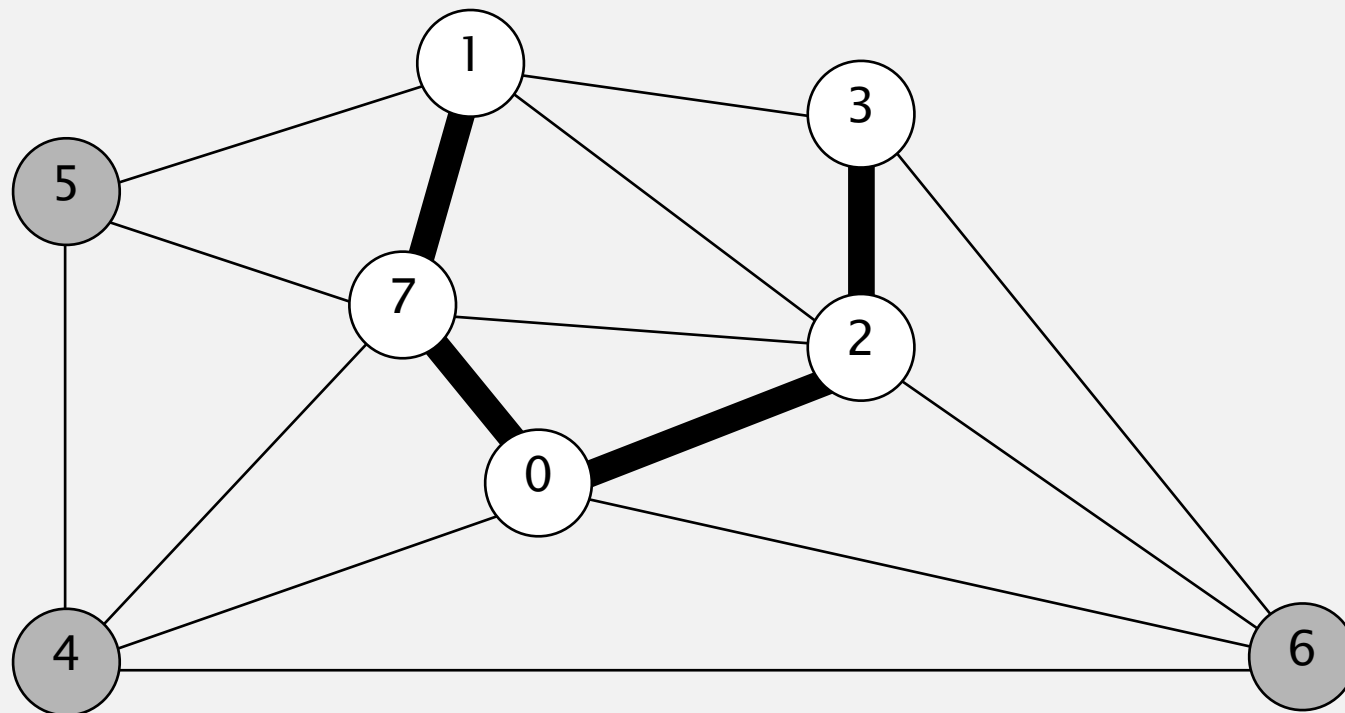
Prim's algorithm demo

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Prim's algorithm demo

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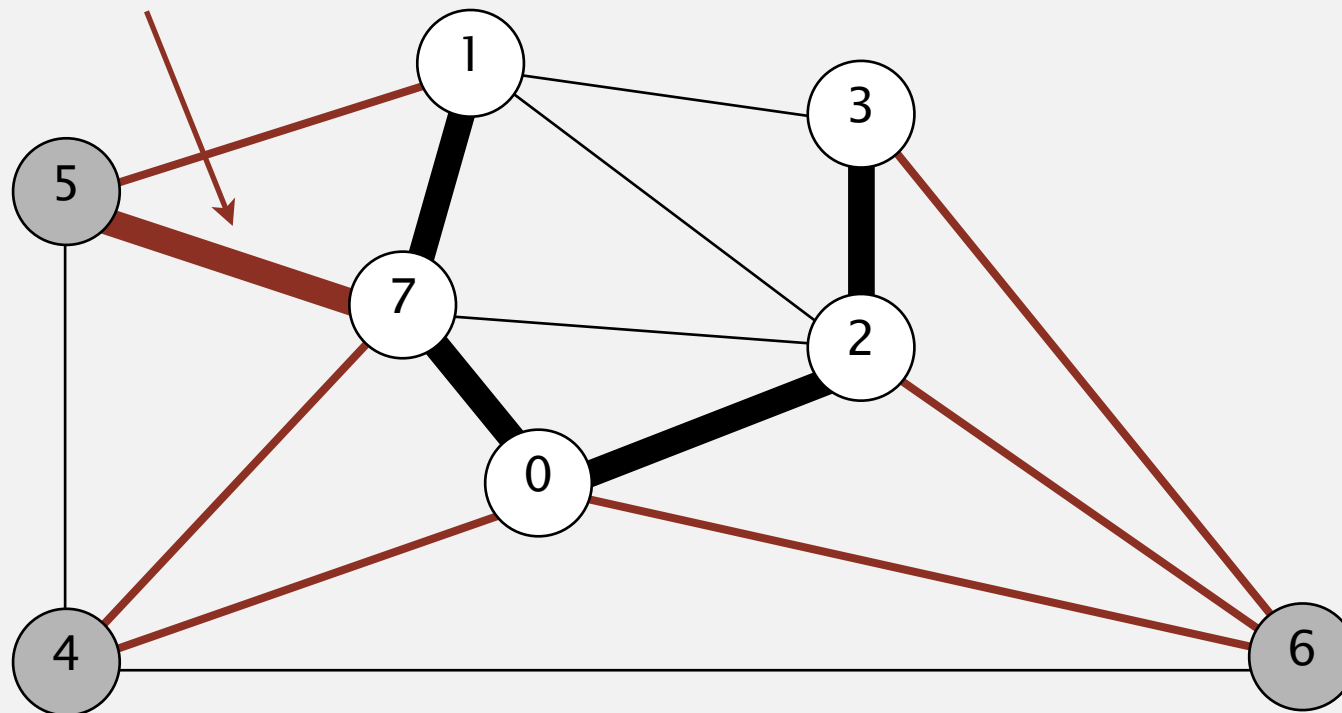
MST edges

0-7 1-7 0-2 2-3

Prim's algorithm demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

min weight edge with
exactly one endpoint in T



edges with exactly
one endpoint in T
(sorted by weight)

in MST →

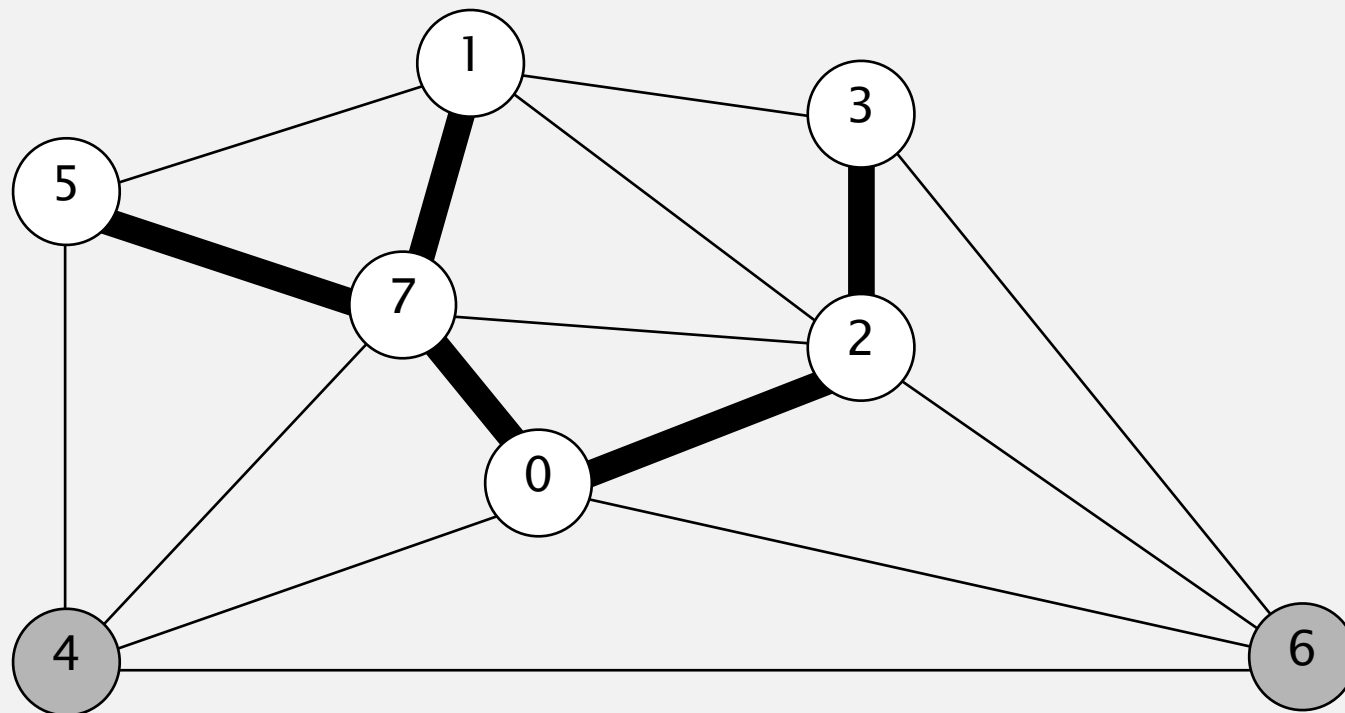
5-7	0.28
1-5	0.32
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

MST edges

0-7 1-7 0-2 2-3

Prim's algorithm demo

- Start with vertex 0 and greedily grow tree T .
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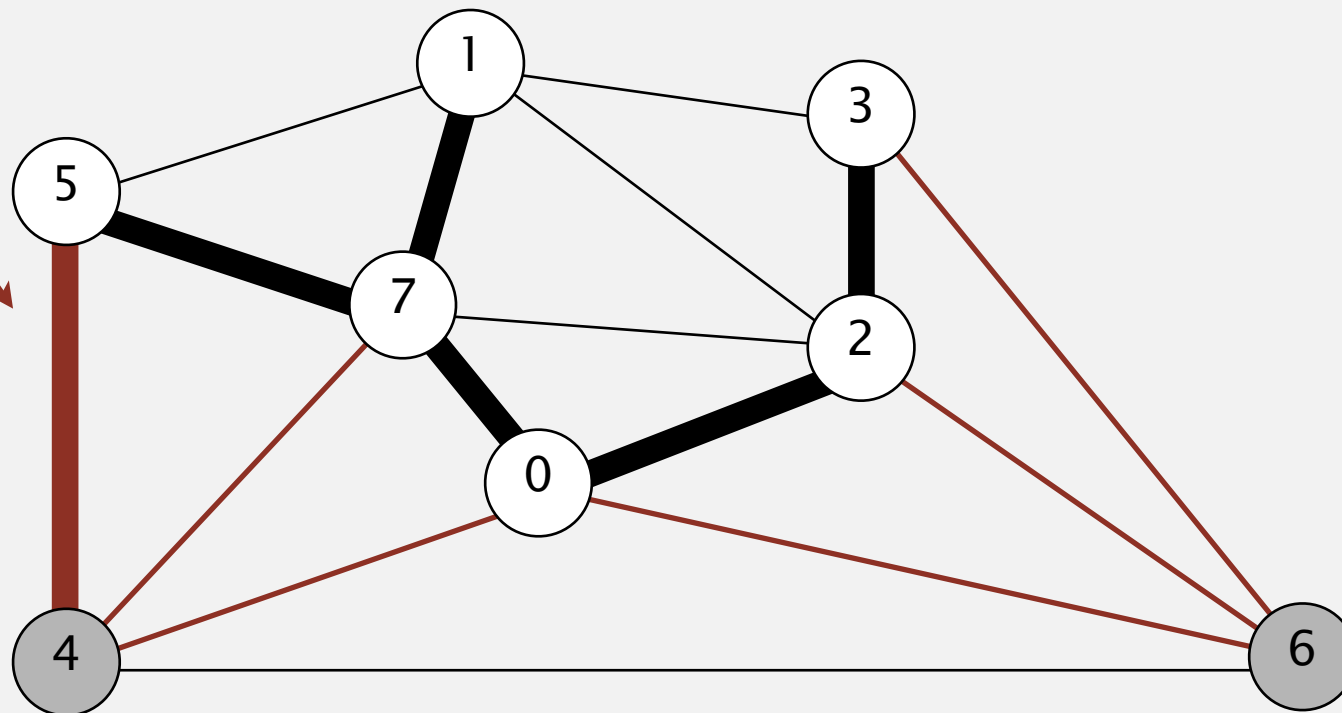
MST edges

0-7 1-7 0-2 2-3 5-7

Prim's algorithm demo

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min weight edge with
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edges with exactly
one endpoint in T
(sorted by weight)

in MST →

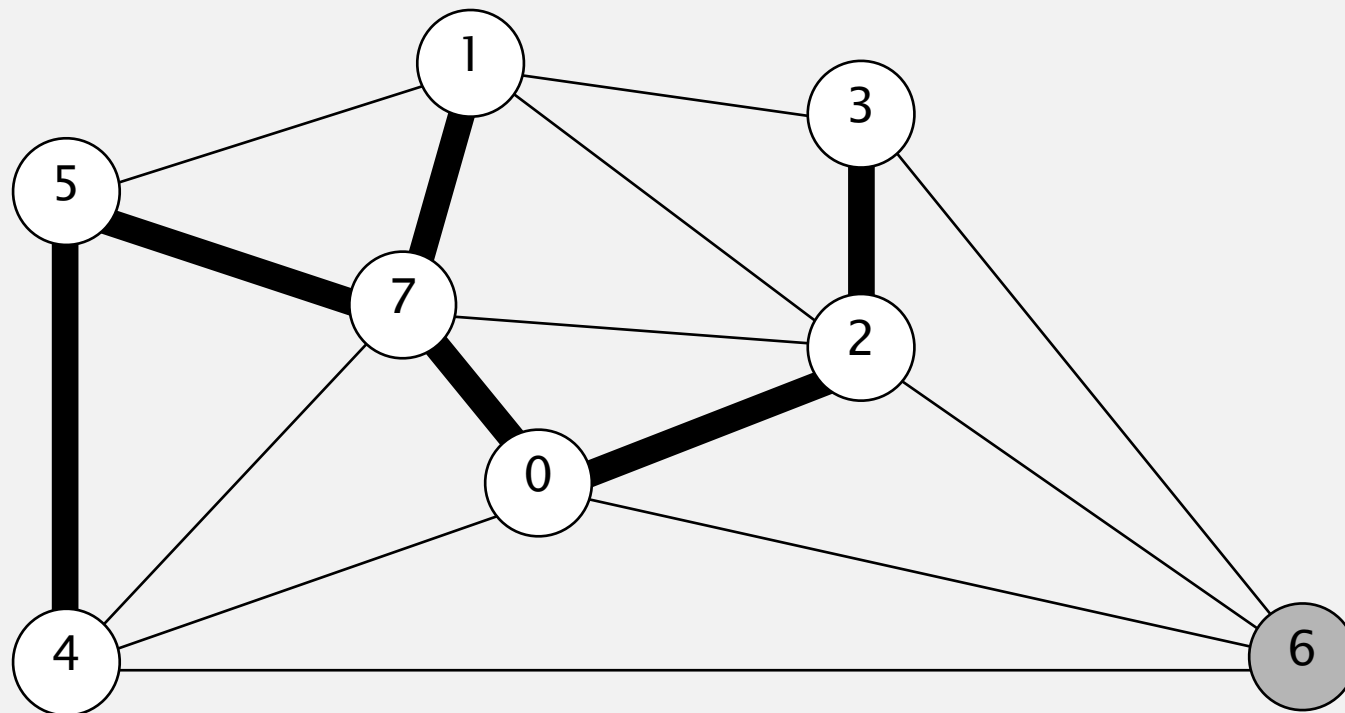
4-5	0.35
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

MST edges

0-7 1-7 0-2 2-3 5-7

Prim's algorithm demo

- Start with vertex 0 and greedily grow tree T .
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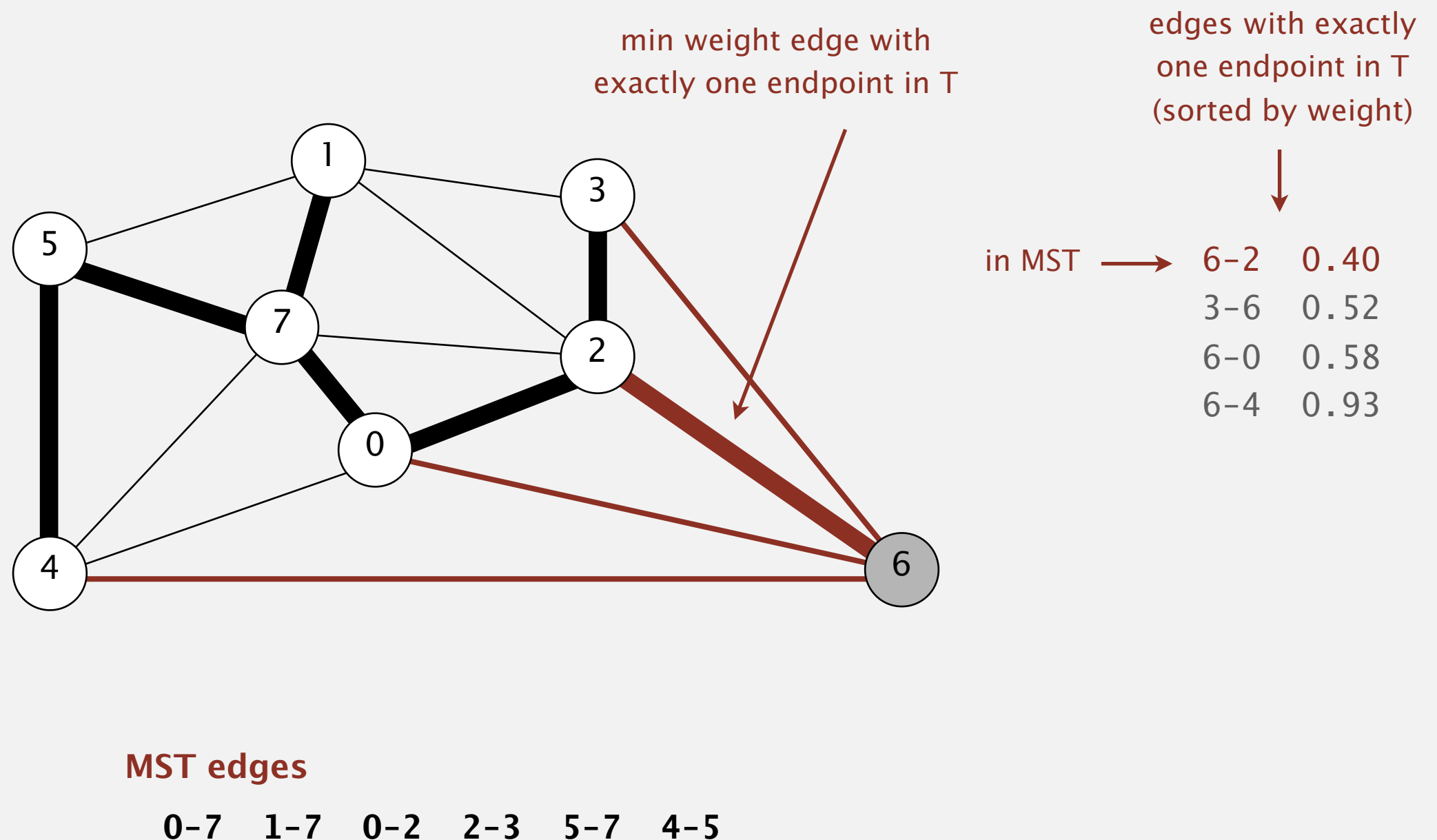


MST edges

0-7 1-7 0-2 2-3 5-7 4-5

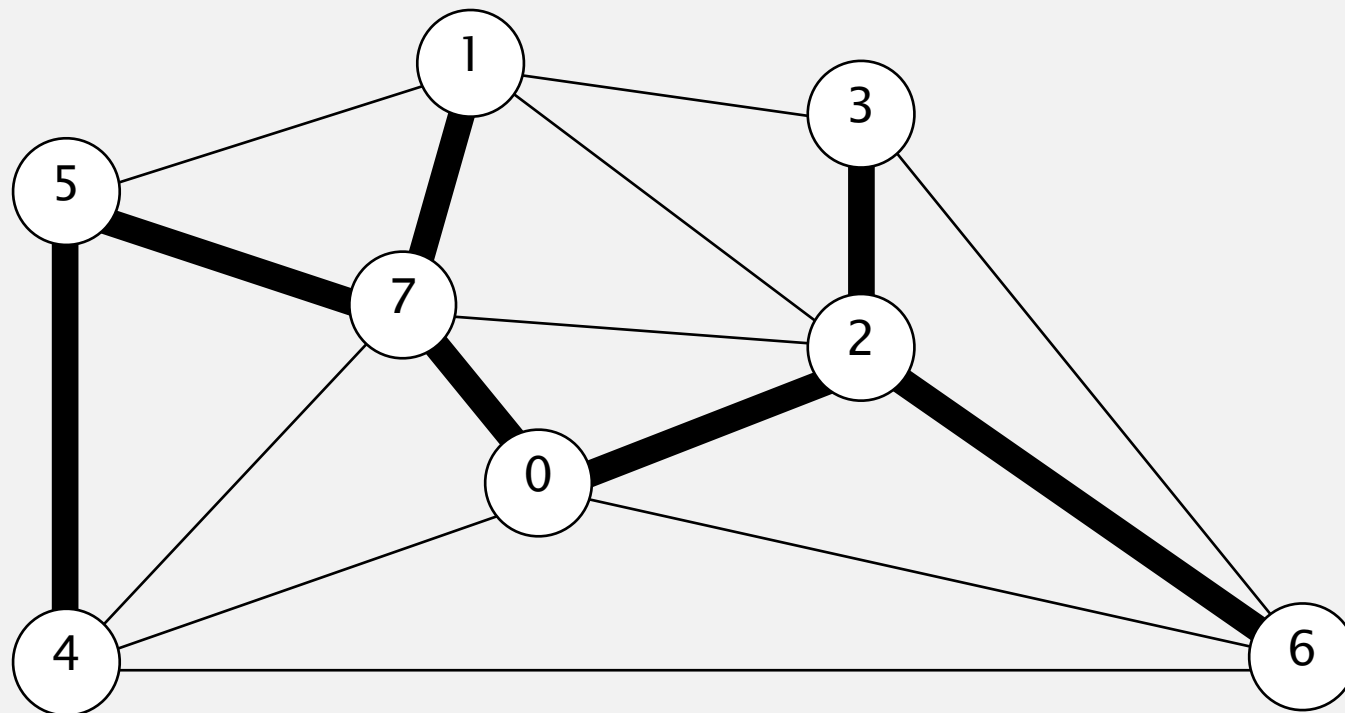
Prim's algorithm demo

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- Repeat until $V - 1$ edges.



Prim's algorithm demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



MST edges

0-7 1-7 0-2 2-3 5-7 4-5 6-2



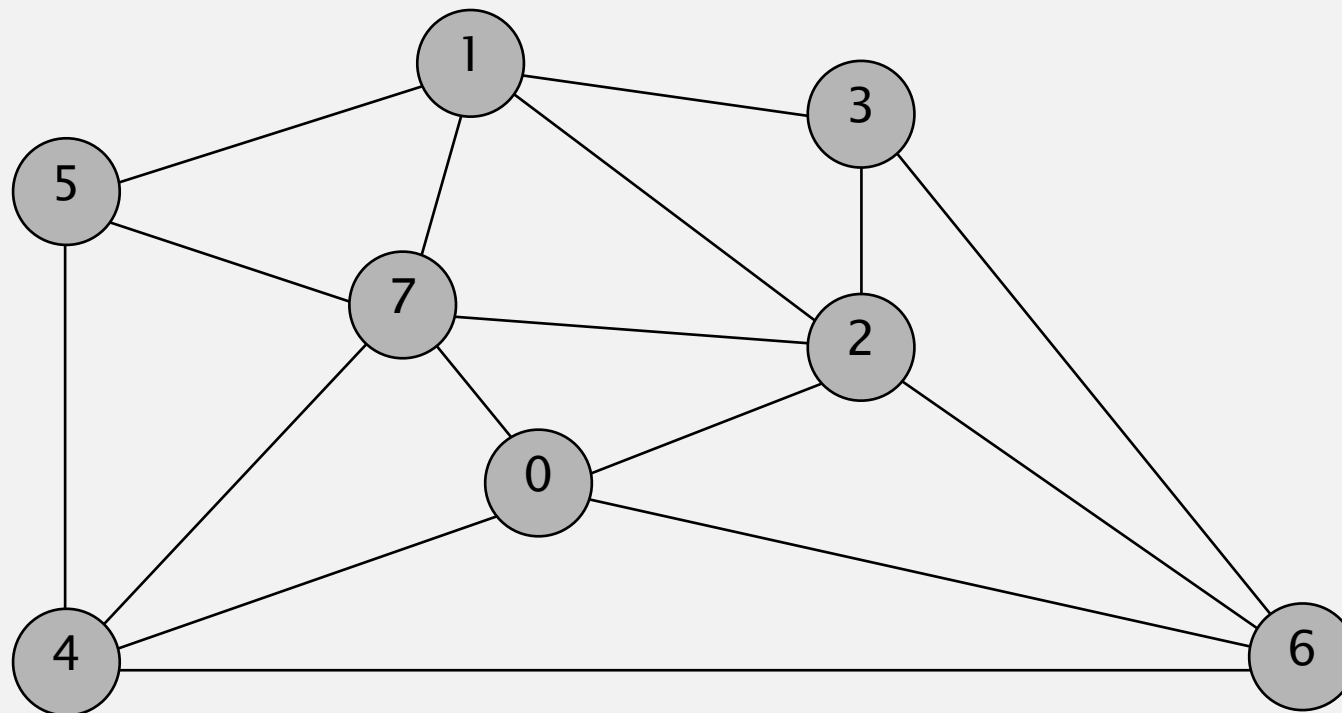
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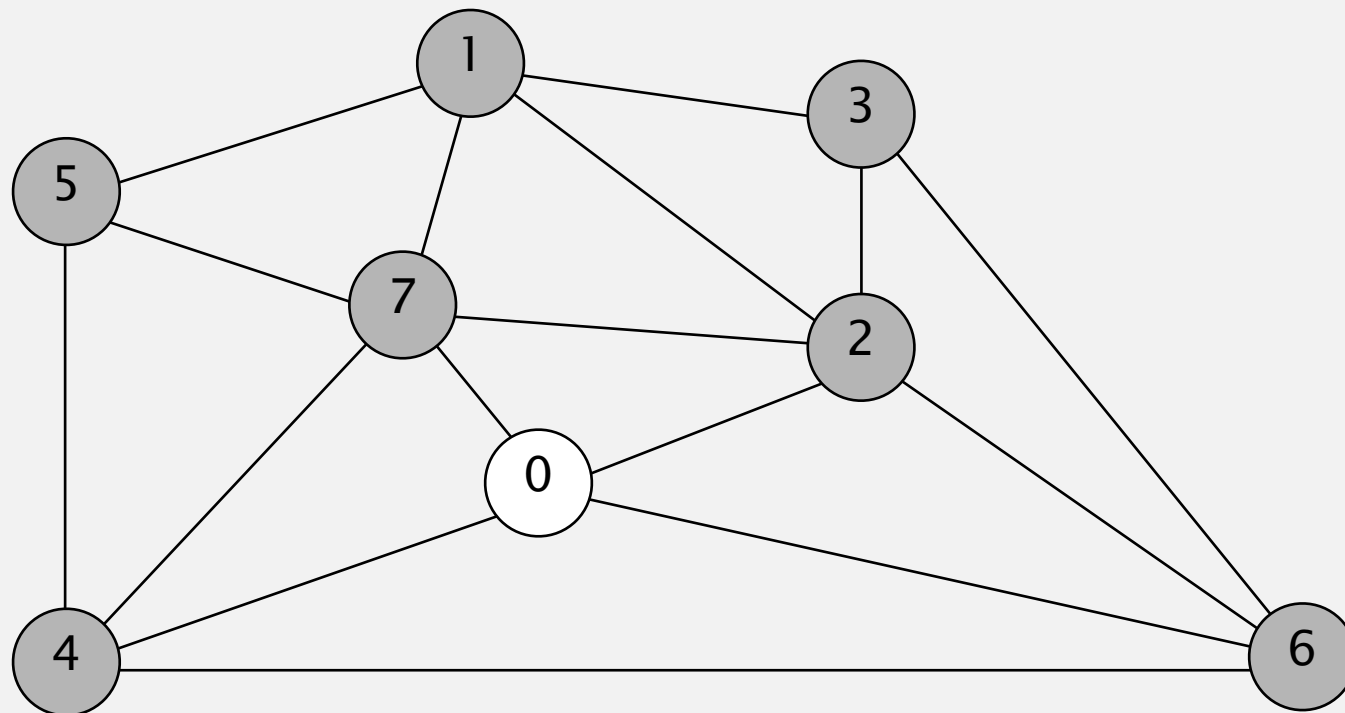


an edge-weighted graph

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4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

Prim's algorithm: lazy implementation demo

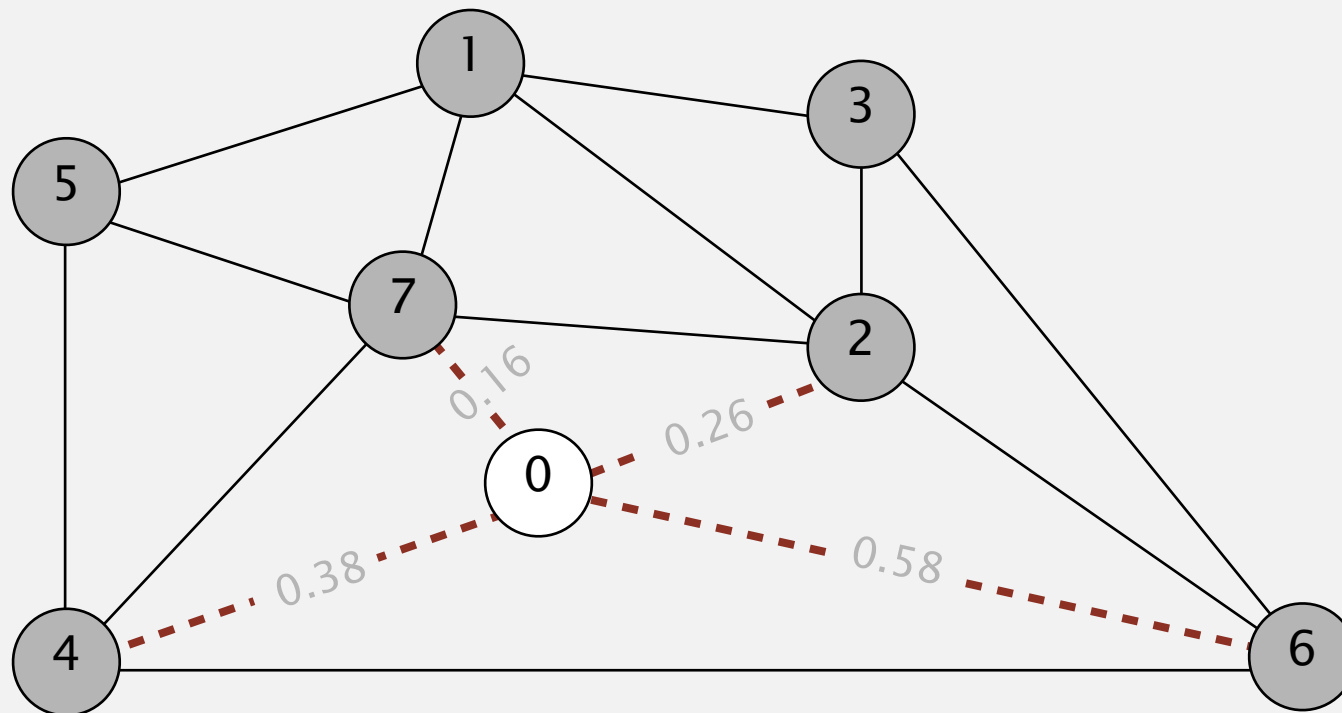
- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 0



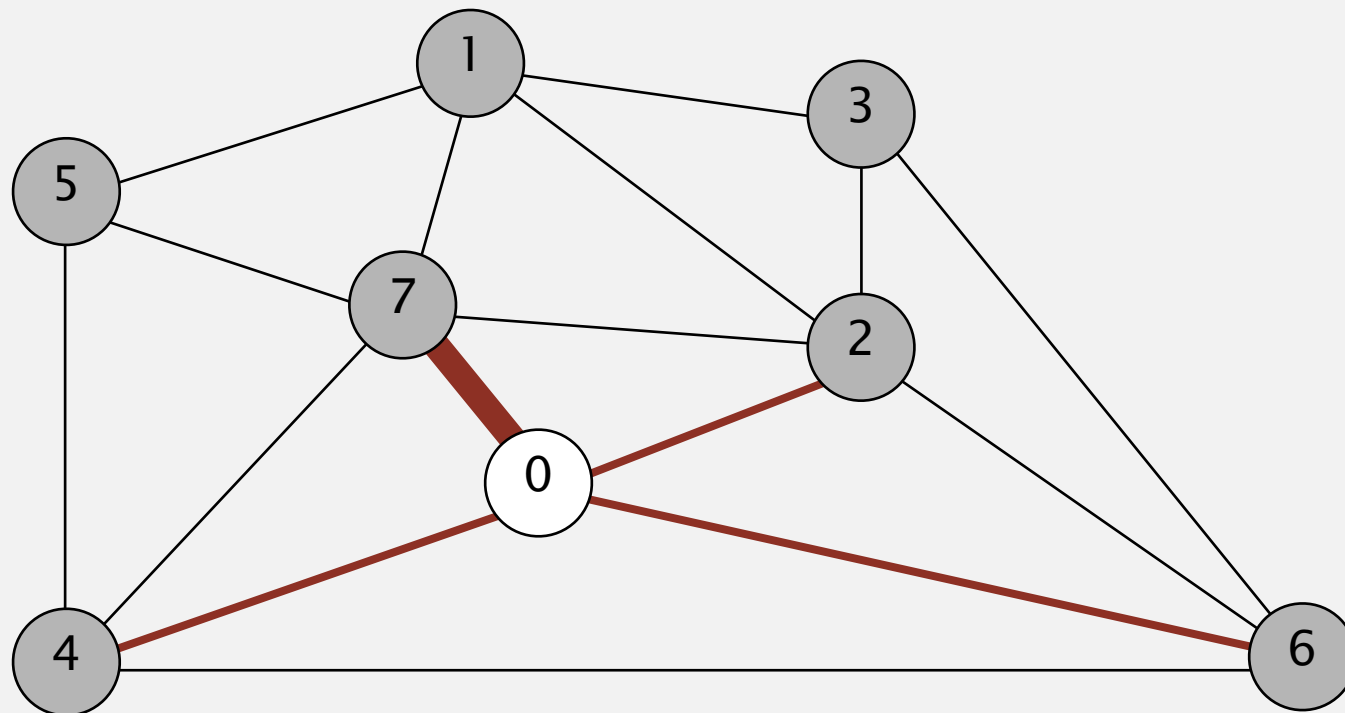
edges on PQ
(sorted by weight)

*	0-7	0.16
*	0-2	0.26
*	0-4	0.38
*	6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 0-7 and add to MST

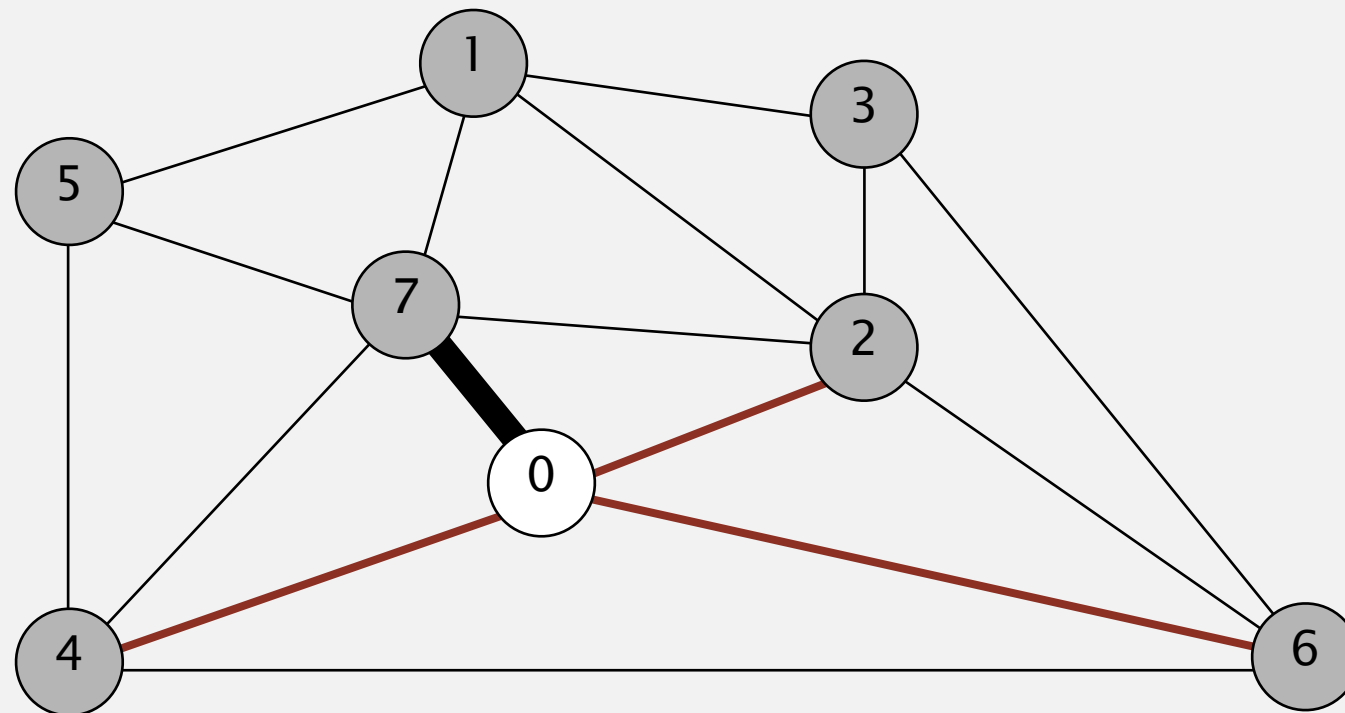


edges on PQ
(sorted by weight)

0-7	0.16
0-2	0.26
0-4	0.38
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



edges on PQ
(sorted by weight)

0-2	0.26
0-4	0.38
6-0	0.58

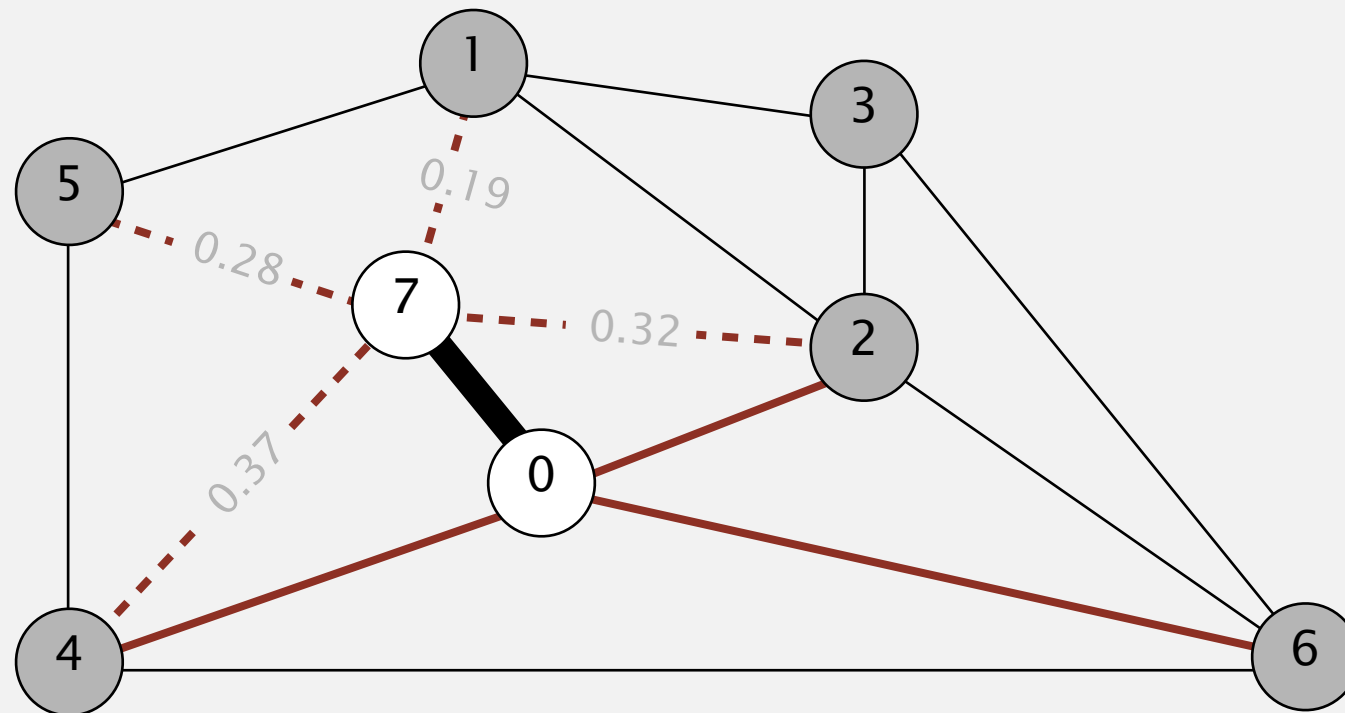
MST edges

0-7

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 7



MST edges

0-7

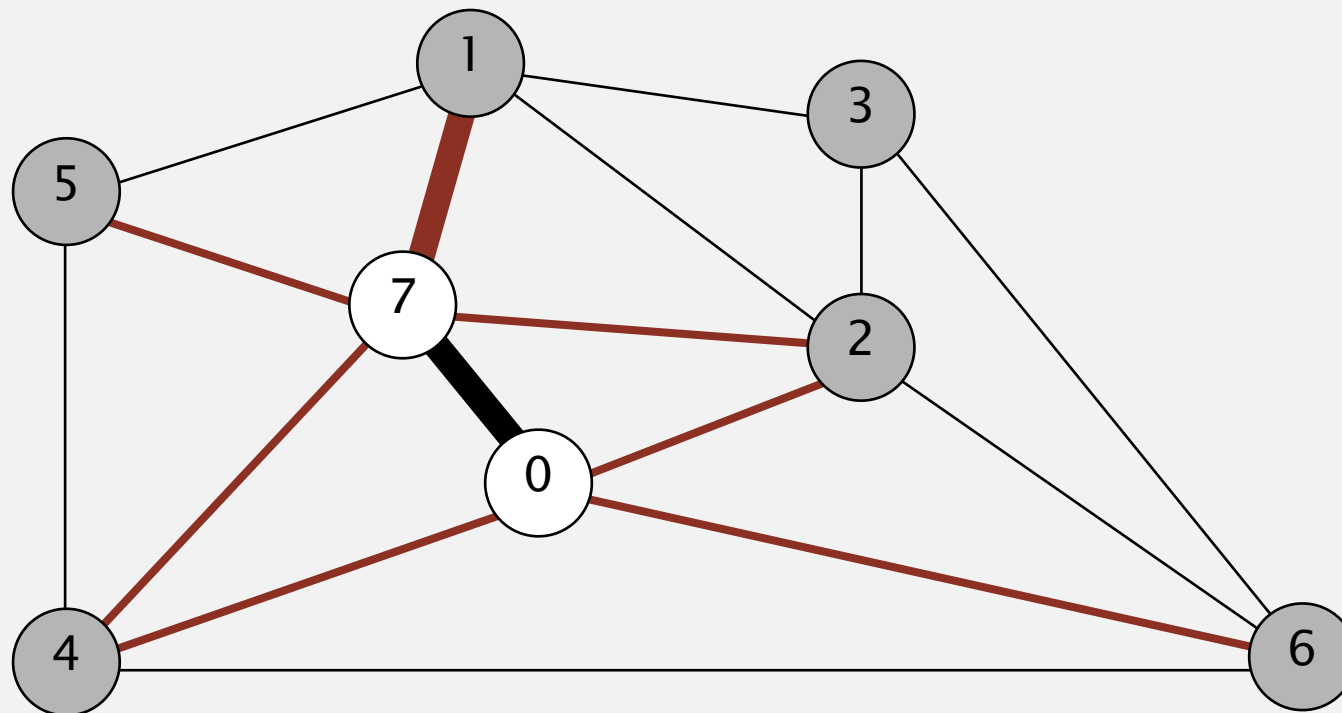
edges on PQ
(sorted by weight)

*	1-7	0.19
	0-2	0.26
*	5-7	0.28
*	2-7	0.34
*	4-7	0.37
	0-4	0.38
	6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 1-7 and add to MST



MST edges

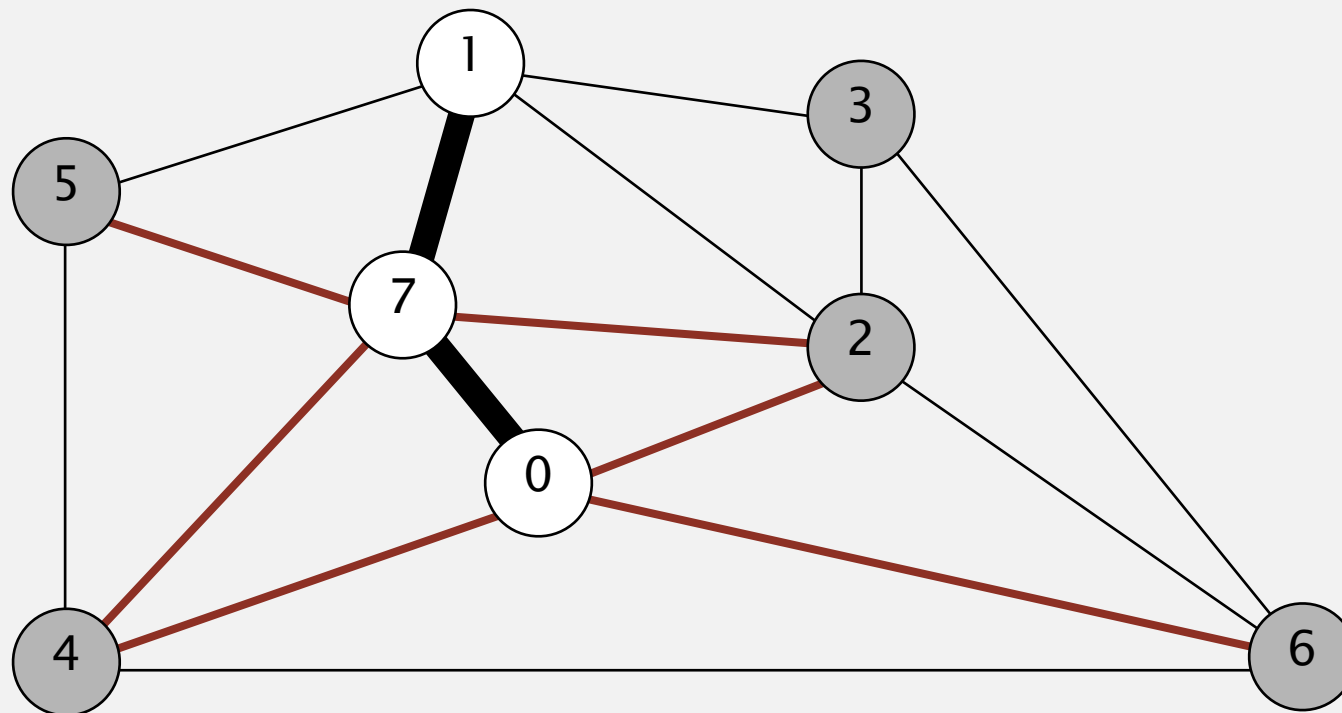
0-7

edges on PQ
(sorted by weight)

1-7	0.19
0-2	0.26
5-7	0.28
2-7	0.34
4-7	0.37
0-4	0.38
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



edges on PQ
(sorted by weight)

0-2	0.26
5-7	0.28
2-7	0.34
4-7	0.37
0-4	0.38
6-0	0.58

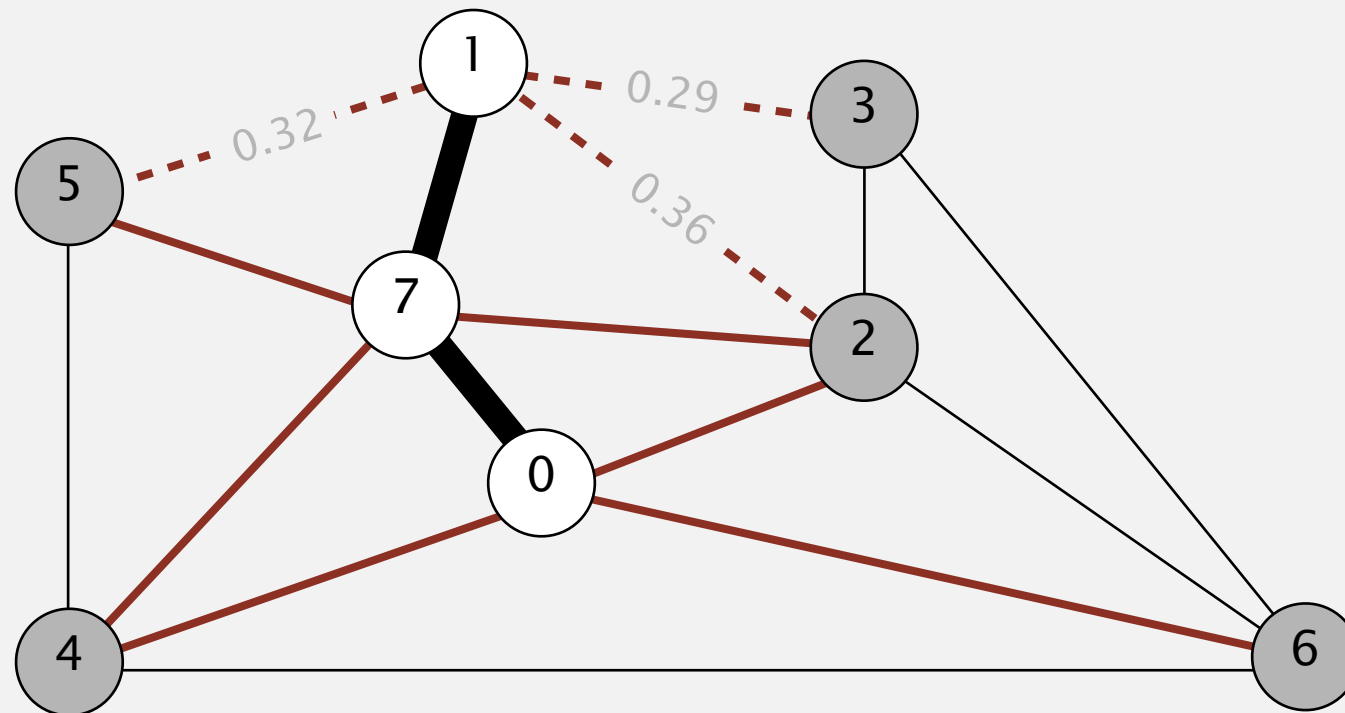
MST edges

0-7 1-7

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 1



MST edges

0-7 1-7

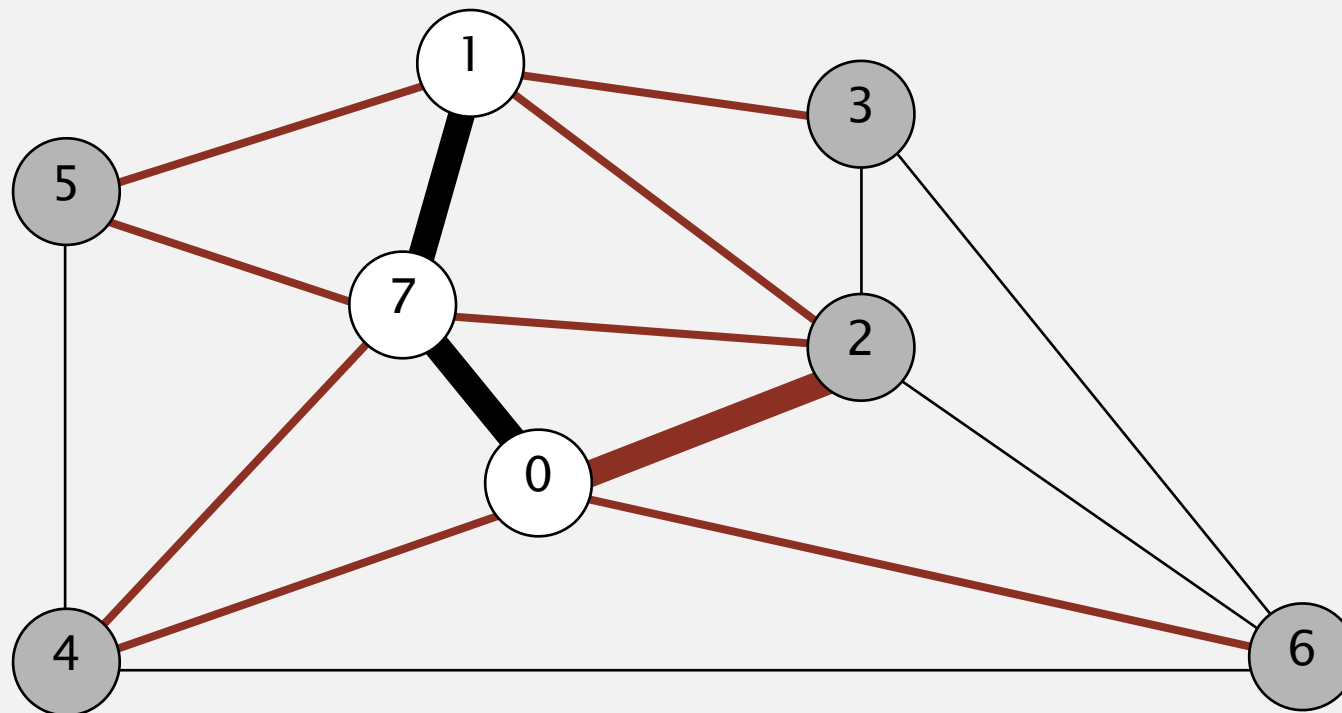
edges on PQ
(sorted by weight)

0-2	0.26
5-7	0.28
* 1-3	0.29
* 1-5	0.32
2-7	0.34
* 1-2	0.36
4-7	0.37
0-4	0.38
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete edge 0-2 and add to MST



MST edges

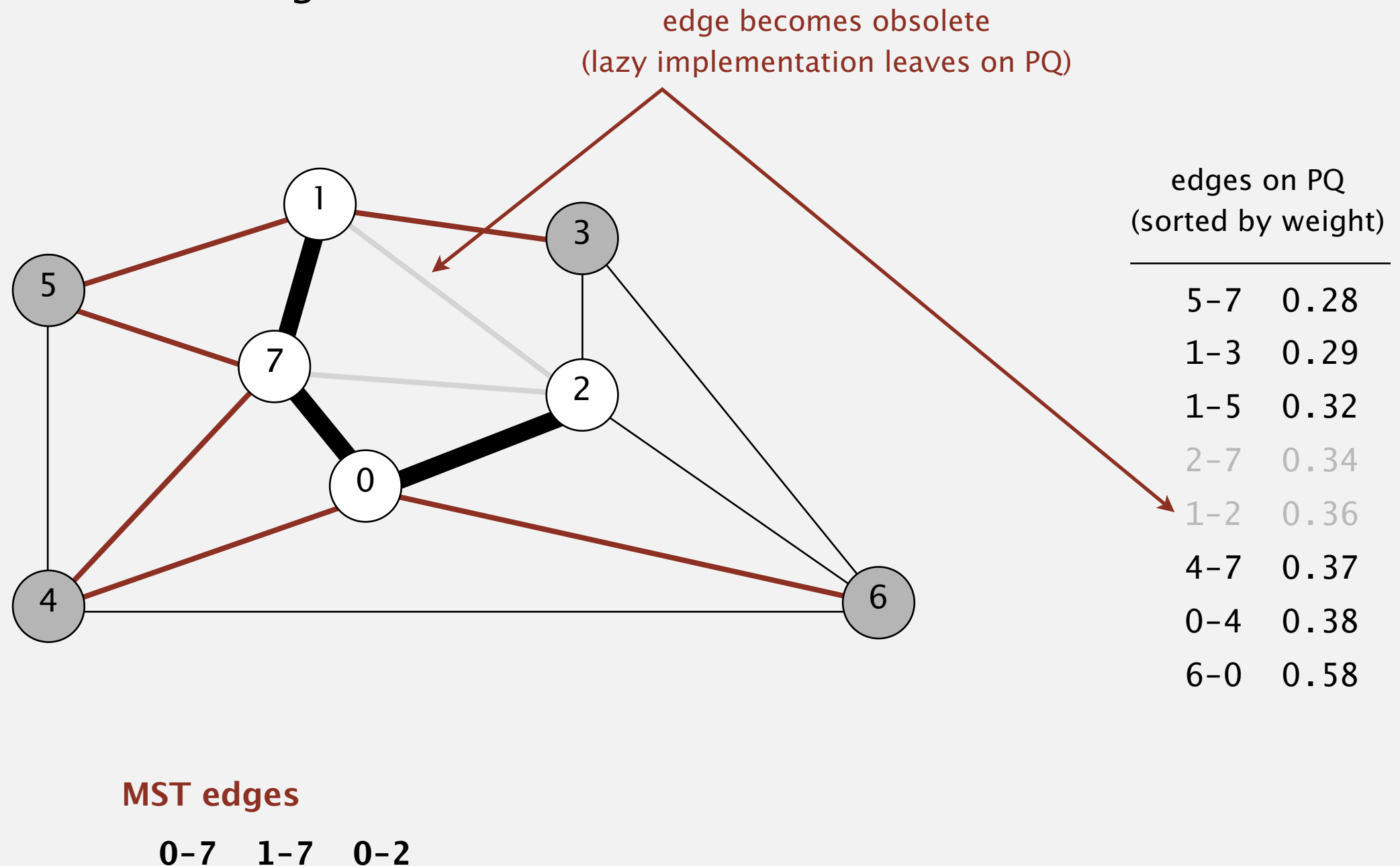
0-7 1-7

edges on PQ
(sorted by weight)

0-2	0.26
5-7	0.28
1-3	0.29
1-5	0.32
2-7	0.34
1-2	0.36
4-7	0.37
0-4	0.38
6-0	0.58

Prim's algorithm: lazy implementation demo

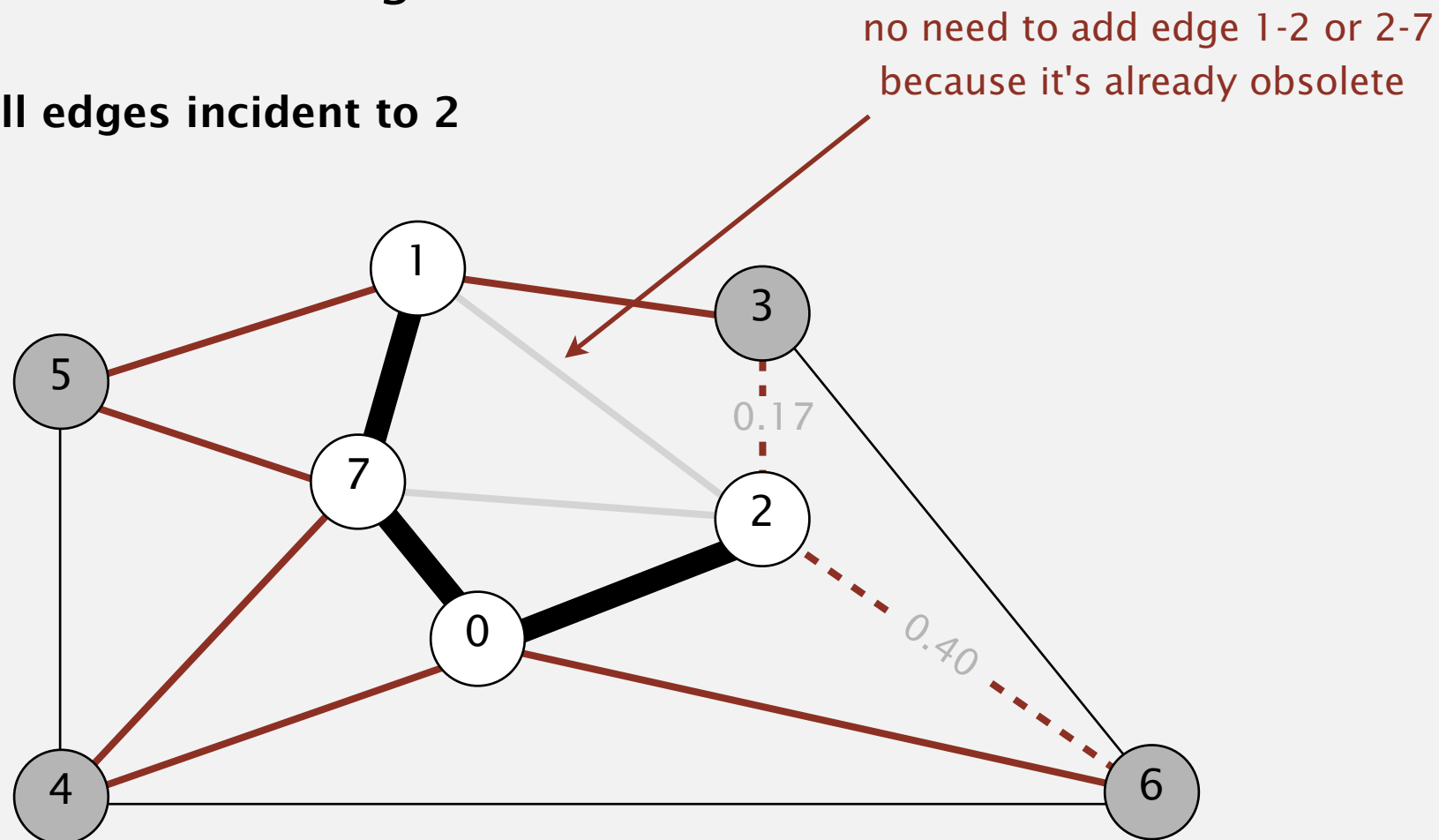
- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 2



MST edges

0-7 1-7 0-2

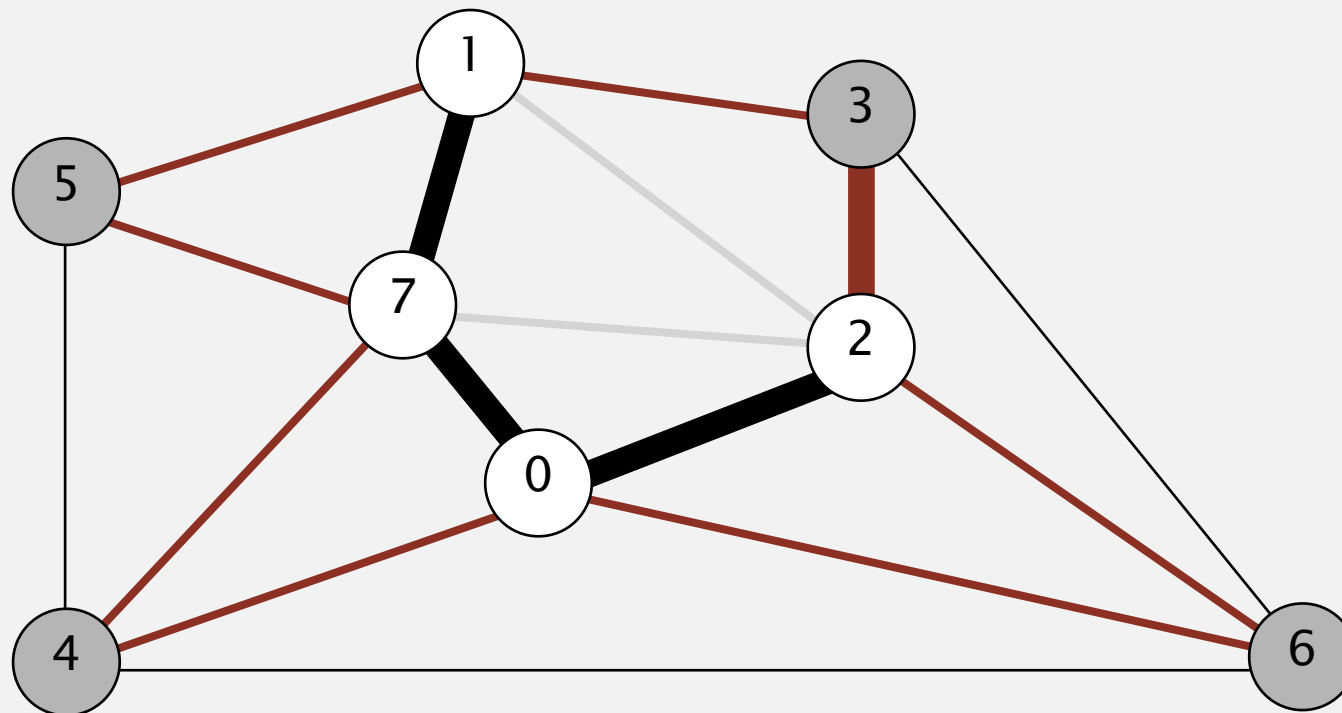
edges on PQ
(sorted by weight)

*	2-3	0.17
	5-7	0.28
	1-3	0.29
	1-5	0.32
	2-7	0.34
	1-2	0.36
	4-7	0.37
	0-4	0.38
*	6-2	0.40
	6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 2-3 and add to MST



MST edges

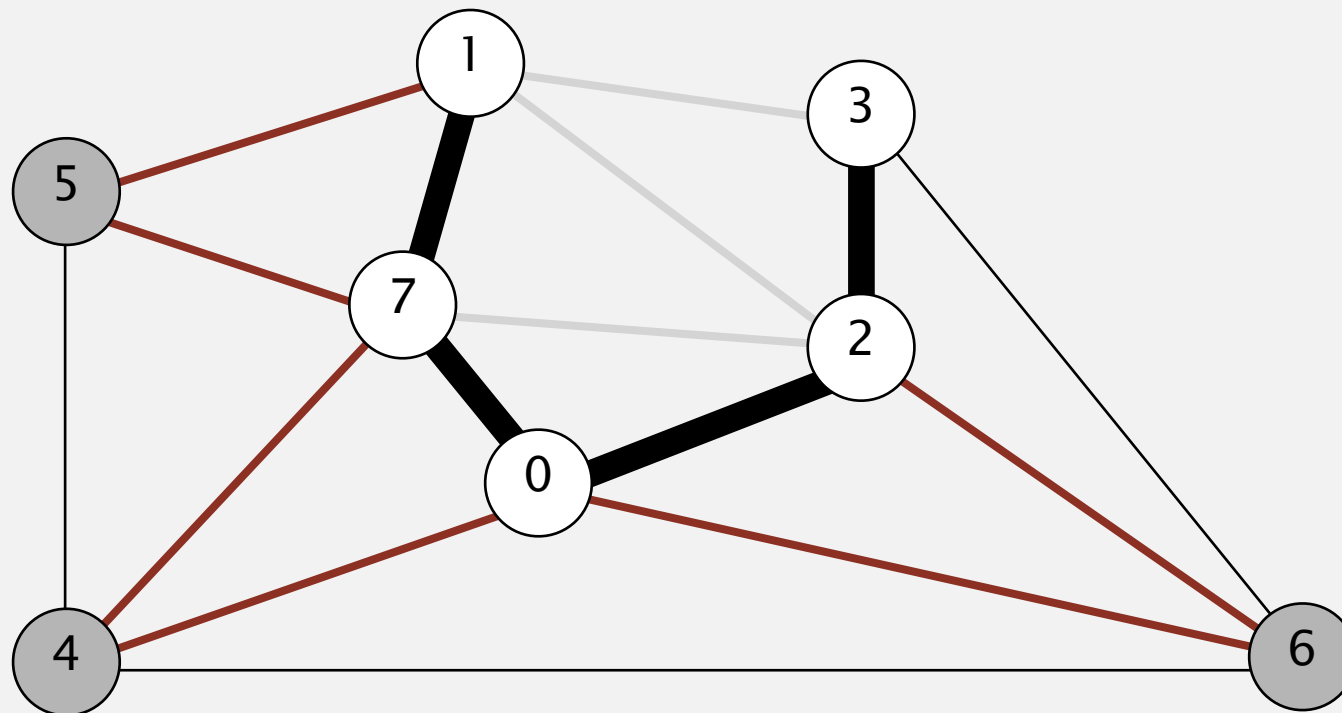
0-7 1-7 0-2

edges on PQ
(sorted by weight)

*	2-3	0.17
	5-7	0.28
	1-3	0.29
	1-5	0.32
	2-7	0.34
	1-2	0.36
	4-7	0.37
	0-4	0.38
*	6-2	0.40
	6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



edges on PQ
(sorted by weight)

5-7	0.28
1-3	0.29
1-5	0.32
2-7	0.34
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
6-0	0.58

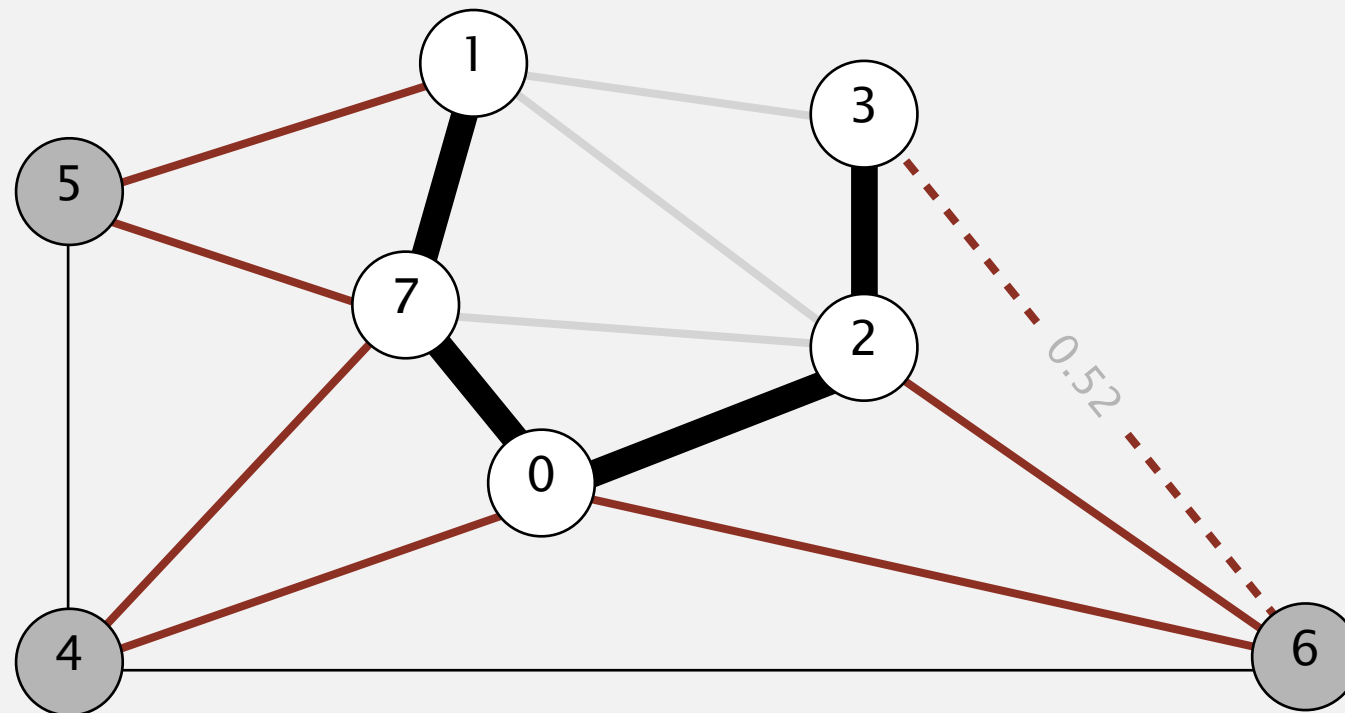
MST edges

0-7 1-7 0-2 2-3

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 3



MST edges

0-7 1-7 0-2 2-3

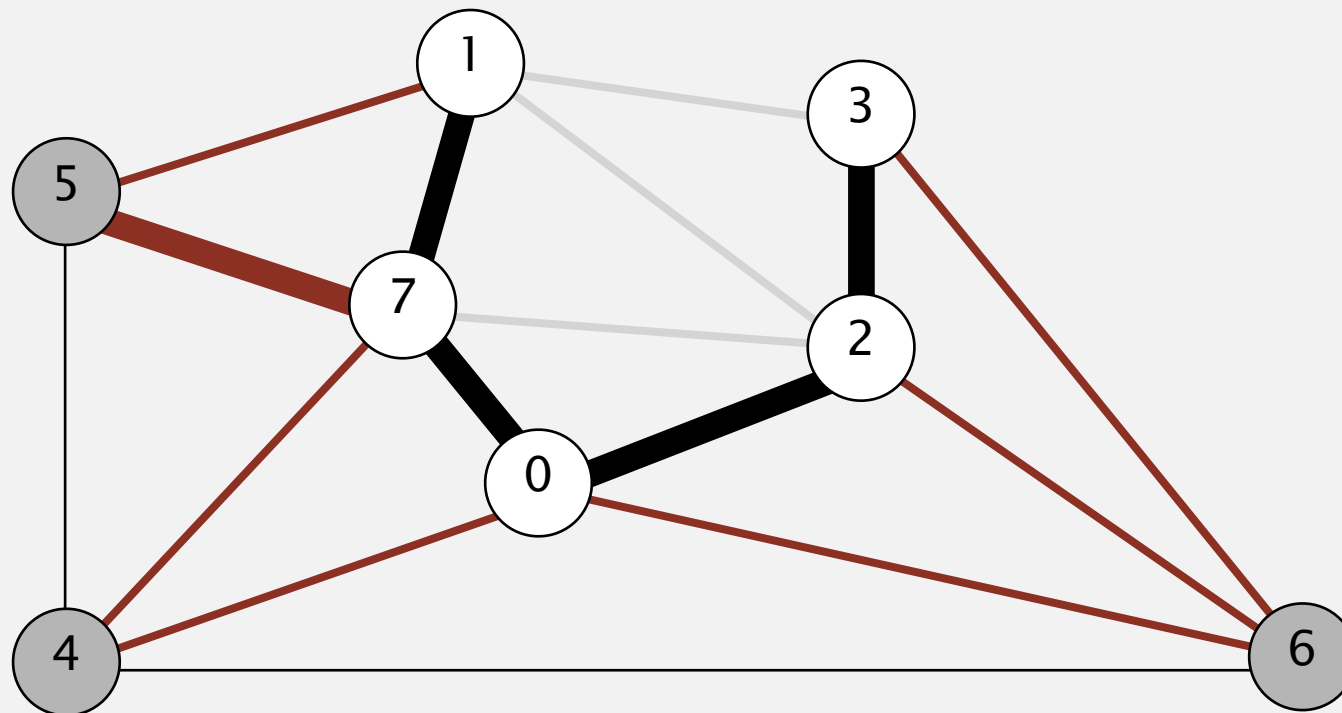
edges on PQ
(sorted by weight)

5-7	0.28
1-3	0.29
1-5	0.32
2-7	0.34
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
* 3-6	0.52
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 5-7 and add to MST



MST edges

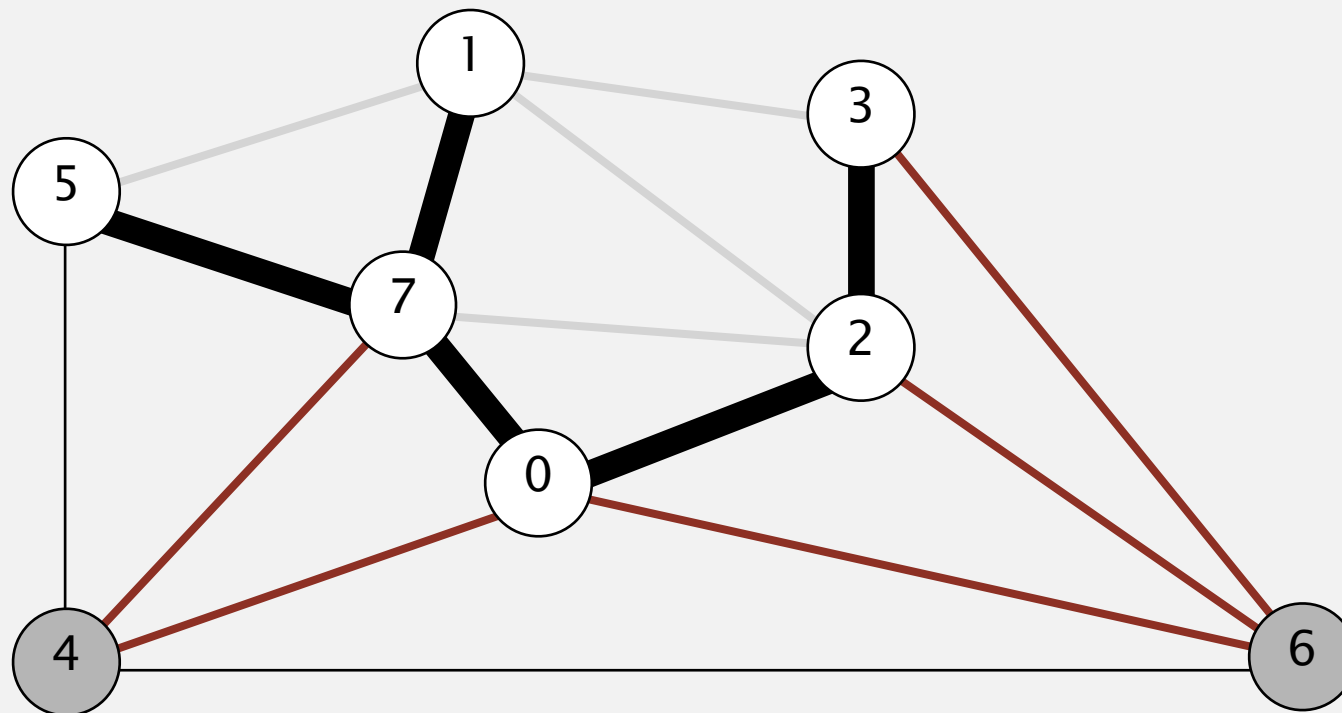
0-7 1-7 0-2 2-3

edges on PQ
(sorted by weight)

5-7	0.28
1-3	0.29
1-5	0.32
2-7	0.34
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



edges on PQ
(sorted by weight)

1-3	0.29
1-5	0.32
2-7	0.34
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

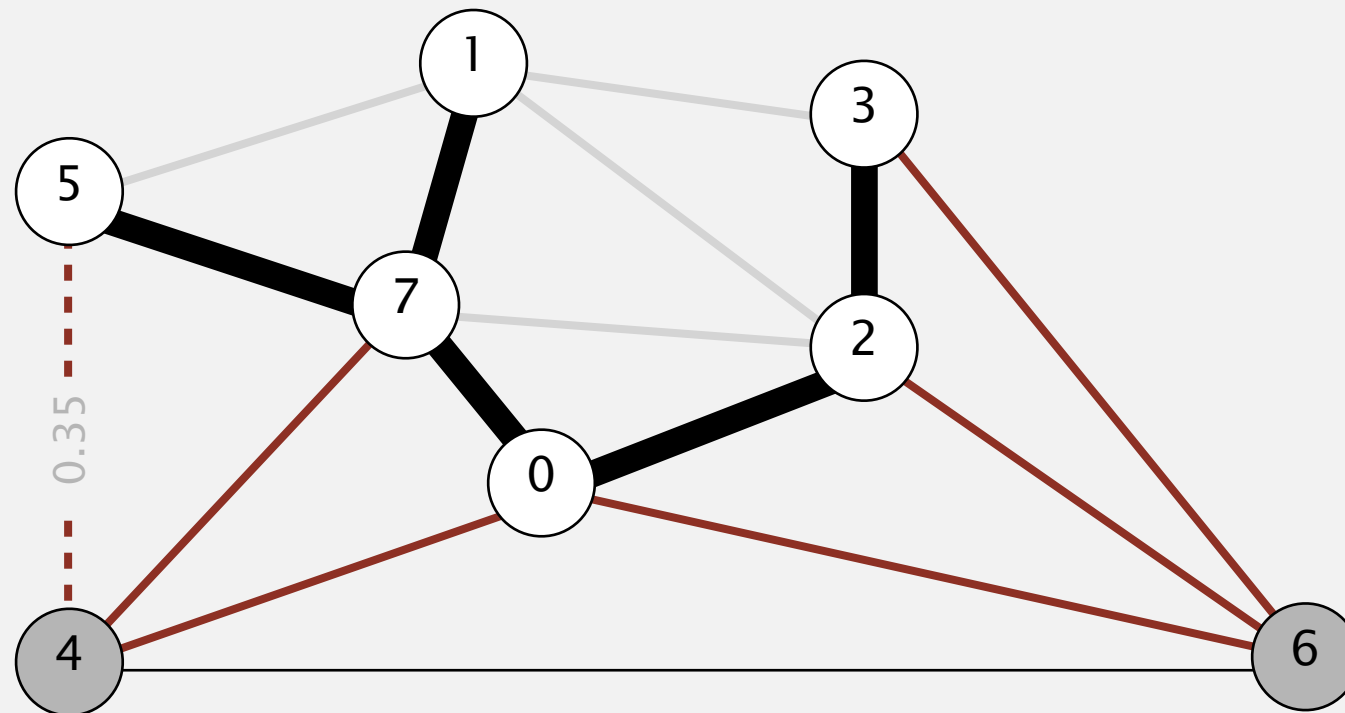
MST edges

0-7 1-7 0-2 2-3 5-7

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 5



0-7 1-7 0-2 2-3 5-7

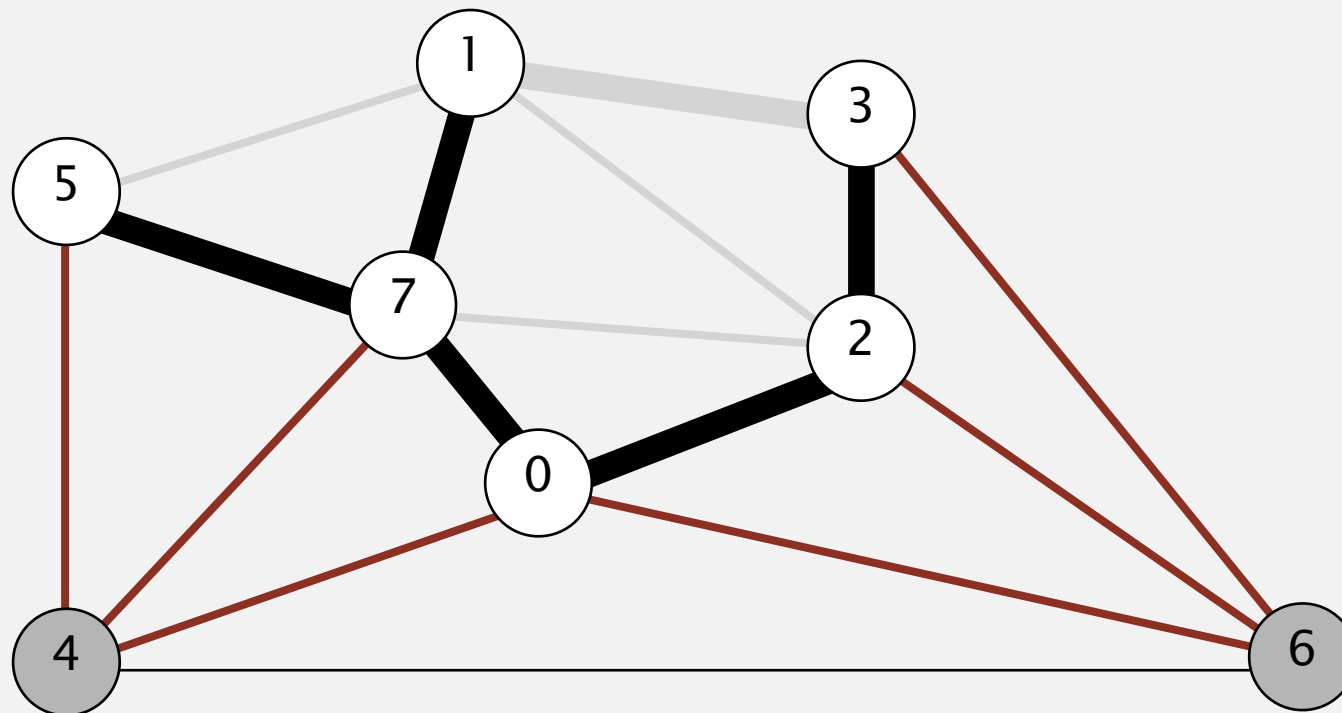
edges on PQ
(sorted by weight)

1-3	0.29
1-5	0.32
2-7	0.34
* 4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 1-3 and discard obsolete edge



MST edges

0-7 1-7 0-2 2-3 5-7

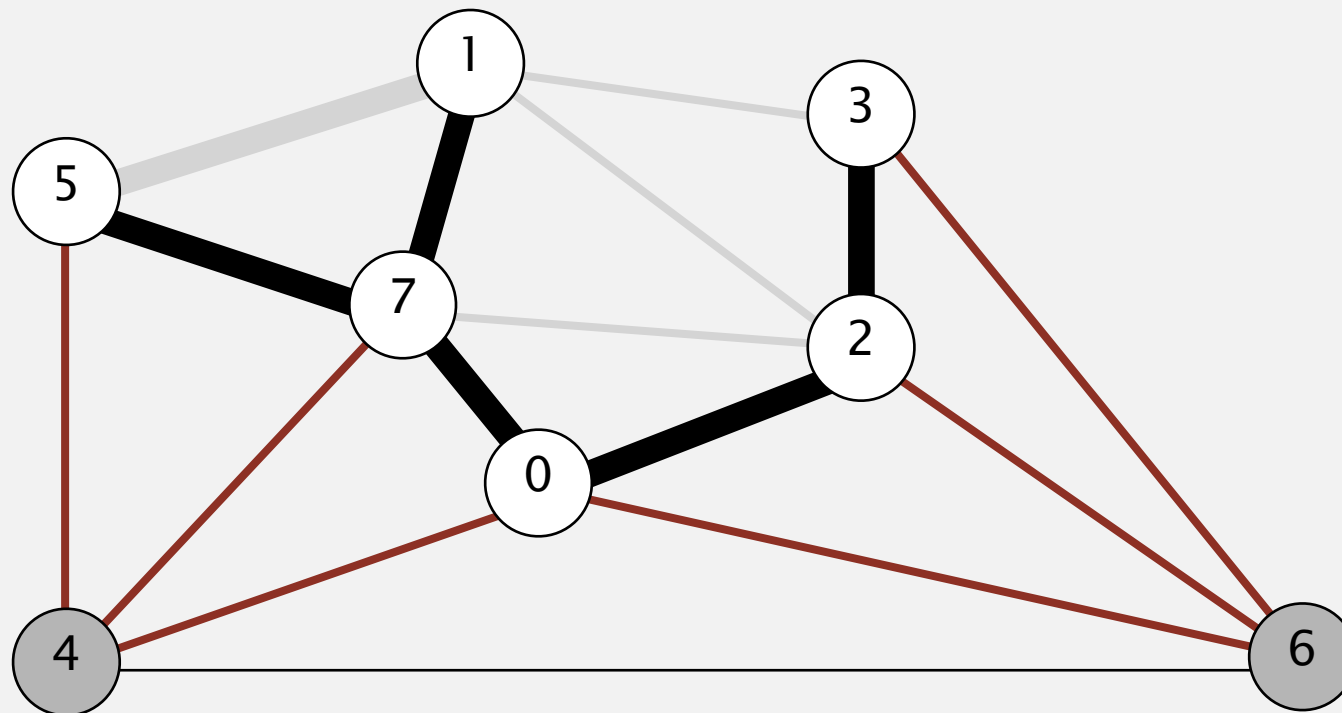
edges on PQ
(sorted by weight)

1-3	0.29
1-5	0.32
2-7	0.34
4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 1-5 and discard obsolete edge



MST edges

0-7 1-7 0-2 2-3 5-7

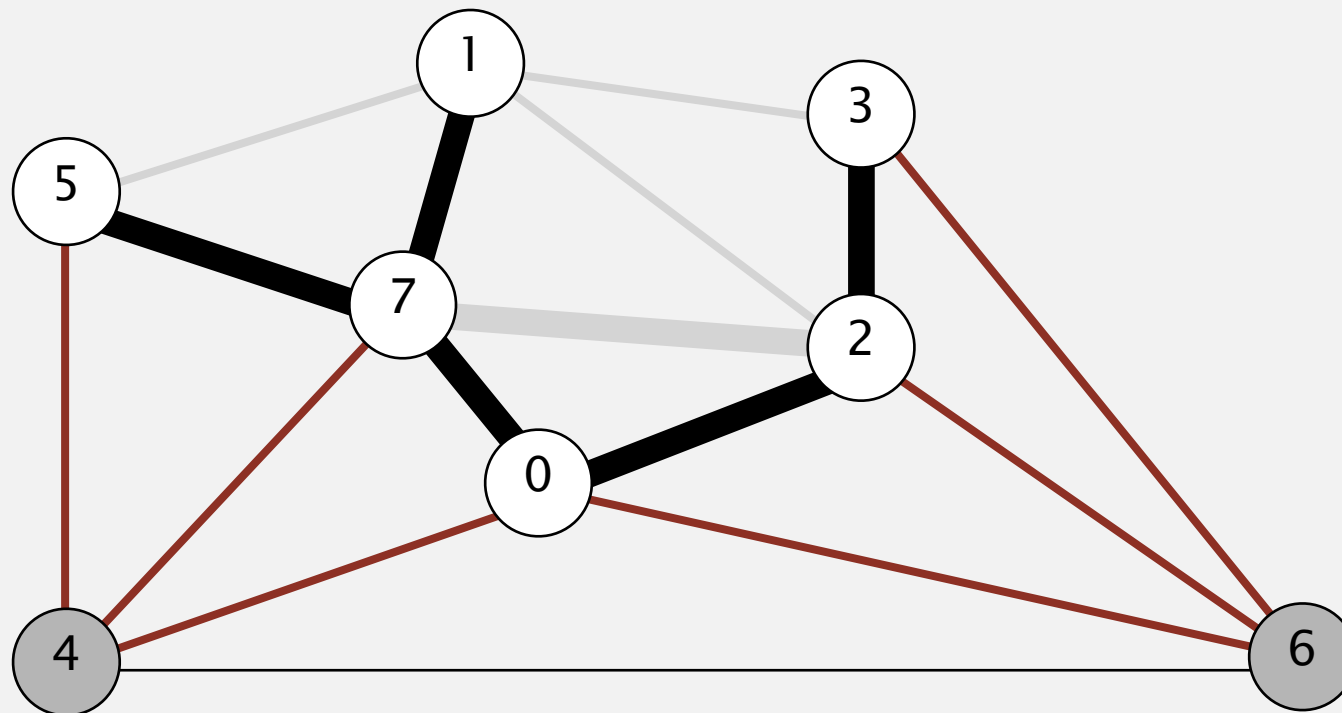
edges on PQ
(sorted by weight)

1-5	0.32
2-7	0.34
4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 2-7 and discard obsolete edge



MST edges

0-7 1-7 0-2 2-3 5-7

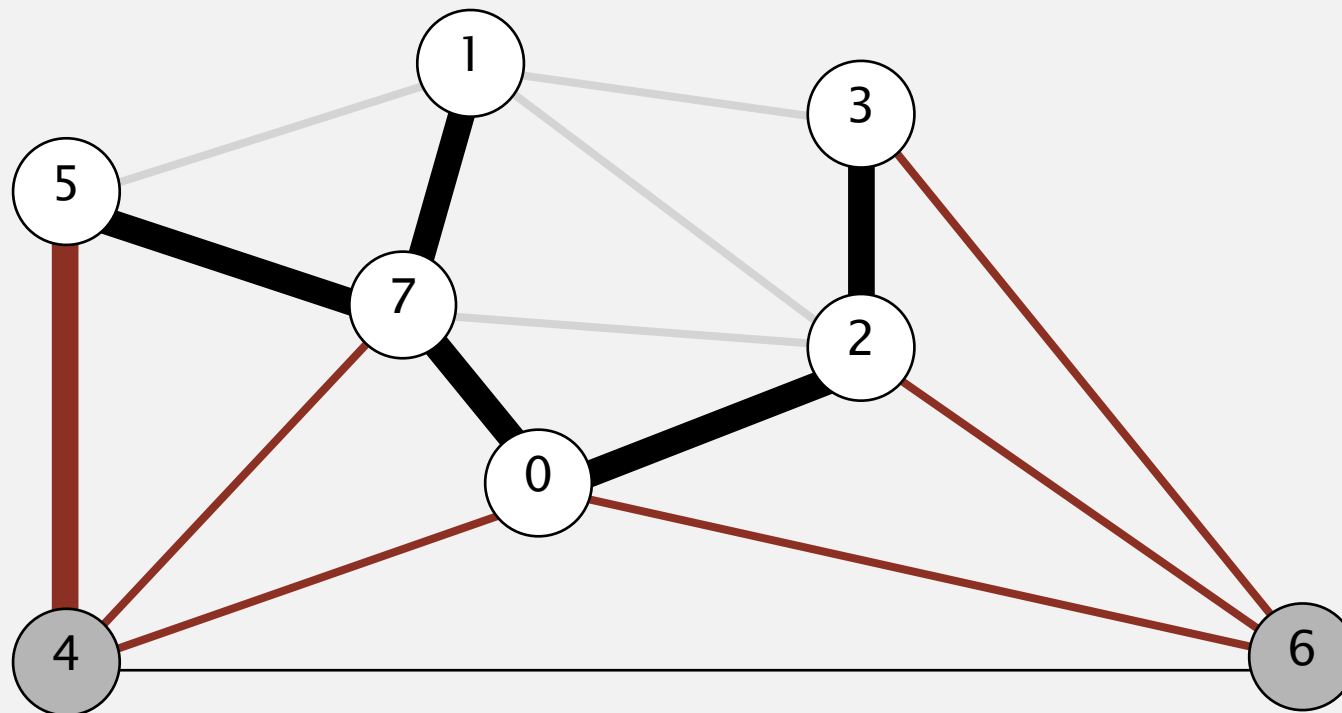
edges on PQ
(sorted by weight)

2-7	0.34
4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 4-5 and add to MST



edges on PQ
(sorted by weight)

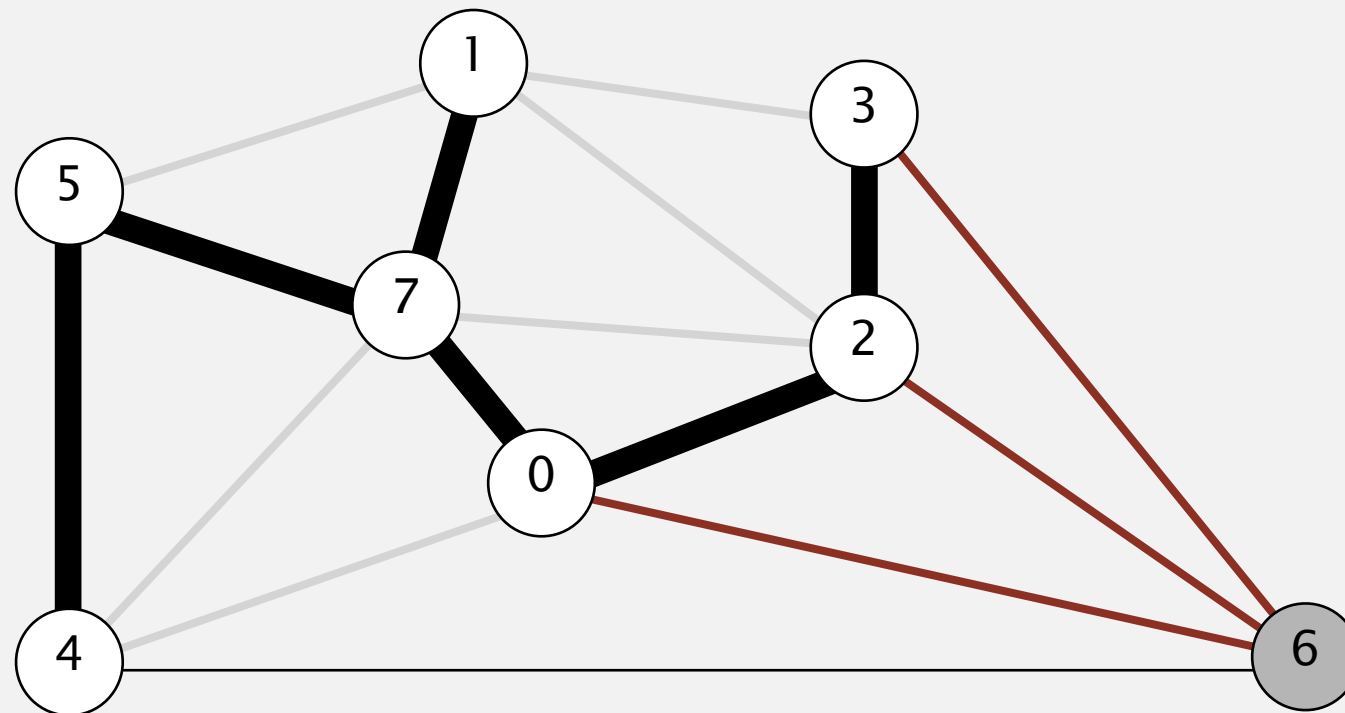
4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

MST edges

0-7 1-7 0-2 2-3 5-7

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



edges on PQ
(sorted by weight)

1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58

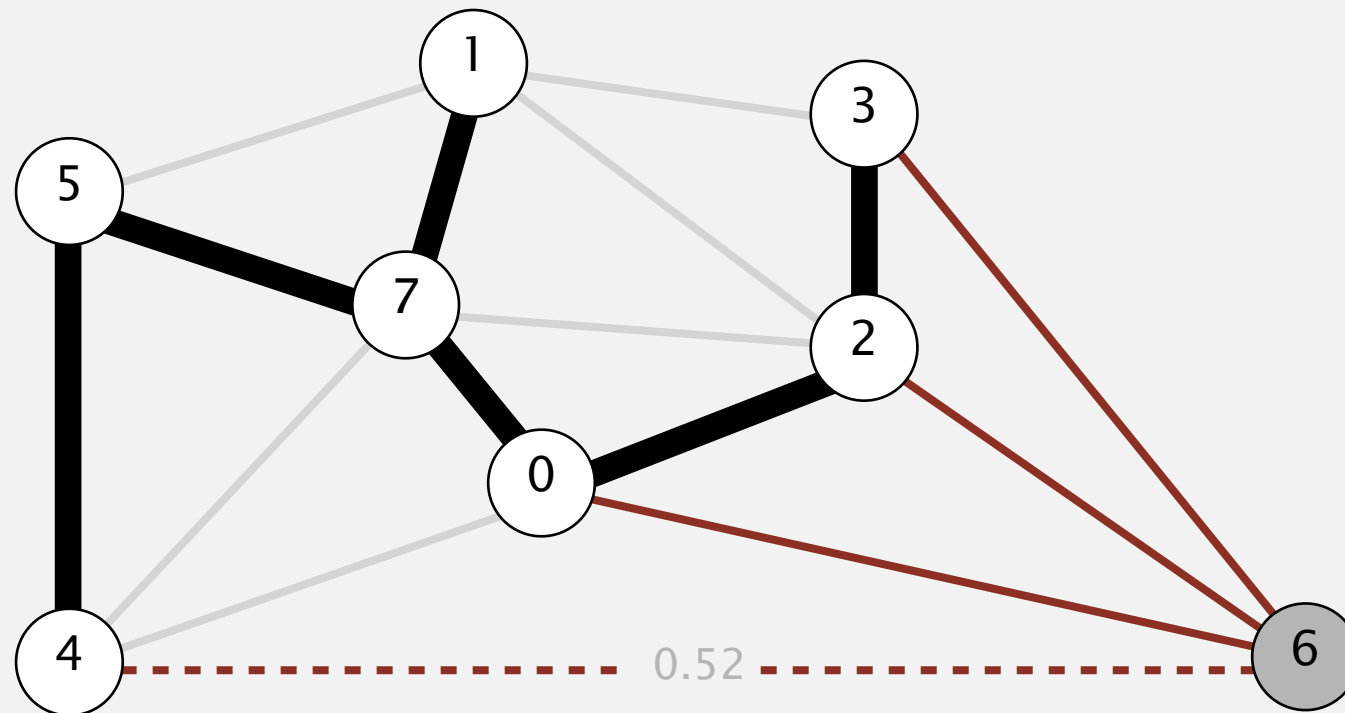
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

add to PQ all edges incident to 4



edges on PQ
(sorted by weight)

1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
* 6-4	0.93

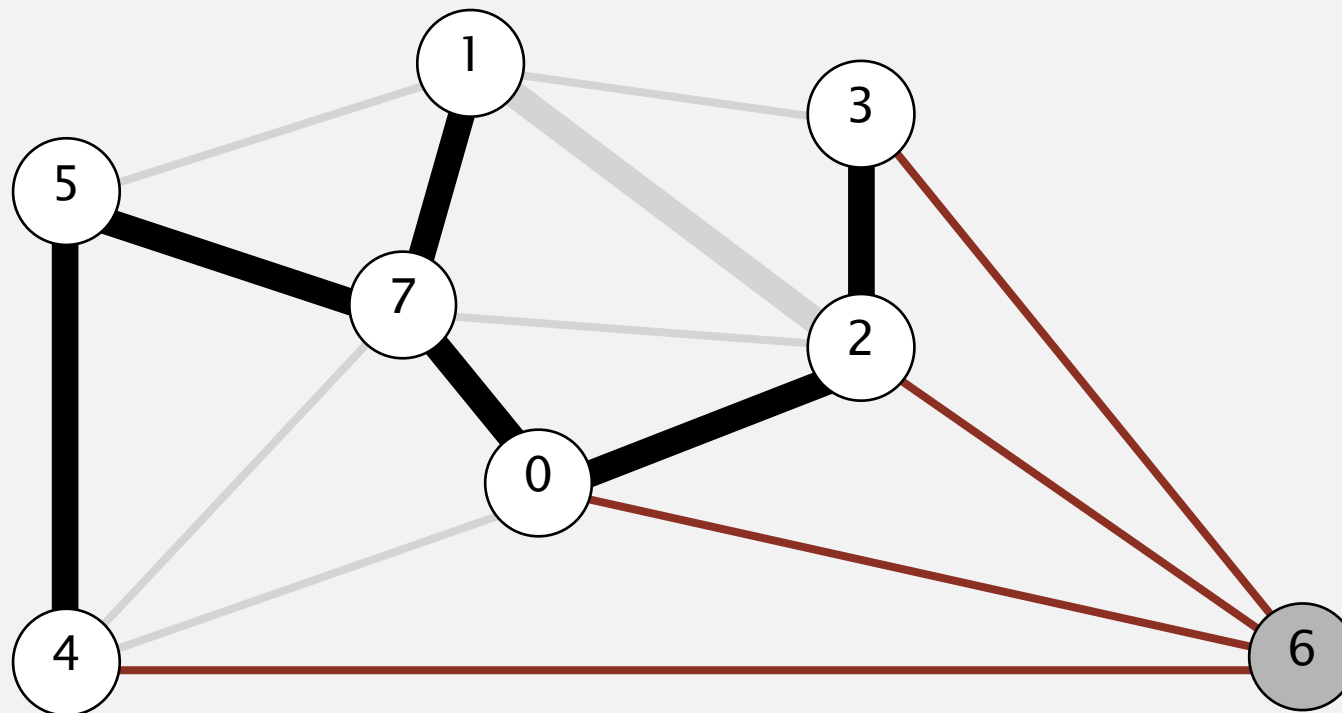
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 1-2 and discard obsolete edge



edges on PQ
(sorted by weight)

1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

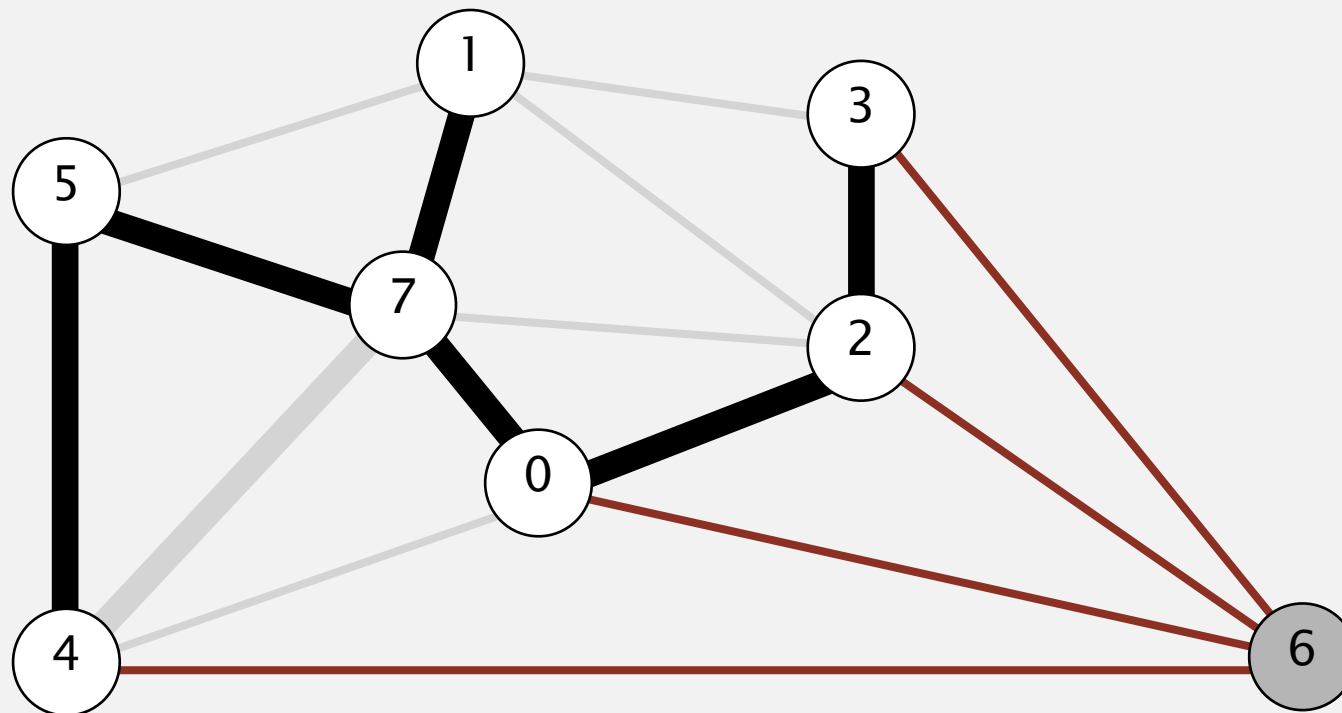
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 4-7 and discard obsolete edge



edges on PQ
(sorted by weight)

4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

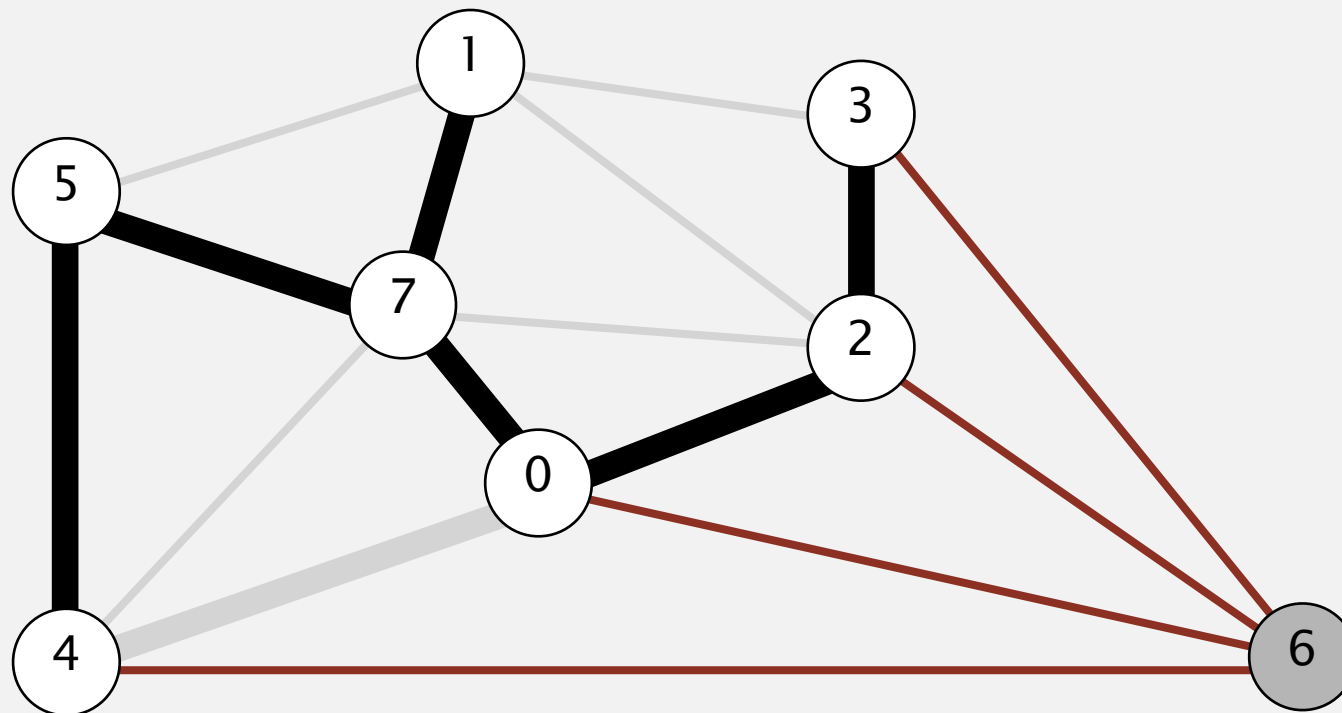
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 0-4 and discard obsolete edge



edges on PQ
(sorted by weight)

0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

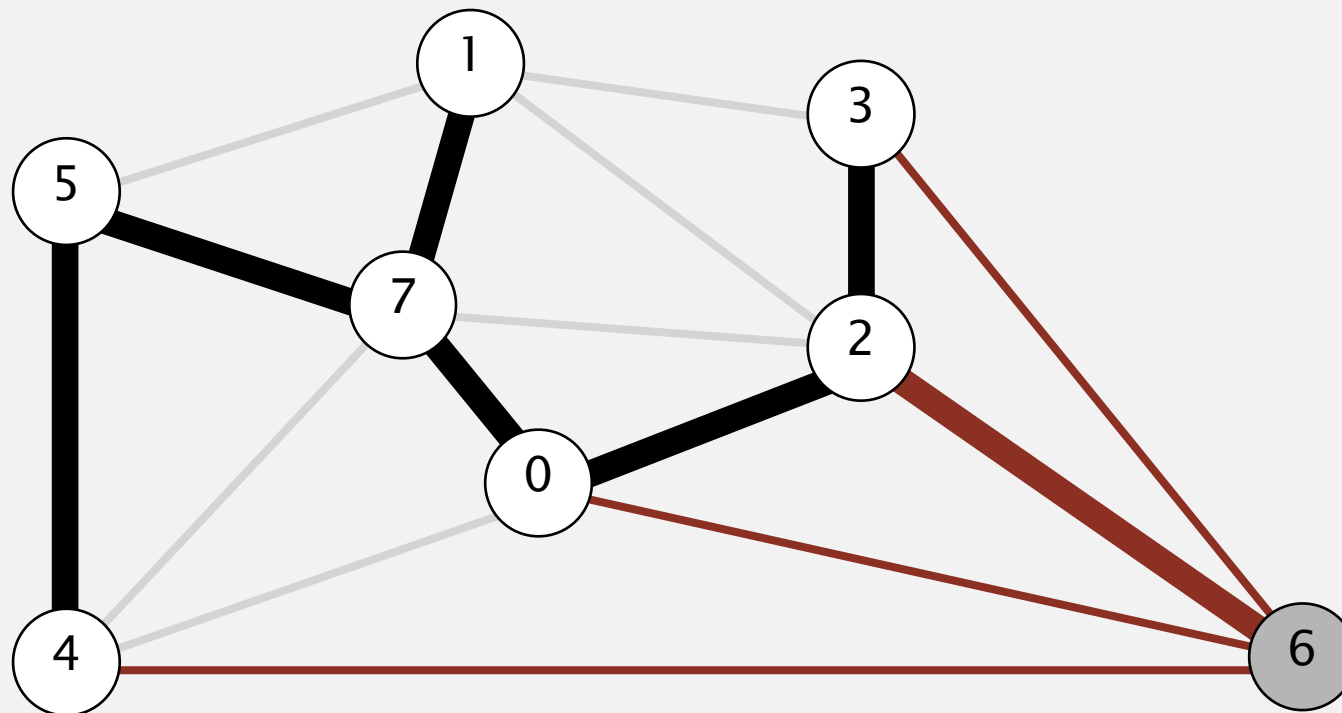
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 6-2 and add to MST



edges on PQ
(sorted by weight)

6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

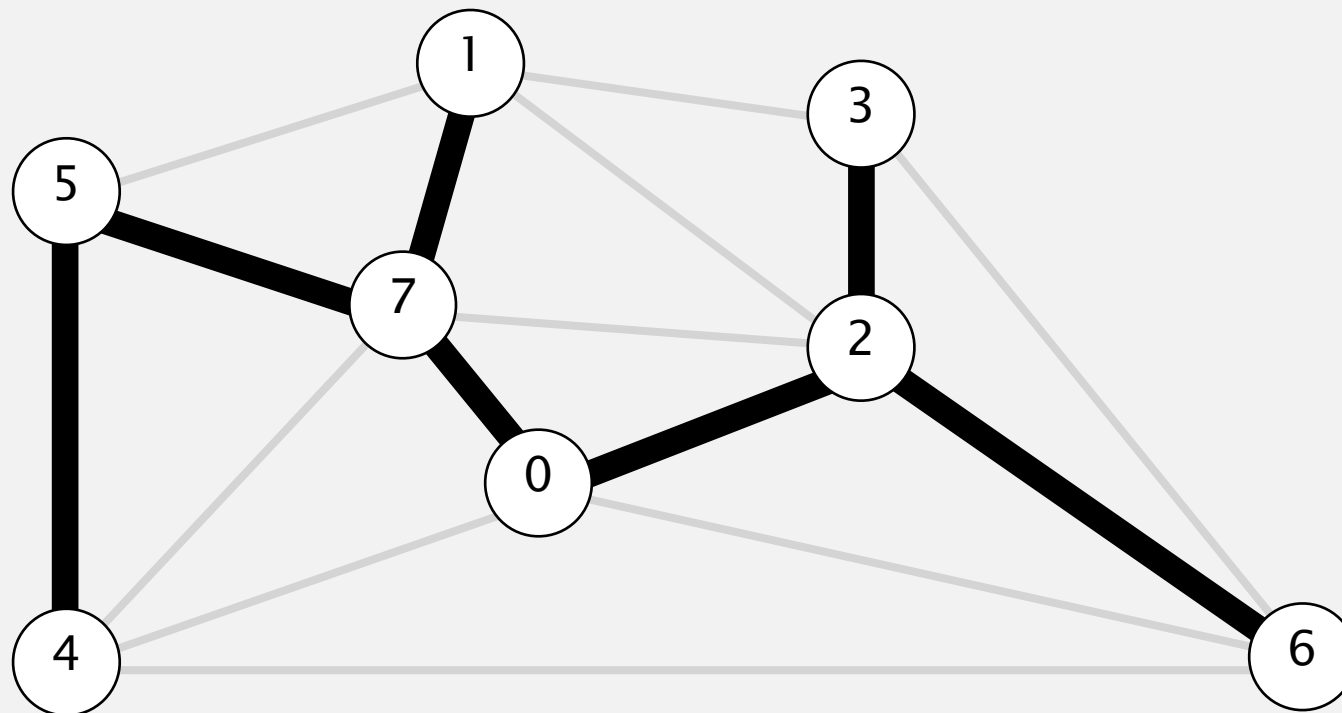
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

delete 6-2 and add to MST



edges on PQ
(sorted by weight)

3-6	0.52
6-0	0.58
6-4	0.93

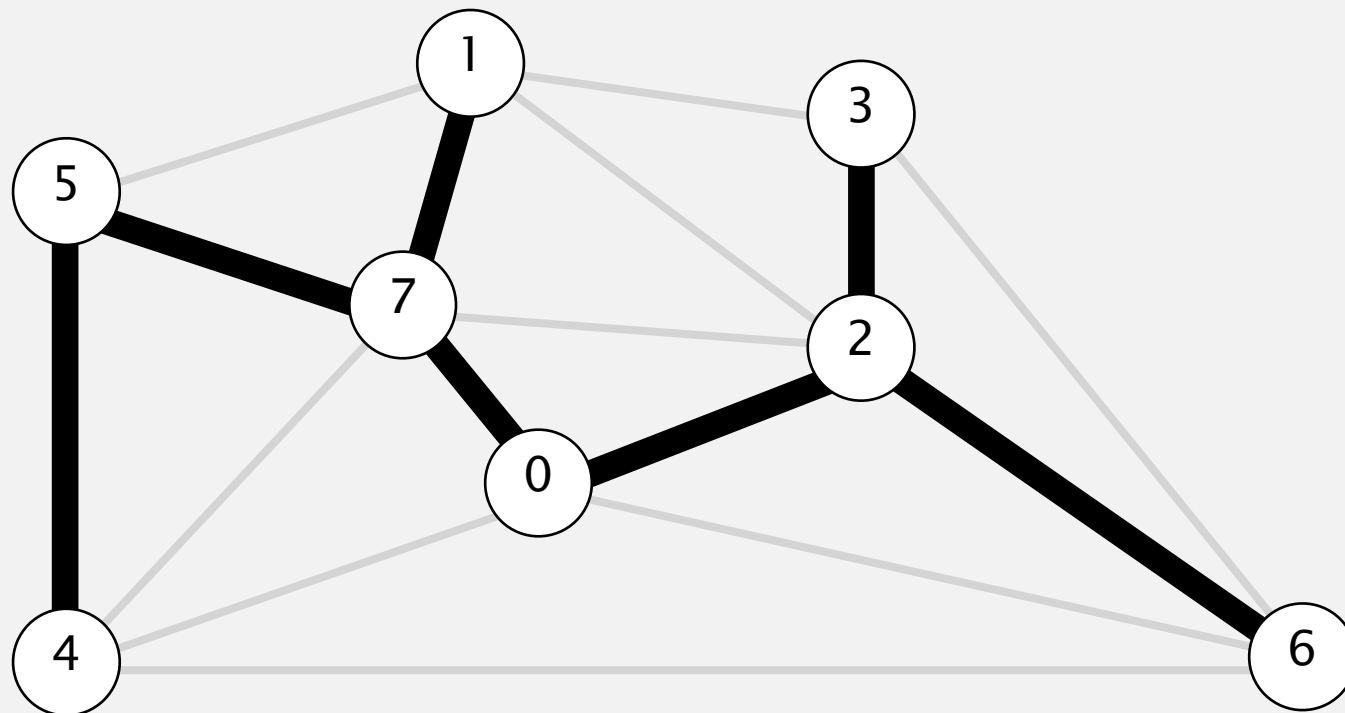
MST edges

0-7 1-7 0-2 2-3 5-7 4-5 6-2

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

stop since $V-1$ edges



MST edges

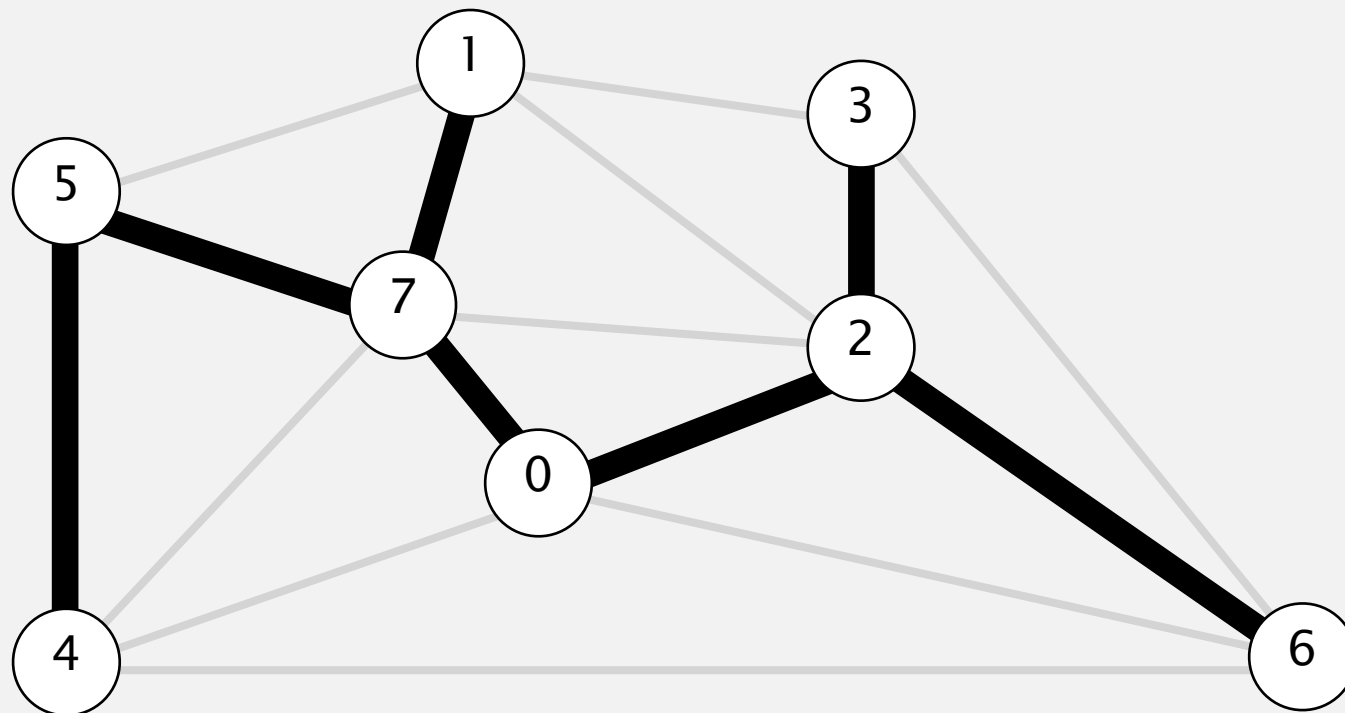
0-7 1-7 0-2 2-3 5-7 4-5 6-2

edges on PQ
(sorted by weight)

3-6	0.52
6-0	0.58
6-4	0.93

Prim's algorithm: lazy implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



MST edges

0-7 1-7 0-2 2-3 5-7 4-5 6-2



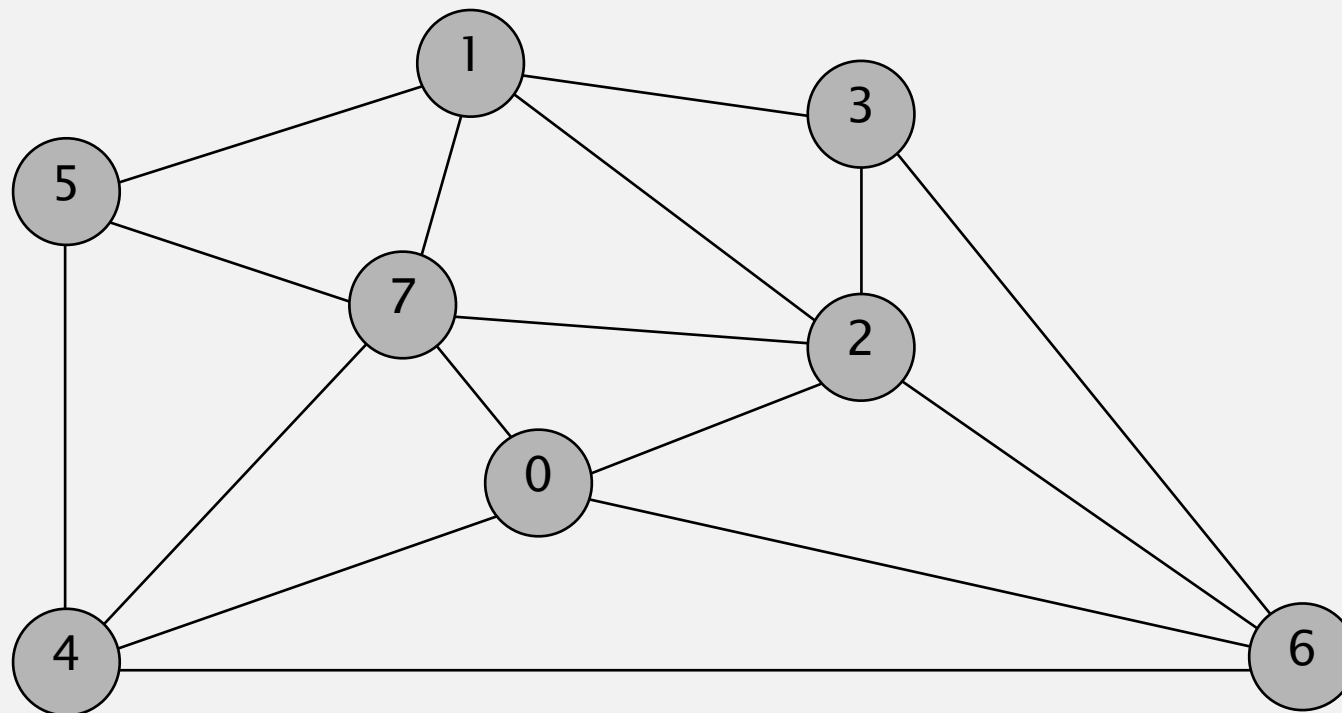
<http://algs4.cs.princeton.edu>

PRIM'S ALGORITHM DEMO

- ▶ *Prim's algorithm*
- ▶ *lazy implementation*
- ▶ *eager implementation*

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

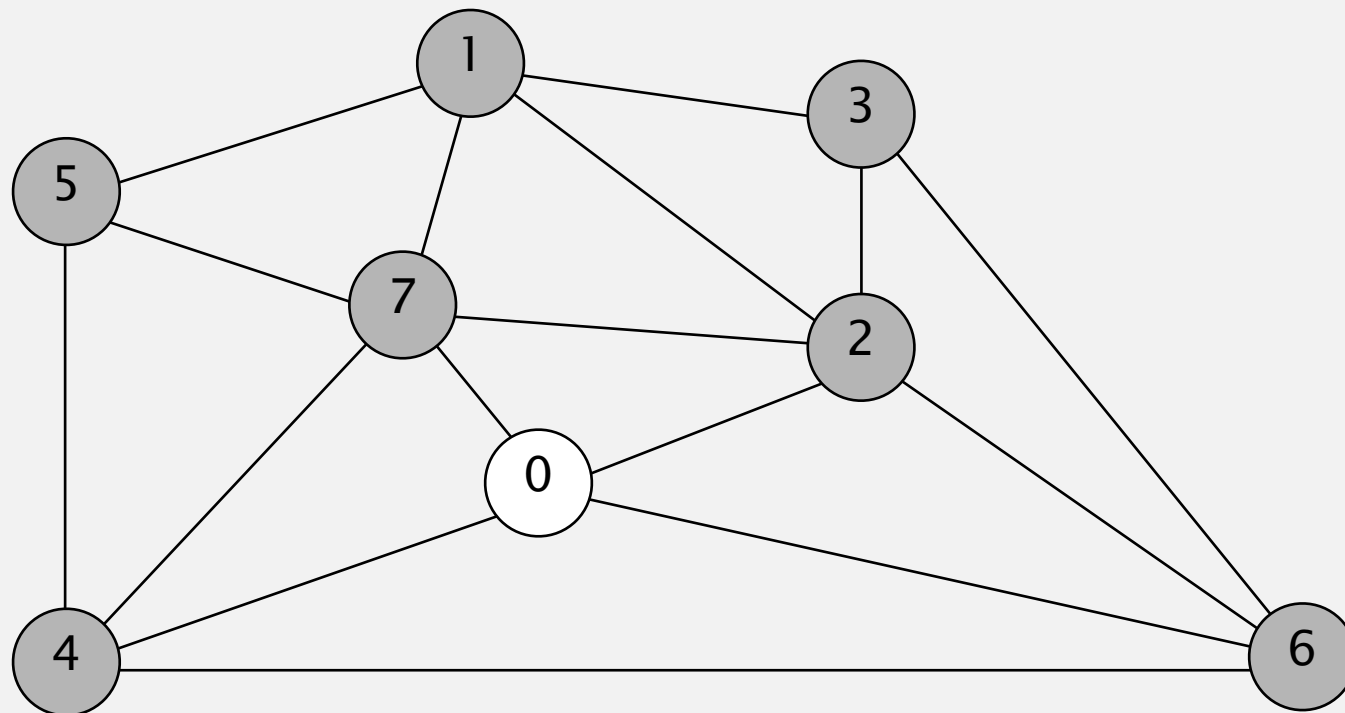


an edge-weighted graph

0-7	0.16
2-3	0.17
1-7	0.19
0-2	0.26
5-7	0.28
1-3	0.29
1-5	0.32
2-7	0.34
4-5	0.35
1-2	0.36
4-7	0.37
0-4	0.38
6-2	0.40
3-6	0.52
6-0	0.58
6-4	0.93

Prim's algorithm: eager implementation demo

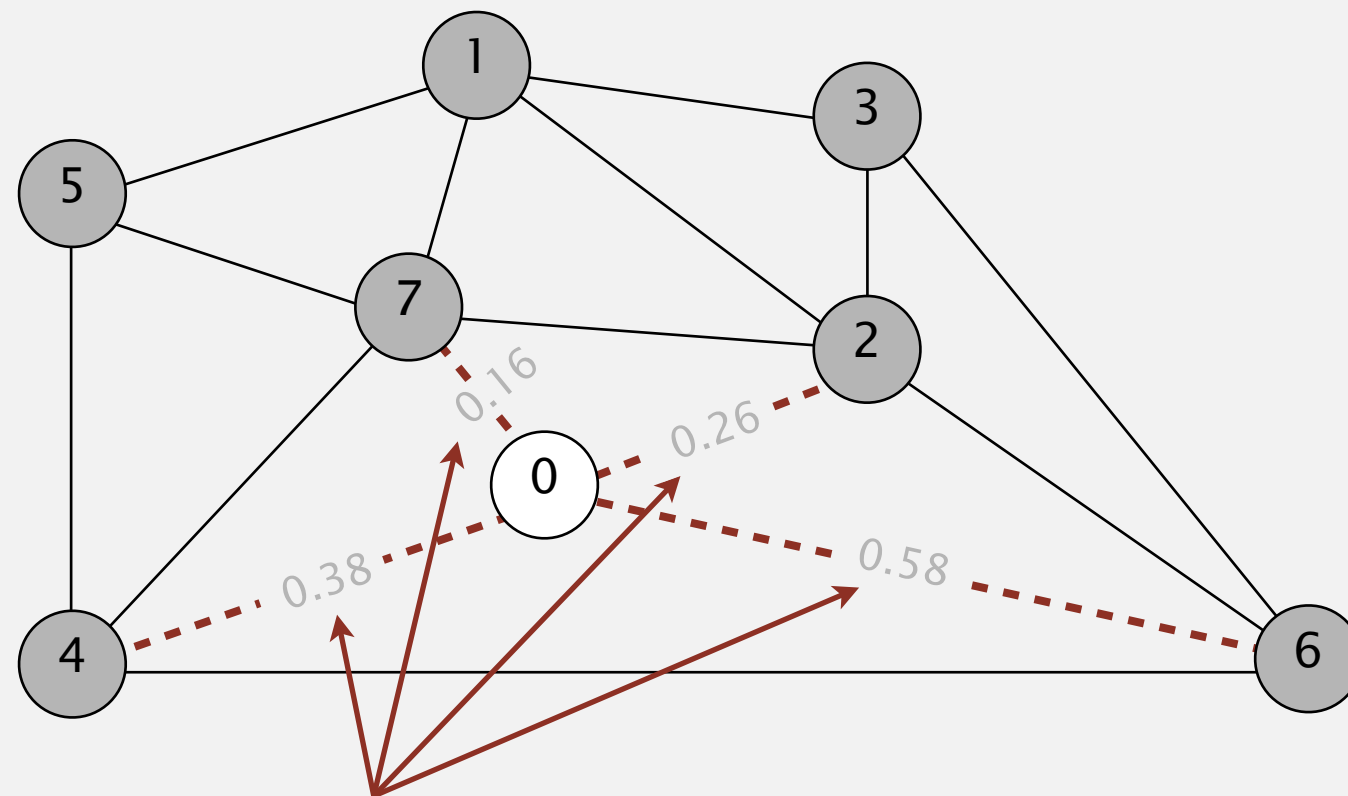
- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



v	edgeTo[]	distTo[]
→ 0	-	-

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



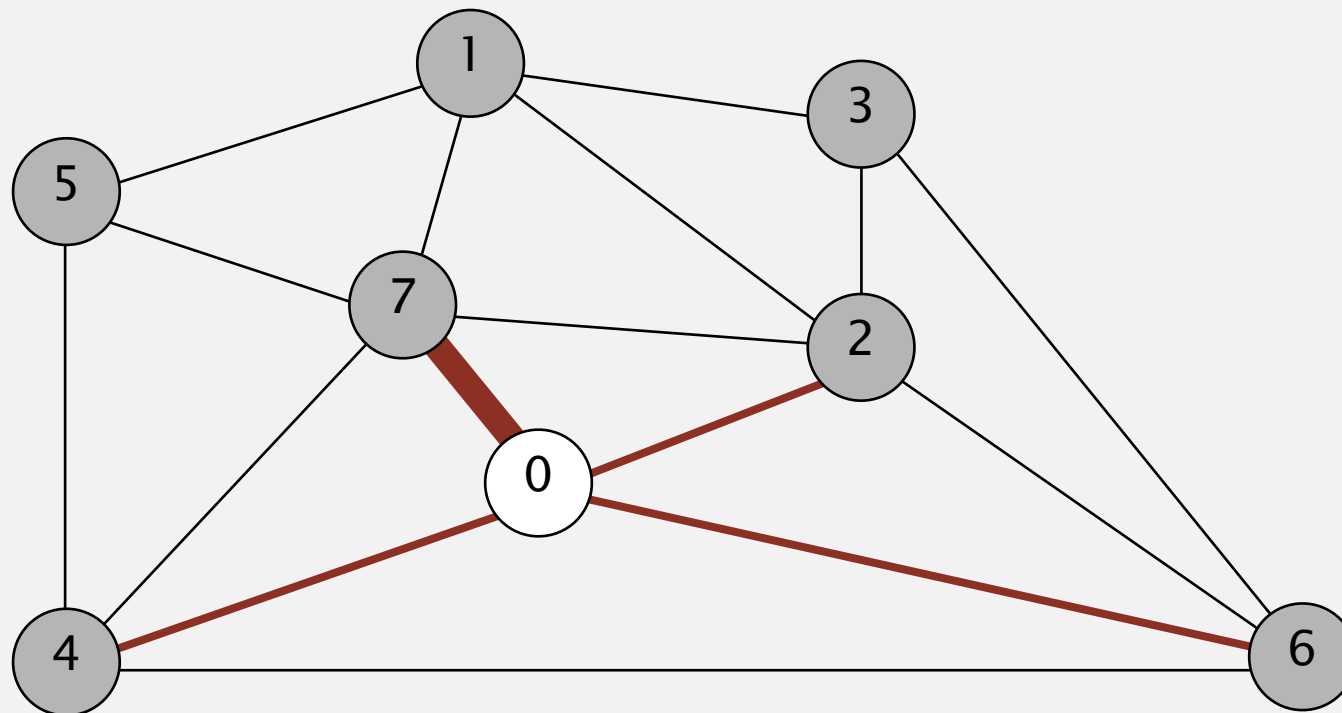
found connections to 7, 2, 4, and 6
(add to PQ)

v	edgeTo[]	distTo[]
→ 0	-	-
7	0-7	0.16
2	0-2	0.26
4	0-4	0.38
6	6-0	0.58

vertices on PQ
(sorted by weight)

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

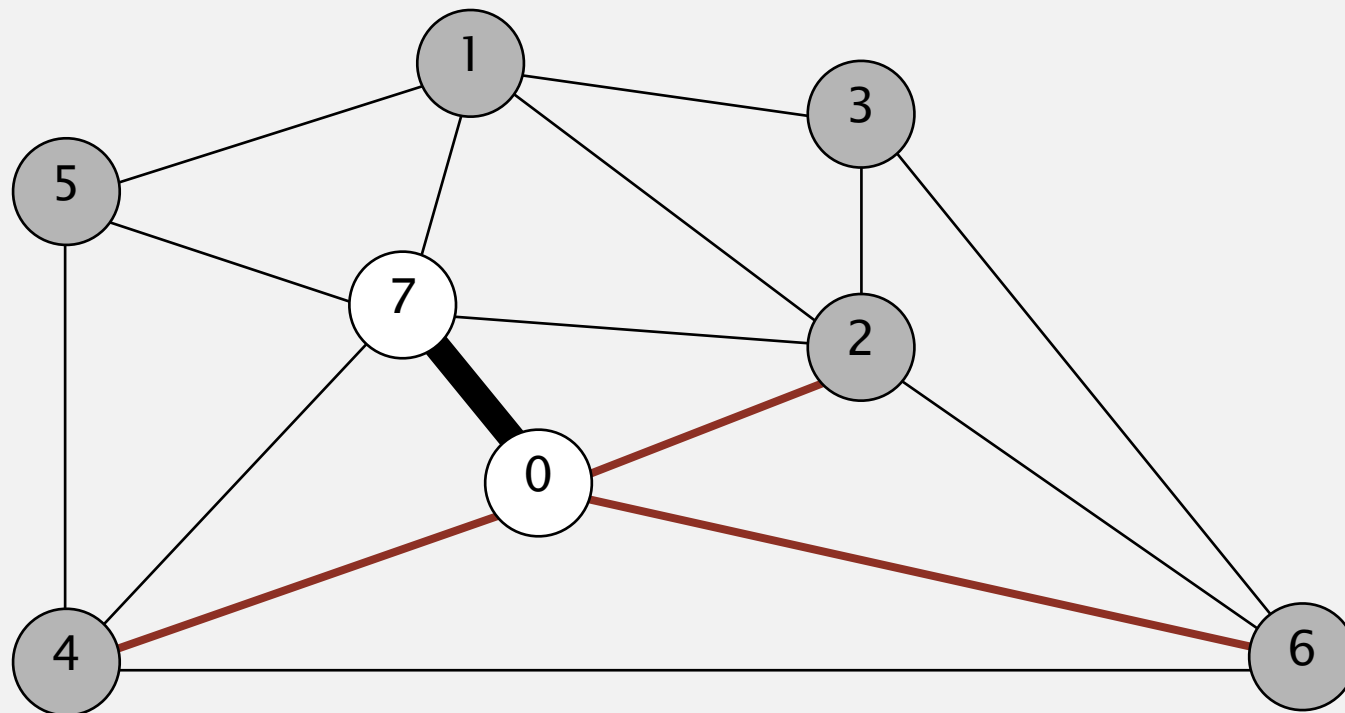


v	edgeTo[]	distTo[]
0	-	-
→ 7	0-7	0.16
2	0-2	0.26
4	0-4	0.38
6	6-0	0.58

vertices on PQ
(sorted by weight)

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



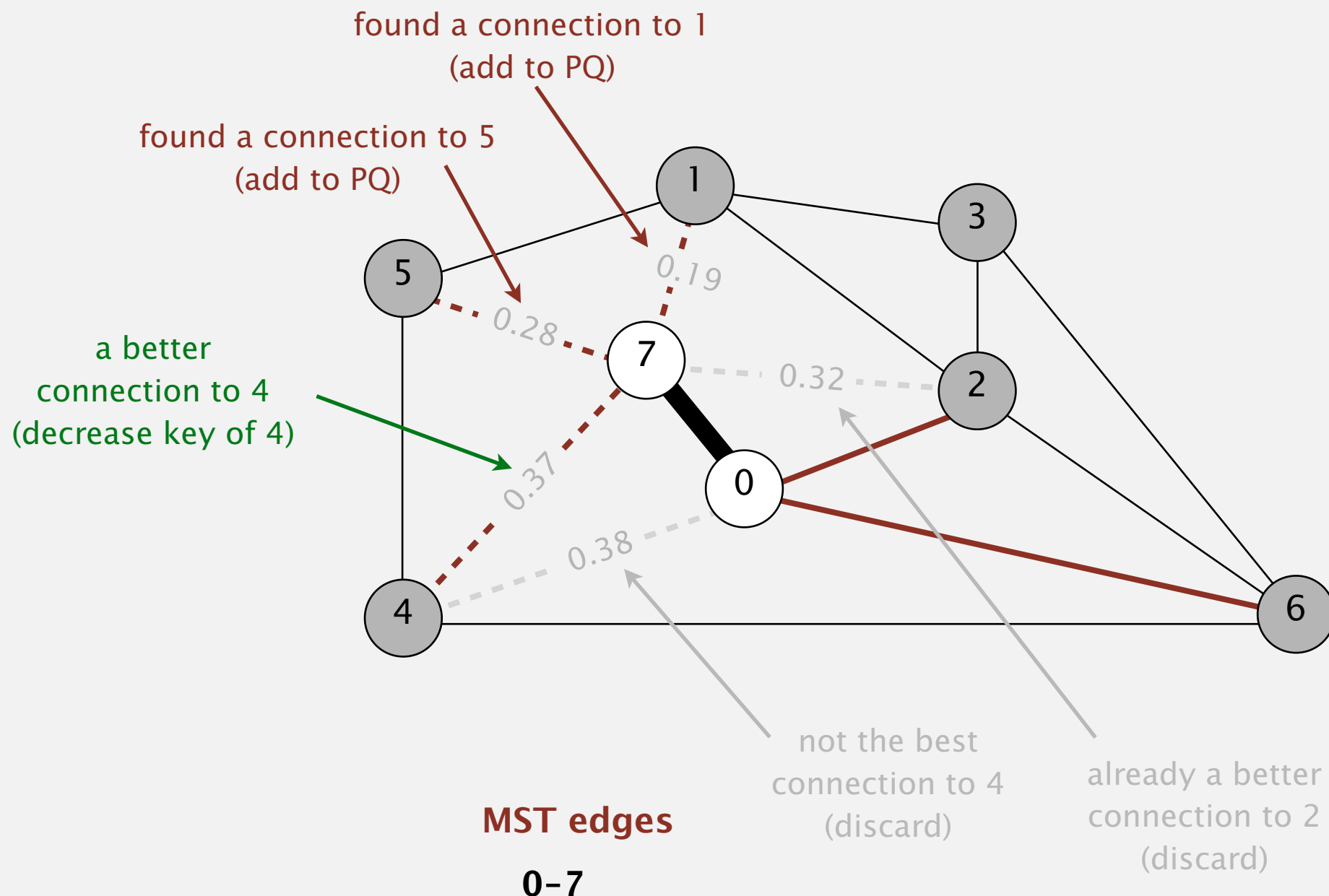
MST edges

0-7

v	edgeTo[]	distTo[]
0	-	-
→ 7	0-7	0.16
2	0-2	0.26
4	0-4	0.38
6	6-0	0.58

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

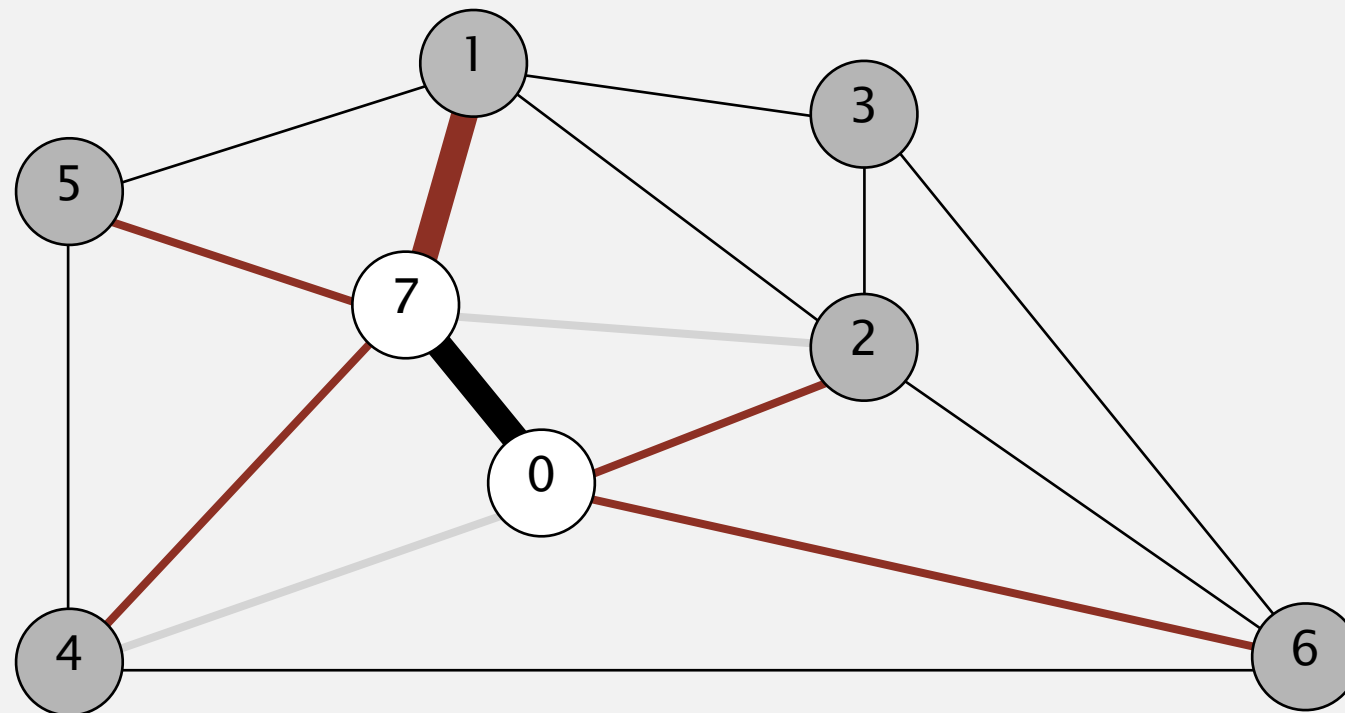


v	edgeTo[]	distTo[]
0	-	-
→ 7	0-7	0.16
①	1-7	0.19
2	0-2	0.26
⑤	5-7	0.28
④	0-4 4-7	0.38 0.37
6	6-0	0.58

vertices on PQ
(sorted by weight)

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



MST edges

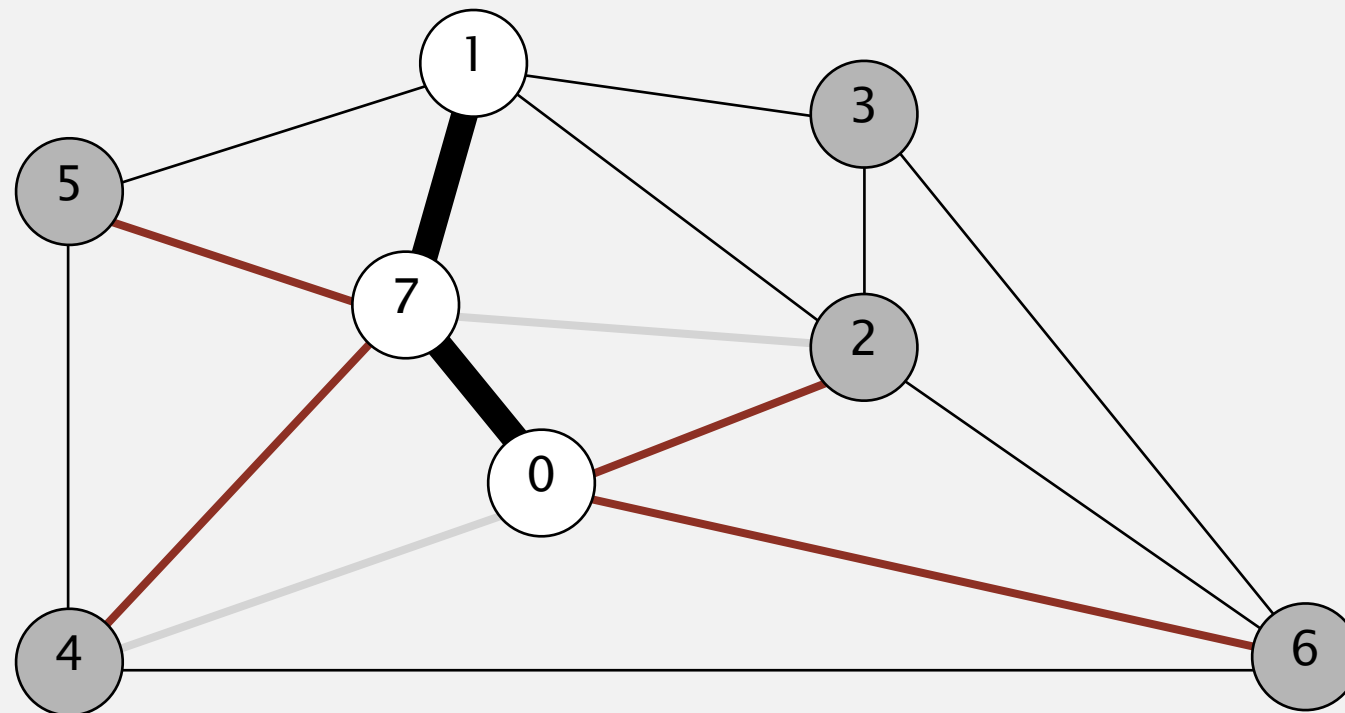
0-7 1-7

v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
→ 1	1-7	0.19
2	0-2	0.26
5	5-7	0.28
4	4-7	0.37
6	6-0	0.58

vertices on PQ
(sorted by weight)

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



MST edges

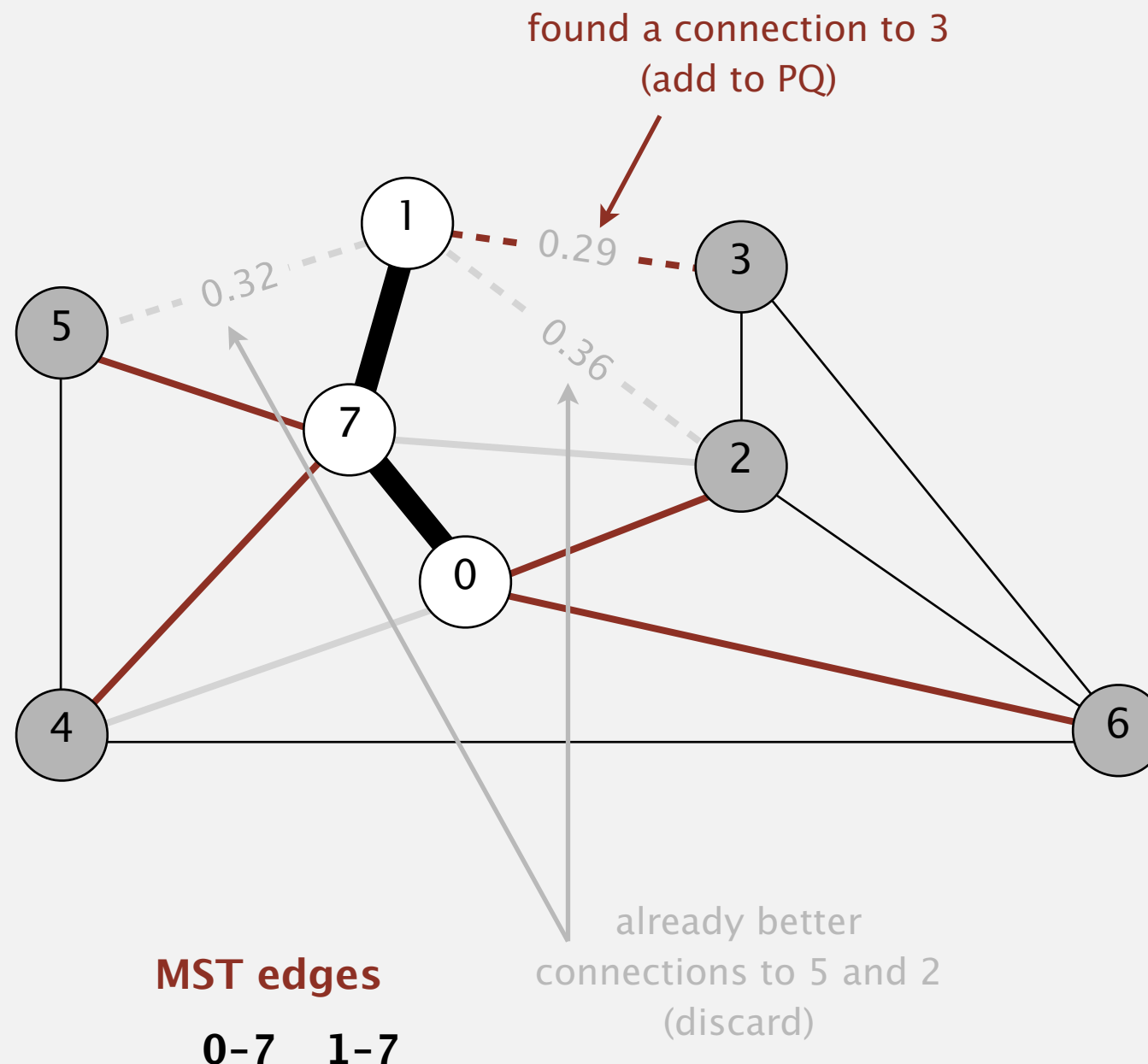
0-7 1-7

v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
→ 1	1-7	0.19
2	0-2	0.26
5	5-7	0.28
4	4-7	0.37
6	6-0	0.58

vertices on PQ
(sorted by weight)

Prim's algorithm: eager implementation demo

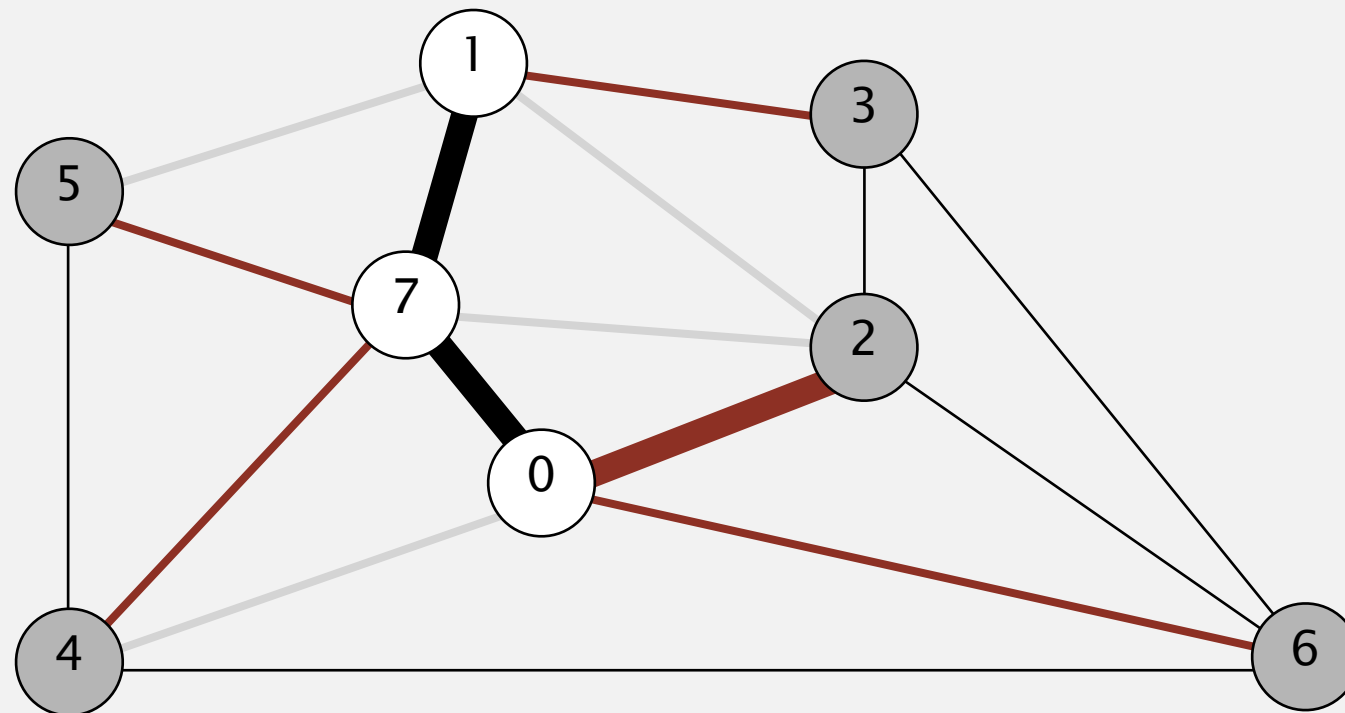
- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
→ 1	1-7	0.19
2	0-2	0.26
5	5-7	0.28
3	1-3	0.29
4	4-7	0.37
6	6-0	0.58

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



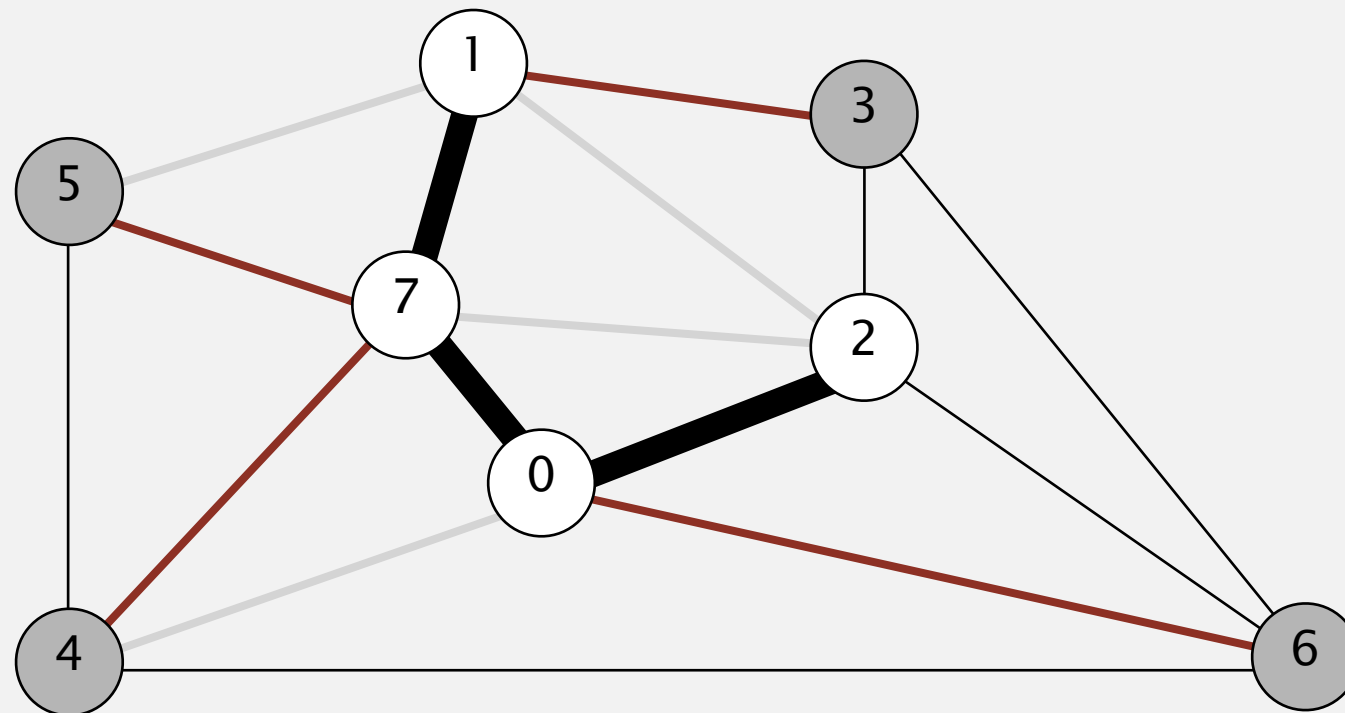
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
→ 2	0-2	0.26
5	5-7	0.28
3	1-3	0.29
4	4-7	0.37
6	6-0	0.58

MST edges

0-7 1-7

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



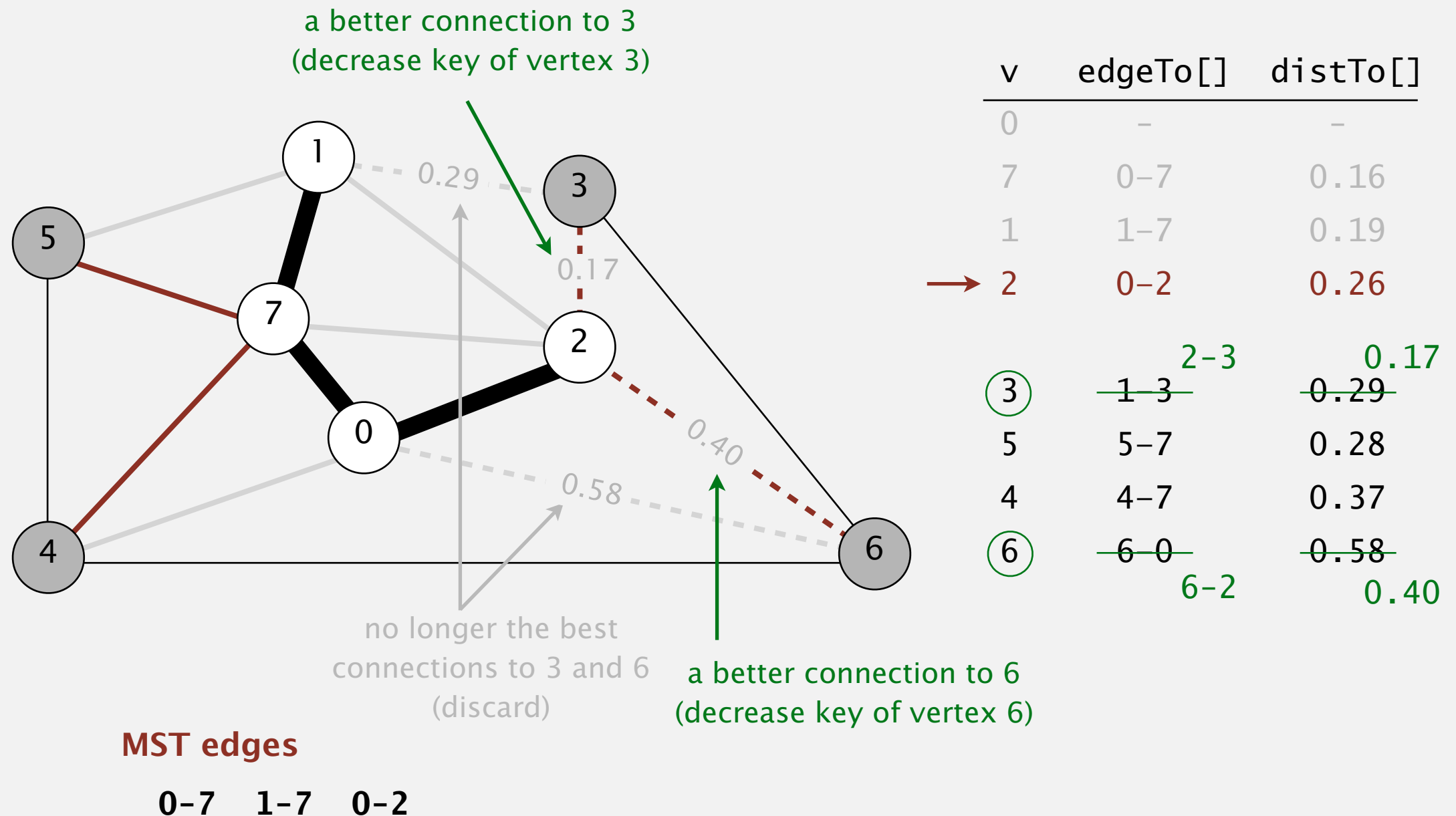
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
→ 2	0-2	0.26
5	5-7	0.28
3	1-3	0.29
4	4-7	0.37
6	6-0	0.58

MST edges

0-7 1-7 0-2

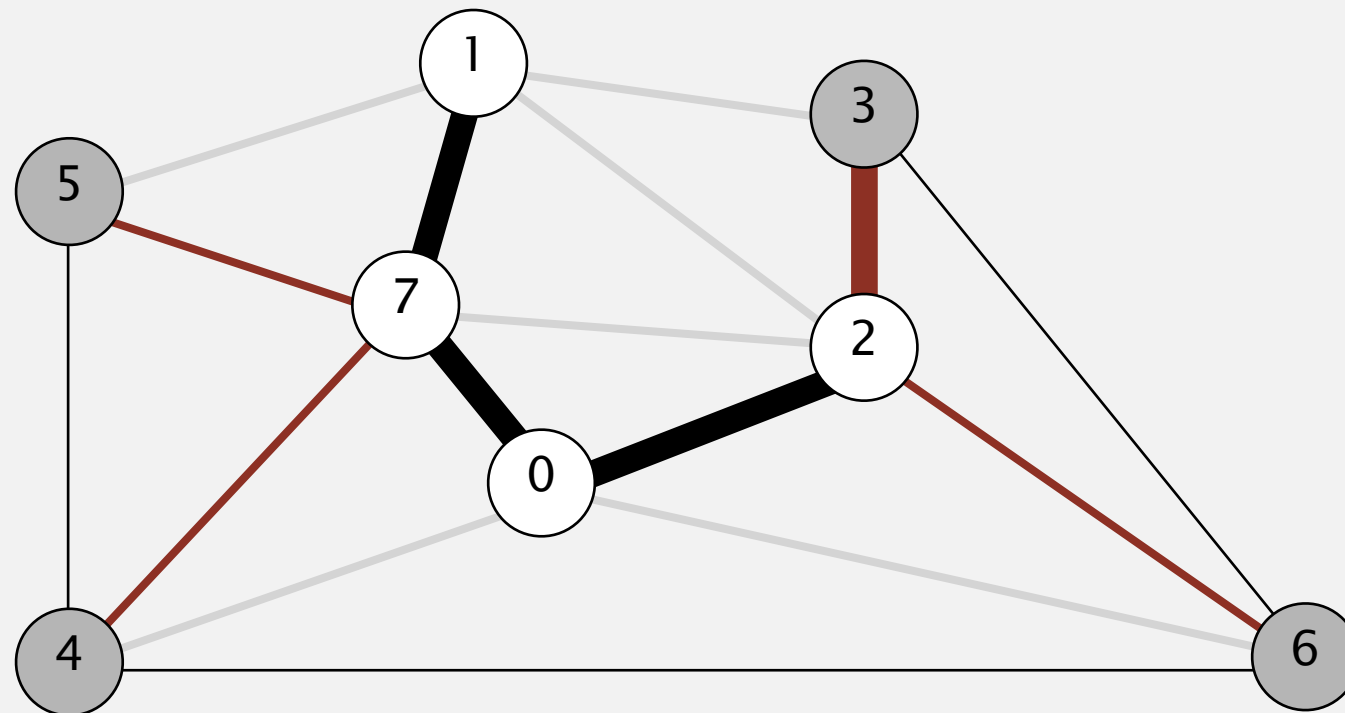
Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



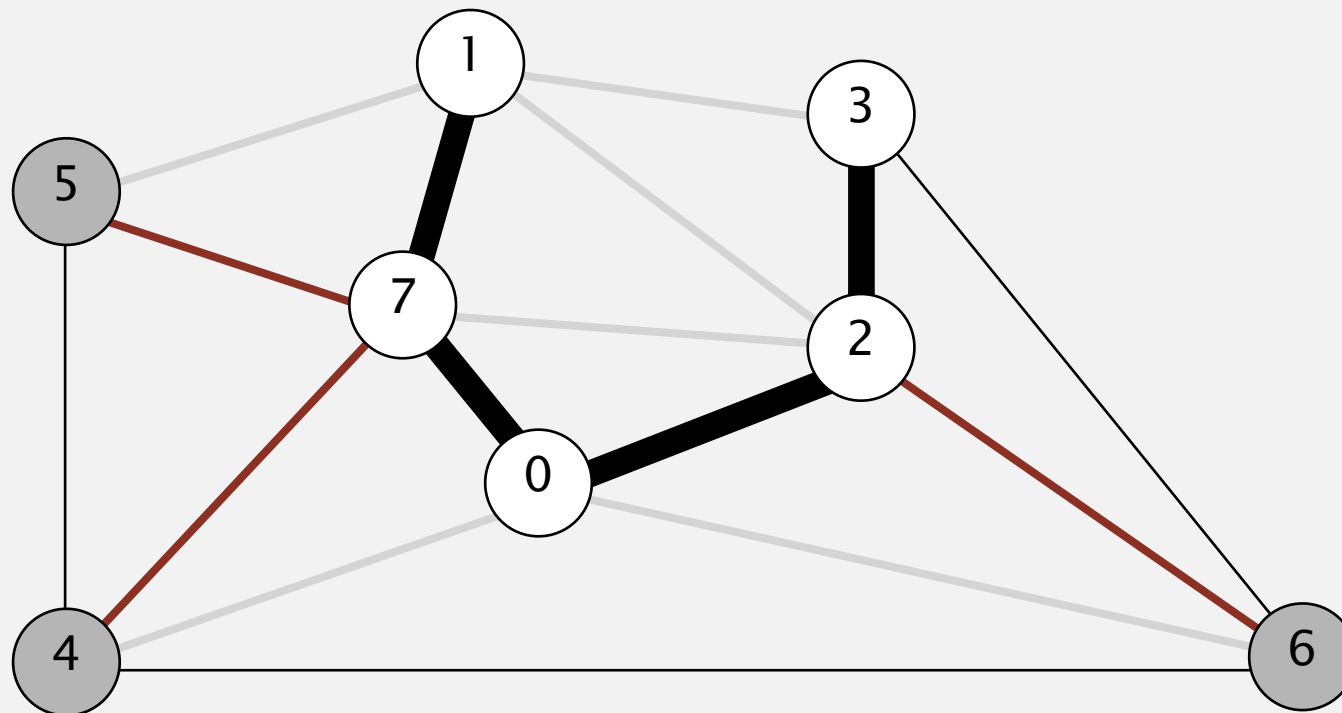
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
→ 3	2-3	0.17
5	5-7	0.28
4	4-7	0.37
6	6-2	0.40

MST edges

0-7 1-7 0-2 2-3

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



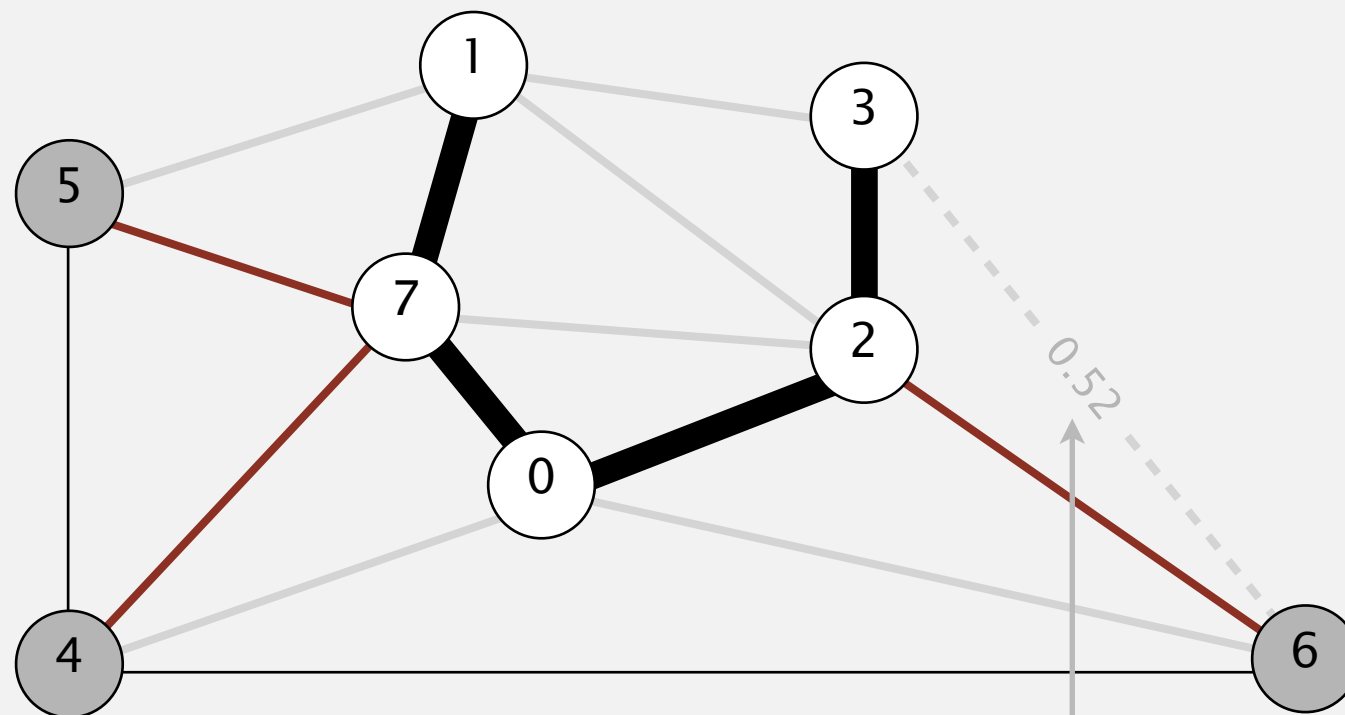
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
→ 3	2-3	0.17
5	5-7	0.28
4	4-7	0.37
6	6-2	0.40

MST edges

0-7 1-7 0-2 2-3

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



MST edges

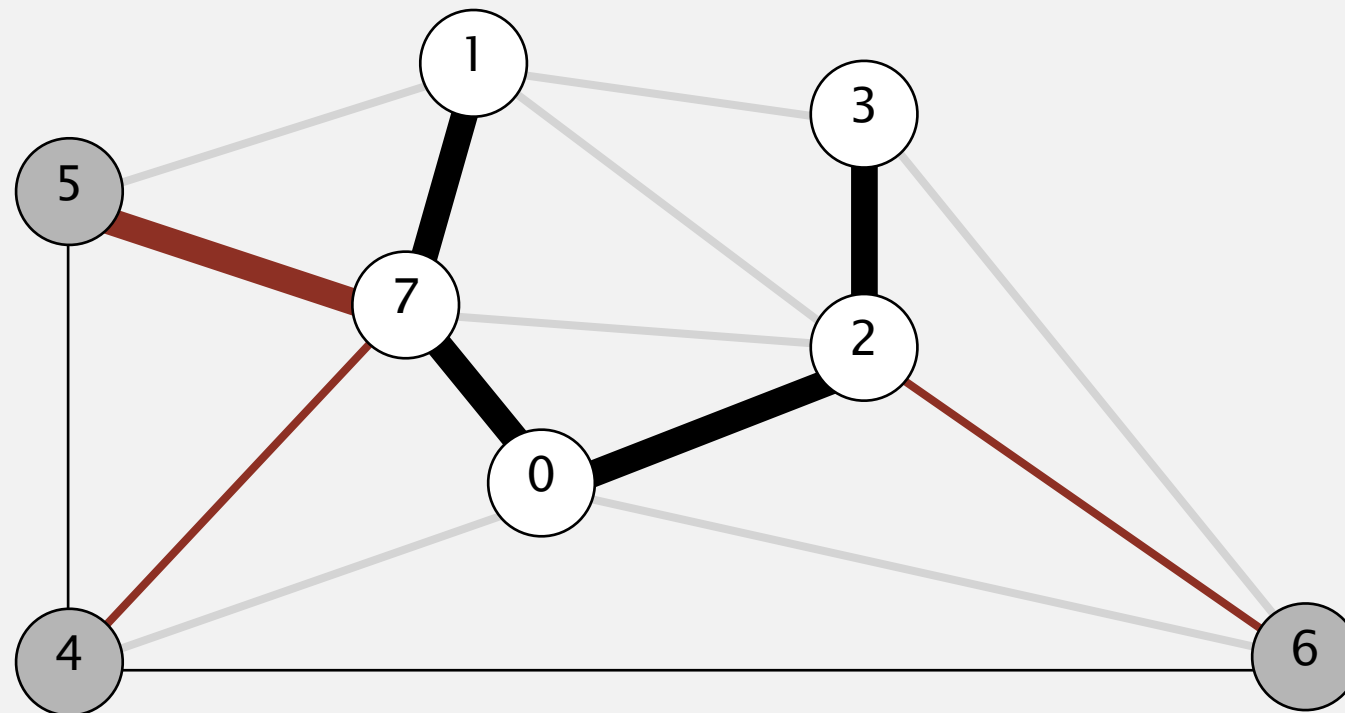
0-7 1-7 0-2 2-3

already a better
connection to 6
(discard)

v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
→ 3	2-3	0.17
5	5-7	0.28
4	4-7	0.37
6	6-2	0.40

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



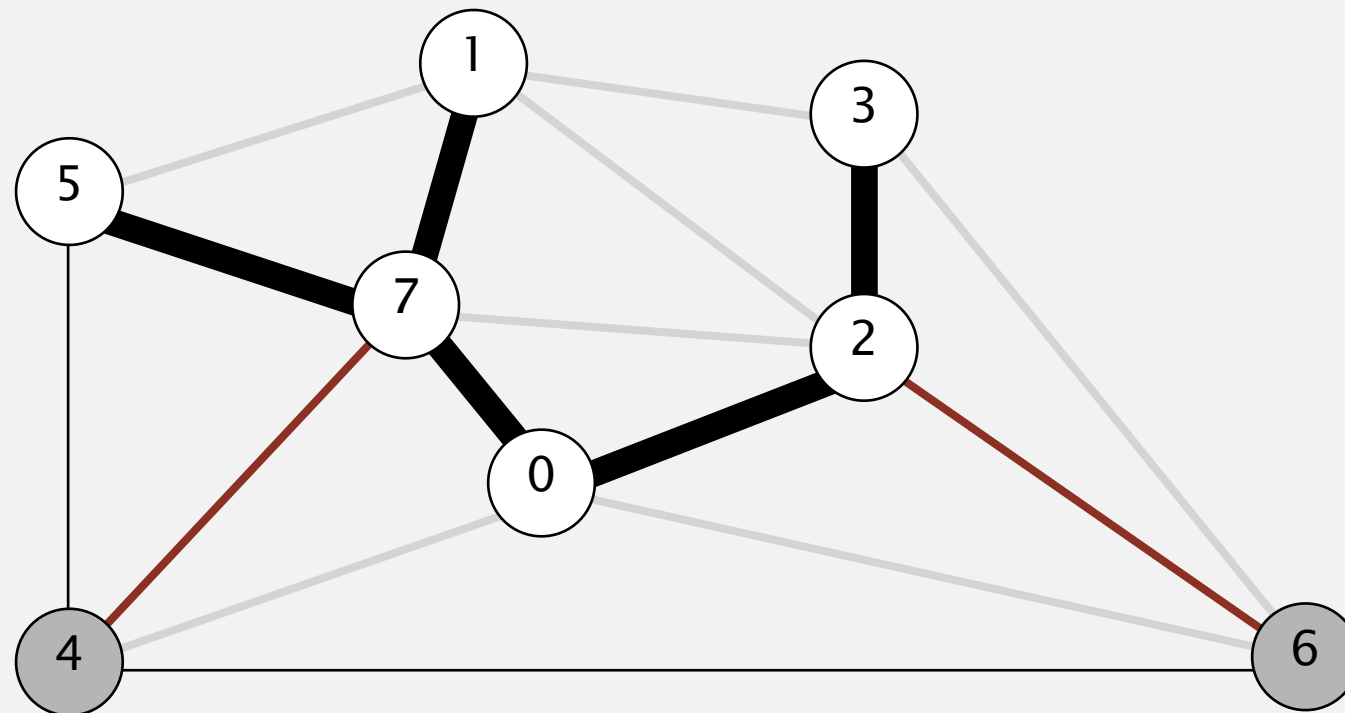
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
→ 5	5-7	0.28
4	4-7	0.37
6	6-2	0.40

MST edges

0-7 1-7 0-2 2-3

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
→ 5	5-7	0.28
4	4-7	0.37
6	6-2	0.40

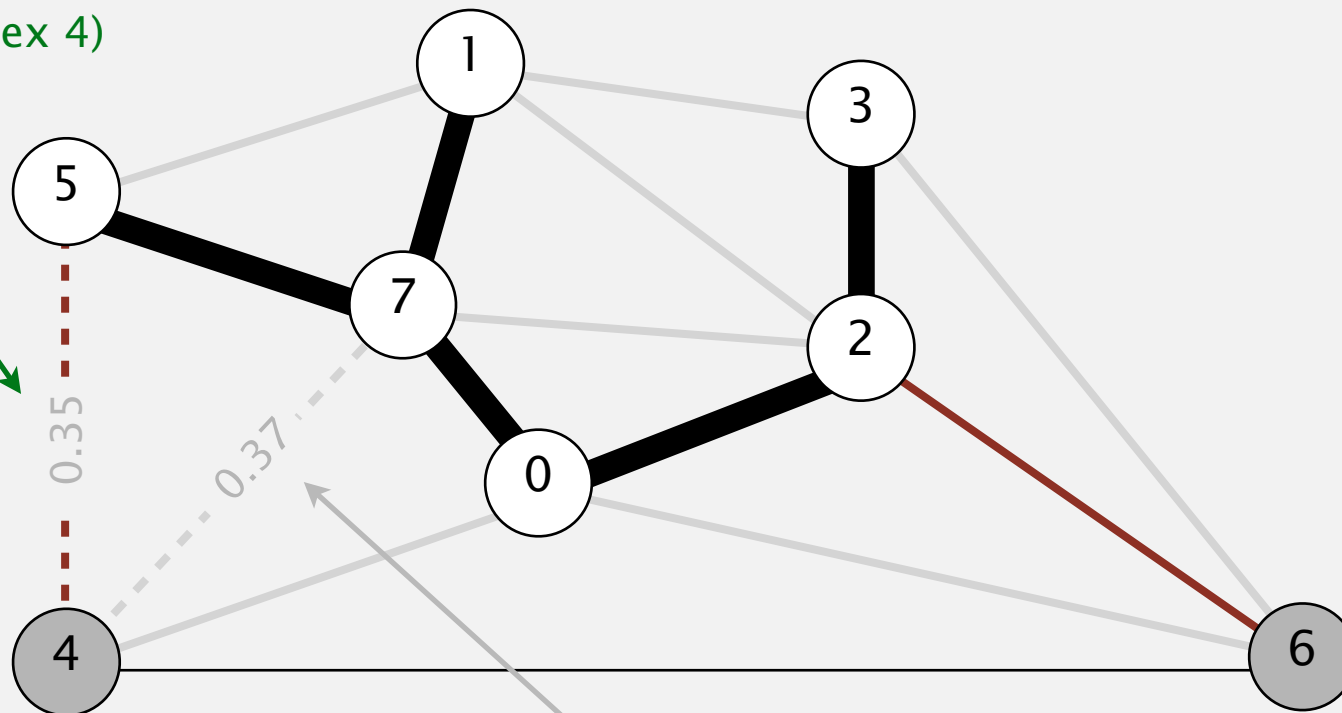
MST edges

0-7 1-7 0-2 2-3 5-7

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.

a better connection to 4
(decrease key of vertex 4)



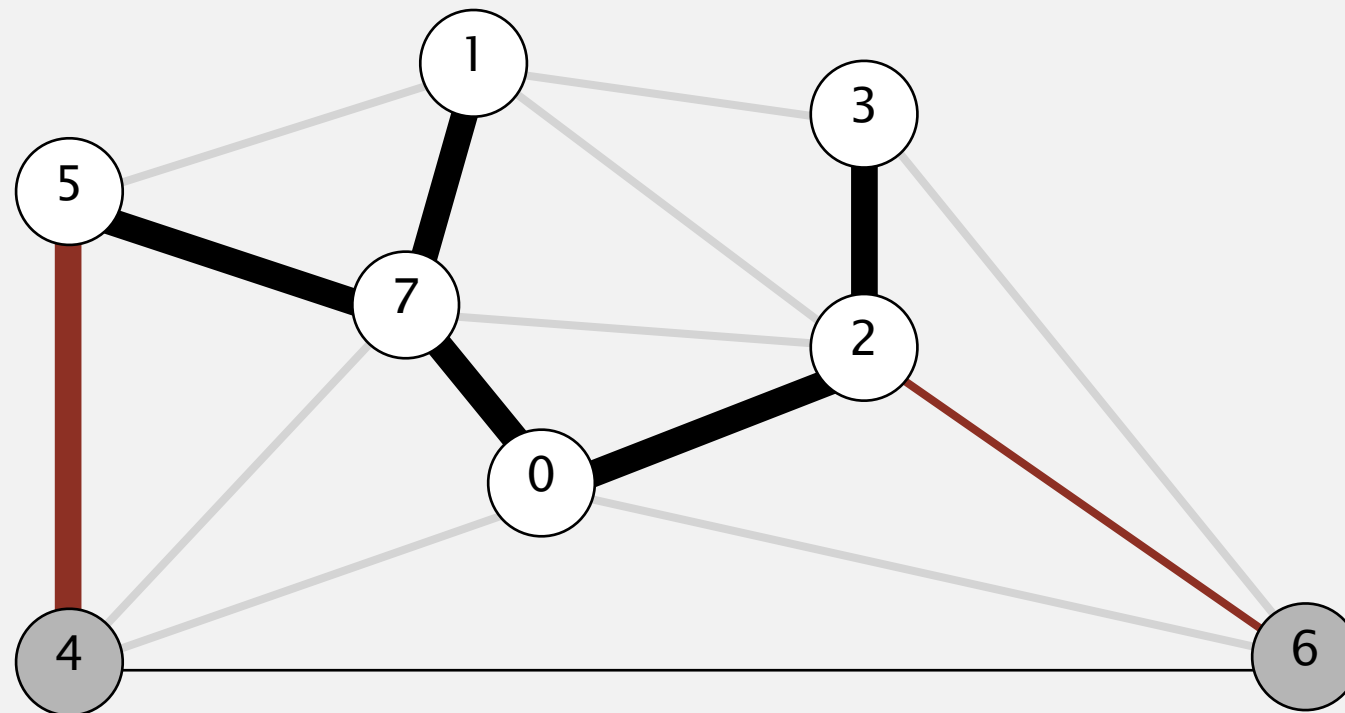
MST edges

0-7 1-7 0-2 2-3 5-7

v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
→ 5	5-7	0.28
4	4-7 ⁴⁻⁵	0.37 ^{0.35}
6	6-2	0.40

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



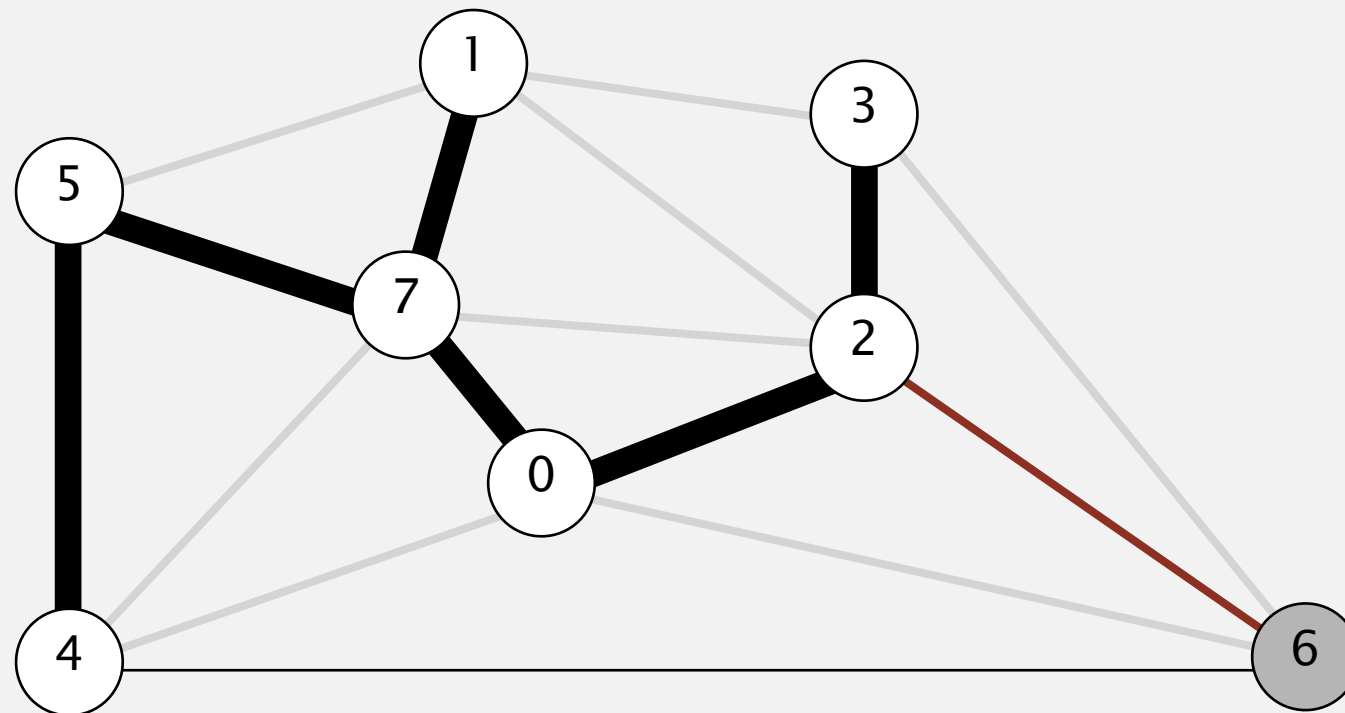
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
5	5-7	0.28
→ 4	4-5	0.35
6	6-2	0.40

MST edges

0-7 1-7 0-2 2-3 5-7

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



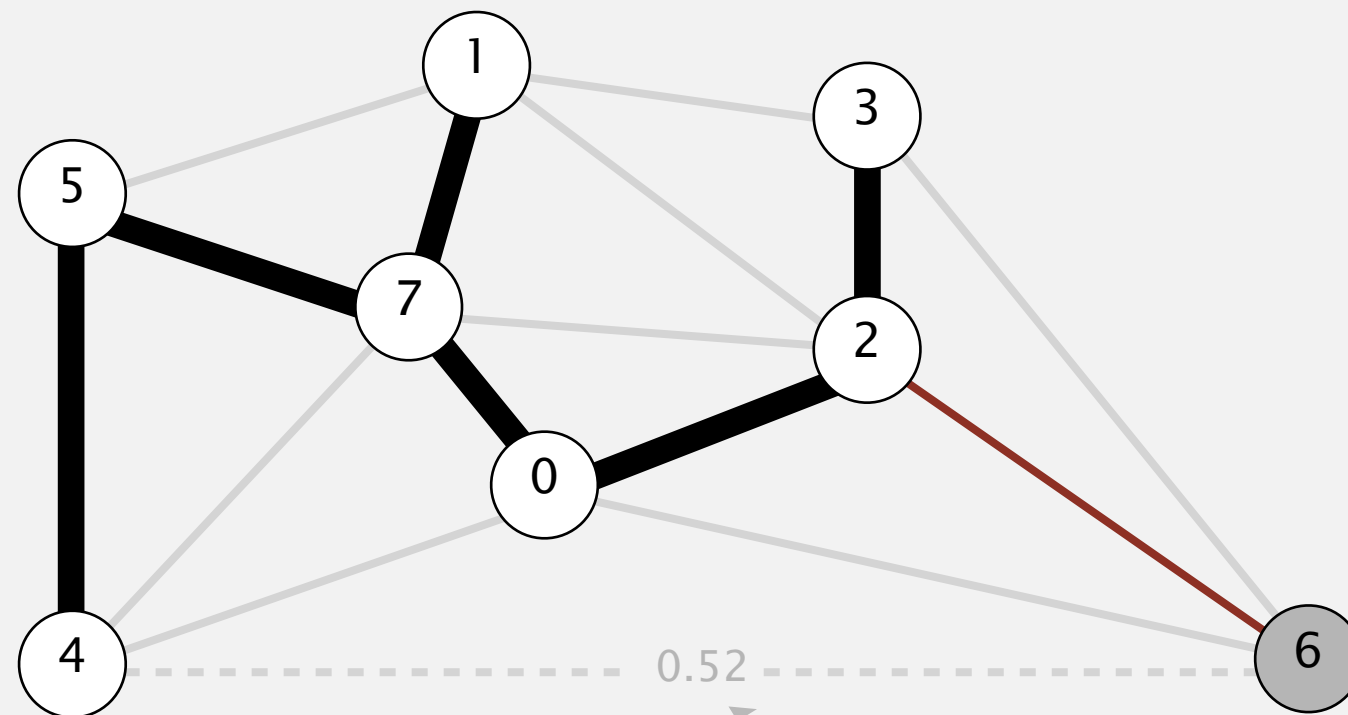
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
5	5-7	0.28
→ 4	4-5	0.35
6	6-2	0.40

MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



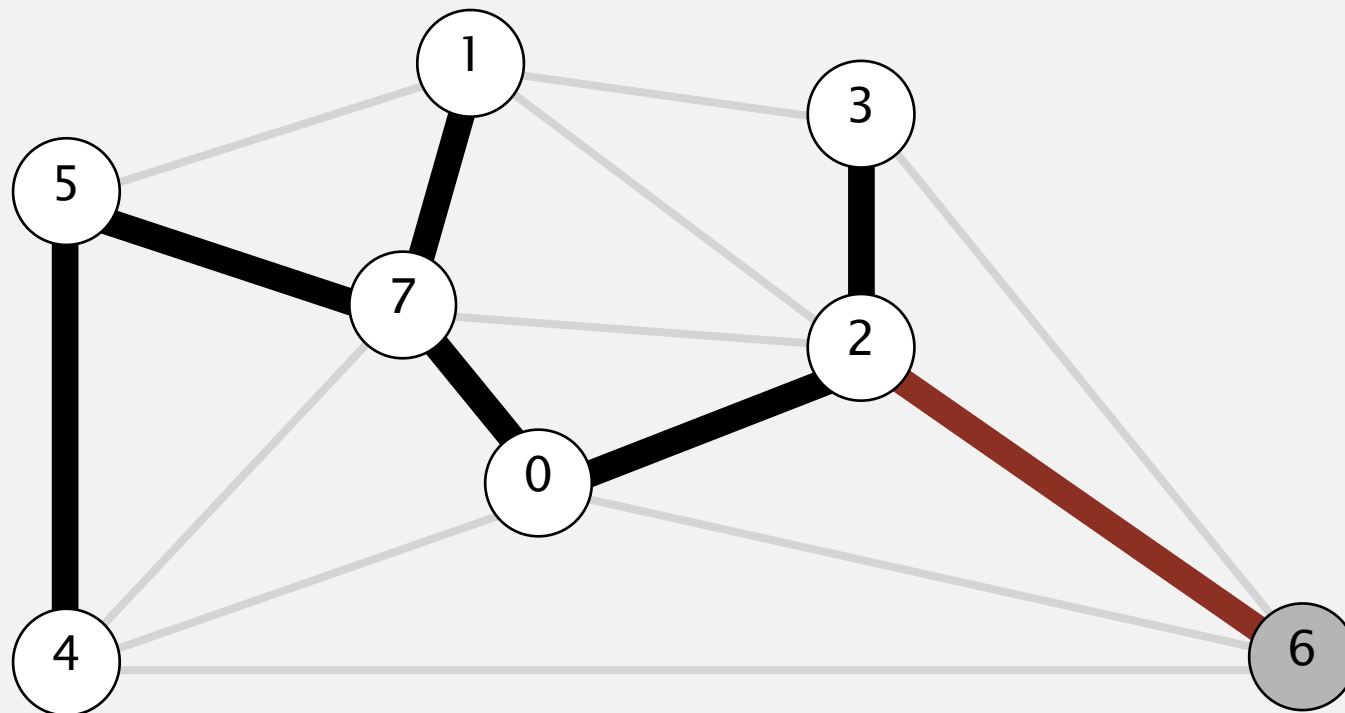
MST edges

0-7 1-7 0-2 2-3 5-7 4-5

v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
5	5-7	0.28
→ 4	4-5	0.35
6	6-2	0.40

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



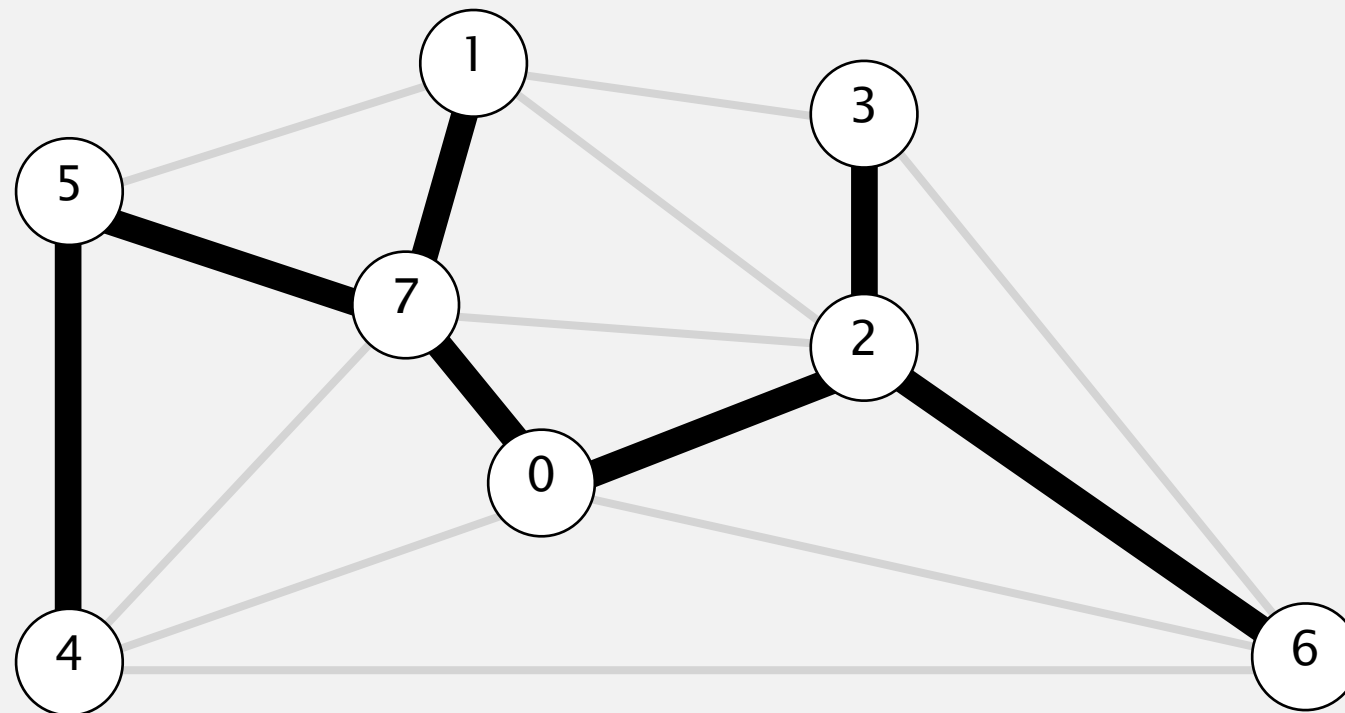
v	edgeTo[]	distTo[]
0	–	–
7	0–7	0.16
1	1–7	0.19
2	0–2	0.26
3	2–3	0.17
5	5–7	0.28
4	4–5	0.35
▶ 6	6–2	0.40

MST edges

0-7 1-7 0-2 2-3 5-7 4-5

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



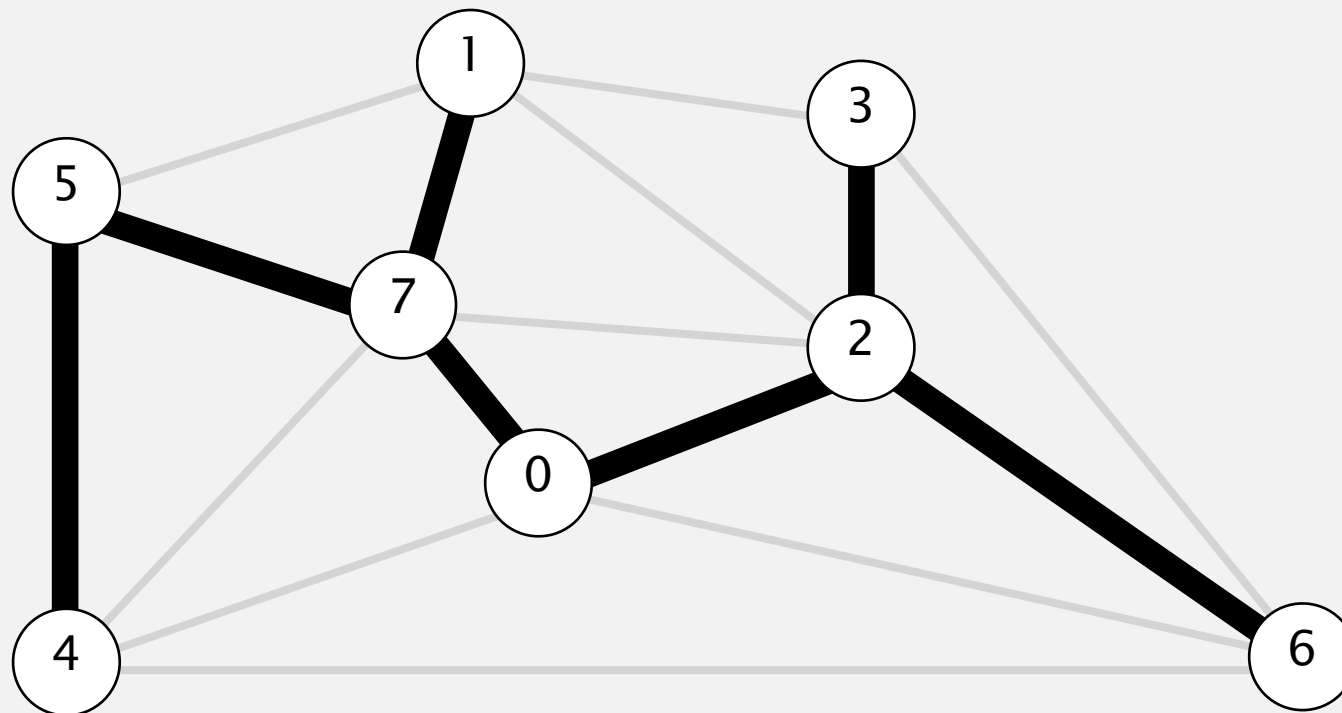
v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
5	5-7	0.28
4	4-5	0.35
→ 6	6-2	0.40

MST edges

0-7 1-7 0-2 2-3 5-7 4-5 6-2

Prim's algorithm: eager implementation demo

- Start with vertex 0 and greedily grow tree T .
- Add to T the min weight edge with exactly one endpoint in T .
- Repeat until $V - 1$ edges.



MST edges

0-7 1-7 0-2 2-3 5-7 4-5 6-2

v	edgeTo[]	distTo[]
0	-	-
7	0-7	0.16
1	1-7	0.19
2	0-2	0.26
3	2-3	0.17
5	5-7	0.28
4	4-5	0.35
6	6-2	0.40